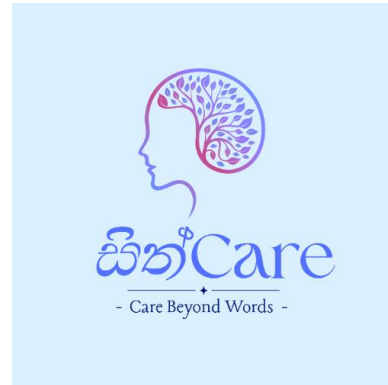




SENG 31242 – System Design Project

Final Design Report

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26/06/2026



Declaration

We, the undersigned members of Group 10, hereby declare that this Final Design Report for SithCare - Mental Health Channeling Application is our own original work prepared in partial fulfilment of the requirements for SENG 31242 - System Design Project. All sources of information and references have been duly acknowledged. This document has not been submitted elsewhere for any academic or professional qualification.

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Our heartfelt thanks go to the mental health professionals, patients, and community stakeholders who participated in our interviews and surveys, providing the essential insights that shaped the requirements and design of SithCare. We also acknowledge the contributions of open-source communities and the developers of the MERN stack technologies that form the foundation of our technical architecture.



Abstract

SithCare ("Care Beyond Words") is a web-based mental health channelling application designed specifically for the Sri Lankan context to address the critical gap in accessible, private, and stigma-free mental health support. The system enables patients to discover qualified mental health specialists, book online or in-person appointments, access telemedicine consultations, and receive guidance through an AI-powered triage chatbot named SithBot. This Final Design Report consolidates the complete system analysis and design artefacts produced during the SENG 31242 System Design Project. The report encompasses a comprehensive Software Requirements Specification (SRS) derived from structured interviews, questionnaires, and competitive analysis; and a detailed Software Design Specification (SDS) covering the Layered MVC architecture, MERN technology stack, class diagrams, sequence diagrams, database design, user interface wireframes, and security design. The system supports distinct role-based dashboards for patients, clinic administrators, platform administrators, volunteer listeners, and community moderators, with multi-clinic management, a live patient queue tracker, anonymous community support, and an emergency response subsystem. All design decisions are traceable to the non-functional requirements through an explicit mapping framework.

Keywords: *mental health, telemedicine, appointment booking, system design, SRS, SDS, Sri Lanka, MERN stack, SithCare, SENG 31242*



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Abbreviations

Abbreviation	Definition
SRS	Software Requirements Specification
SDS	Software Design Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
RBAC	Role-Based Access Control
JWT	JSON Web Token
OTP	One-Time Password
MFA	Multi-Factor Authentication
LKR	Sri Lankan Rupee
API	Application Programming Interface
TLS	Transport Layer Security
AES	Advanced Encryption Standard
MVP	Minimum Viable Product
MoSCoW	Must-have, Should-have, Could-have, Won't-have
PDPA	Personal Data Protection Act
SLMC	Sri Lanka Medical Council
SLPA	Sri Lanka Psychological Association
CBT	Cognitive Behavioural Therapy
MERN	MongoDB, Express.js, React.js, Node.js
ODM	Object Data Modeling
SPA	Single Page Application
REST	Representational State Transfer
CDN	Content Delivery Network
CI/CD	Continuous Integration / Continuous Deployment
WCAG	Web Content Accessibility Guidelines
OWASP	Open Web Application Security Project
SQL	Structured Query Language
NoSQL	Not Only SQL

Table 0.1: Abbreviations



1. INTRODUCTION

1.1. Background

Mental health care in Sri Lanka faces significant challenges that limit access to professional support for millions of citizens. According to the WHO Mental Health Atlas 2024, Sri Lanka has only 2.5 mental health workers per 100,000 population, indicating a severe shortage of qualified professionals. The National Mental Health Policy 2020-2030 emphasises community support, early detection, continuing care, and stigma reduction as national priorities. Despite these policy directives, the treatment gap remains substantial, with many individuals who suffer from mental disorders not seeking professional help due to barriers including stigma, poor awareness, transport limitations, language barriers, and financial constraints.

The current landscape of mental health service delivery relies primarily on outpatient department queues at government hospitals and manual telephone bookings at private psychiatric clinics. These traditional methods create significant pain points: the fear of being overheard during phone bookings, lack of digital queue tracking resulting in waiting times of one to four hours at clinic premises, absence of telemedicine options for rural patients, and limited knowledge of available crisis support helplines. The Sri Lankan Digital Health Blueprint 2023 supports the adoption of digital tools to improve healthcare accessibility and timely service delivery, creating a favourable policy environment for technology-enabled mental health solutions.

SithCare ("Care Beyond Words") is proposed as a comprehensive web-based mental health channelling application specifically designed for the Sri Lankan context. The platform aims to bridge the gap between patients and mental health professionals by providing a secure, private, and stigma-free digital environment for discovering specialists, booking appointments, accessing telemedicine consultations, and receiving community support.

1.2. Problem Statement

The mental health care ecosystem in Sri Lanka is characterised by several interconnected problems that collectively limit access to professional support:

- No centralised directory of licensed mental health professionals is publicly accessible, making it difficult for patients to find verified specialists.



- Telephone-only booking creates stigma, as patients fear being overheard by family members or colleagues when making appointments.
- No digital queue tracking exists; patients wait at clinic premises for extended periods without information about estimated wait times.
- Payment records are maintained in paper ledgers, with no digital receipts or payment history available to patients.
- No telemedicine option exists for rural patients or those with mobility issues, effectively excluding a significant portion of the population.
- Crisis support helplines are unknown to most patients, and numbers are not widely publicised or easily accessible during emergencies.

SithCare addresses these problems through a unified digital platform that combines specialist discovery, appointment management, telemedicine, emergency support, and anonymous community features tailored specifically for the Sri Lankan context.

1.3. Aims and Objectives

The primary aim of this project is to design a scalable, secure, and user-friendly mental health channelling platform that improves access to mental health services in Sri Lanka. The specific objectives are:

1. To conduct comprehensive requirements engineering through structured interviews, surveys, and competitive analysis to identify user needs and system requirements.
2. Producing a professional-grade Software Requirements Specification (SRS) documenting all functional and non-functional requirements.
3. To design a scalable software architecture and produce a detailed Software Design Specification (SDS) including class diagrams, sequence diagrams, database design, and user interface wireframes.
4. To plan and manage the project using GitHub Projects, adhering to industry-standard agile practices.
5. Producing professional documentation artefacts including the final design report, meeting notes, and architectural decision records.

1.4. Scope

The scope of the SithCare platform for the design phase (SENG 31242) and subsequent development phase (SENG 34213) encompasses the following:



1.4.1. In Scope

- Patient registration and authentication with OTP verification and role-based access control.
- Mental health specialist search and discovery with filtering by category, language, location, and availability.
- Appointment booking, rescheduling, and cancellation with SMS and email notifications.
- Online payment processing through PayHere gateway in Sri Lankan Rupees (LKR).
- Telemedicine video consultations with one-tap phone fallback.
- Live patient queue tracking with proximity notifications.
- Emergency support subsystem with helpline display and consent-based location sharing.
- Anonymous community consultation forum with pseudonymous identities.
- AI-powered triage chatbot (SithBot) for specialist category recommendations.
- Multi-language support (Sinhala, Tamil, English) and Adult Mode for elderly users.

1.4.2. Out of Scope

- Native mobile applications for Android and iOS (future consideration).
- Integration with government electronic health record (EHR) systems.
- Insurance claim processing and complex fundraising automation.
- AI clinical diagnosis (SithBot provides triage guidance only, not diagnoses).
- Automatic emergency service dispatch without explicit user confirmation.

1.5. Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 presents the System Analysis, encompassing fact-finding techniques, existing system analysis, use case modelling, functional and non-functional requirements, and feasibility assessment. Chapter 3 details the System Design, including architectural decisions, technology stack selection, detailed design artefacts, user interface design, security design, and non-functional design decision mapping. Chapter 4 provides the Conclusion, summarising objectives met, identifying limitations, and outlining future work. References and Appendices follow the main chapters.



2. SYSTEM ANALYSIS

2.1. Fact-Finding Techniques

The requirements for SithCare were elicited using multiple formal fact-finding methods as mandated by the SENG 31242 course guidelines. The techniques employed and their key outcomes are summarised below:

- **Technique: Structured Interviews:** Details: 5 interviews with mental health patients | Key Findings: Stigma is the primary barrier; patients prefer an anonymous search; the elderly need simplified UX
- **Online Questionnaire:** 150 respondents via Google Forms | Quantitative data on service gaps, preferred features, willingness to use digital platforms
- **Specialist Survey:** 6 interviews with mental health specialists | Identified pain points and requirements for digital professional activities
- **Document Analysis:** WHO Atlas 2024, MoH Policy 2020-2030, Digital Health Blueprint 2023 | 2.5 workers per 100,000; significant treatment gap; government digital health push
- **Competitive Analysis:** BetterHelp, Talkspace, Wysa, NalandaCare | No existing platform provides full Sri Lankan localisation with multi-clinic management

2.2. Existing System Analysis

2.2.1. Current State and Pain Points

Mental health care access in Sri Lanka currently relies on outpatient department queues at government hospitals, manual telephone bookings at private psychiatric clinics, and informal WhatsApp groups for community support. Competitive analysis revealed significant gaps in existing solutions. No existing platform combines full Sri Lankan localisation (Sinhala/Tamil/English, LKR currency, PayHere payments), multi-clinic management, live queue tracking, AI triage, emergency support, anonymous community features, and volunteer listener support in a single integrated solution.

2.3. Use Case Analysis

The use case analysis identified twelve primary use cases spanning patient engagement, clinical operations, administrative functions, and community support. The system boundary



encompasses interactions between six primary actor categories and four external system actors.

2.3.1. System Actors

- **Patient:** Registers, searches specialists, books appointments, joins consultations, uses emergency/community features
- **Elderly Patient / Adult Mode User:** Uses simplified interface for essential functions: booking, guidance, and emergency access
- **Mental Health Specialist / Doctor:** Manages profile, availability, appointments, telemedicine, session notes
- **Clinic Administrator:** Manages clinic branches, schedules, queues, payments, appointment approvals
- **Platform Administrator:** Manages users, roles, clinics, content, reports, compliance
- **Volunteer Listener:** Provides supportive listening sessions following code of conduct
- **Community Moderator:** Reviews reported content, volunteer sessions, recovery stories
- **AI Chatbot (SithBot):** Automated triage interacts with patients to suggest specialist categories
- **Payment Gateway (PayHere):** Processes online payments and returns transaction status
- **Video Consultation Service:** Hosts and manages telemedicine video rooms
- **SMS/Email Notification Service:** Delivers appointment reminders, OTPs, confirmations

2.3.2. Use Case List

Twelve use cases were identified and fully documented with descriptions, activity diagrams, and sequence diagrams:

1. **UC-01 - Register and Login:** Account creation with OTP, secure login, and role-based routing.
2. **UC-02 - Manage User Profile and Consent:** Profile updates, privacy consent, emergency contacts.
3. **UC-03 - Search Mental Health Specialist:** Filter by category, language, location, and availability.
4. **UC-04 - Book Appointment:** Slot selection, payment, confirmation, notifications.
5. **UC-05 - Manage Doctor Availability:** Define, update, and block available time slots.



6. **UC-06 - Approve/Reschedule Appointment:** Review, approve, reject, reschedule requests.
7. **UC-07 - Join Telemedicine Consultation:** Secure video room with phone fallback.
8. **UC-08 - Process Payment and Generate Receipt:** PayHere integration, cash tracking, PDF receipts.
9. **UC-09 - Track Live Patient Queue:** Real-time token display, proximity notifications.
10. **UC-10 - Request Emergency Support:** Helpline display, crisis contact, location sharing.
11. **UC-11 - Use Anonymous Community:** Pseudonymous posts, responses, and moderation.
12. **UC-12 - Complete AI Triage (SithBot):** Guided questions, specialist recommendation.

2.3.3. Activity Diagrams

Five activity diagrams were produced:

- (AD-01) User Authentication covering role-based login and registration with OTP
- (AD-02) Appointment Booking encompassing specialist search, slot selection, payment flow, and confirmation
- (AD-03) Emergency Support detailing button activation, disclaimer display, consent confirmation, and helpline fallback
- (AD-04) Telemedicine Consultation modelling video room link generation, join flow, and phone fallback
- (AD-05) Community Posting depicting anonymous post creation, pseudonymous identity masking, and moderation rules.

All diagrams are committed to the repository in source (.drawio) and exported (SVG) formats.

All the activity diagrams are included in section 2.7.3.6.2.

2.4. Functional Requirements

Functional requirements are specified with unique identifiers (FR-XX), MoSCoW priority, source reference, and verification method. Fifty-two functional requirements were identified across eleven categories. Key categories include:



- **Authentication and Access Control (6 requirements):** Registration with OTP, password complexity, RBAC, password reset, auto logout, MFA
- **Appointment Management (6 requirements):** Booking, cancellation, approval workflow, emergency same-day, reminders, availability management
- **Payment Processing (4 requirements):** PayHere integration, cash payment tracking, PDF receipt generation, refund processing
- **Emergency Support (5 requirements):** Persistent emergency button, consent-based location sharing, helpline display, non-substitution disclaimer, crisis event logging
- **AI Chatbot - SithBot (3 requirements):** Triage without login, crisis keyword detection, non-diagnosis disclaimer
- **Community Features (2 requirements):** Anonymous posting with pseudonymous IDs, content moderation tools
- **Localisation and Communication (4 requirements):** Sinhala/Tamil/English support, LKR currency, local emergency numbers, multilingual notifications

2.5. Non-Functional Requirements

2.5.1. Performance Requirements

1. **NFR-P-01:** Page load ≤ 3 seconds on 4G for 95% of requests
2. **NFR-P-02:** Booking transaction ≤ 5 seconds (excluding PayHere redirect)
3. **NFR-P-03:** Support minimum 500 concurrent users
4. **NFR-P-04:** Queue update propagation within 60 seconds
5. **NFR-P-05:** SithBot response ≤ 2 seconds for 90% of queries

2.5.2. Security Requirements

1. **NFR-S-01:** TLS 1.3 for all data in transit
2. **NFR-S-02:** AES-256 for patient data at rest
3. **NFR-S-03:** JWT access: 15 minutes, refresh: 7 days
4. **NFR-S-04:** OWASP Top 10 mitigations
5. **NFR-S-05:** Health records restricted to authoring doctor
6. **NFR-S-06:** Pseudonymous community IDs
7. **NFR-S-07:** No card data stored; PayHere handles PCI-DSS
8. **NFR-S-08:** Auth events logged with timestamp and IP



2.6. Alternative Solutions and Feasibility

2.6.1. Alternatives Considered

- **Alternative A** - Manual Phone/Walk-in Booking: Simple but causes delays, no transparency, no digital records, significant stigma barriers.
- **Alternative B** - Open-Source Platform (OpenMRS): Primarily for hospital EMR; lacks patient-facing UX, community features, requires substantial customisation.
- **Alternative C** - Healthcare SaaS (Medplum): Optimised for US market; Sri Lankan requirements need significant custom development with vendor dependency.
- **Alternative D** - Social Media/WhatsApp Groups: Poor privacy, misinformation, no moderation, no professional connection or appointment management.

2.6.2. Feasibility Assessment

- **Technical Feasible:** Team skills include React, Node.js, SQL, REST APIs. Jitsi is open-source. PayHere provides sandbox.
- **Economic Feasible:** Development cost is team effort only. Free-tier cloud hosting available. PayHere sandbox has no monthly fee.
- **Operational Feasible:** Clinic admins capable of web dashboards per interviews. Patient UI designed for low-literacy users.
- **Schedule Feasible with MoSCoW:** 8-week design sufficient for MVP. Advanced features deferred to SENG 34213.
- **Legal/Ethical Feasible with constraints:** Requires PDPA controls, RBAC, data minimisation, consent recording, medical disclaimers.

2.6.3. MoSCoW Scope Prioritisation

- **Must Have:** Authentication, role-based dashboards, specialist search, appointment booking with PayHere/cash, availability management, emergency support, SMS/email notifications, multilingual support.
- **Should Have:** Telemedicine, PDF receipts, live queue tracker, doctor verification, content moderation, Adult Mode, multi-branch clinic management.
- **Could Have:** Anonymous community forum, volunteer listener portal, AI chatbot (SithBot), specialist ratings, analytics dashboard.
- **Won't Have (v1.0):** AI clinical diagnosis, automatic emergency dispatch, government EHR integration, native mobile apps, insurance processing.



2.7. Software Requirements Specifications Document

2.7.1. Introduction

2.7.1.1. Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for SithCare, a mental health channelling web application intended for deployment in Sri Lanka. The document is prepared in accordance with the SENG 31242 – System Design Project guidelines issued by the Software Engineering Teaching Unit, University of Kelaniya, and follows the IEEE 830-1998 Recommended Practice for Software Requirements Specifications.

The primary audience of this document is: the project supervisor (Eng. Dr. Tiroshan Madushanka), the development team (Group 10), and any future stakeholders involved in the SENG 34213 development phase. This is a draft version and will be updated after questionnaire analysis, stakeholder interviews, and supervisor feedback are incorporated.

2.7.1.2. Scope

SithCare ("Care Beyond Words") is a web application that:

- Enables patients across Sri Lanka to discover, filter, and book appointments with verified mental health professionals.
- Provides clinic administrators with tools to manage doctor schedules, patient records, payment tracking, and communications.
- Provides a platform administrator with oversight of all registered clinics, user management, and compliance reporting.
- Includes a telemedicine module for video consultations, an AI triage chatbot (SithBot), an emergency support subsystem, and an anonymous community forum.
- Is localised for the Sri Lankan context: Sinhala, Tamil, and English language support; LKR currency; local emergency helplines; and PayHere payment gateway integration.

Out of Scope: native mobile applications (Android/iOS), integration with government electronic health record (EHR) systems, and insurance claim processing are explicitly excluded from this version.



2.7.1.2.1. Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
SDS	Software Design Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
RBAC	Role-Based Access Control
JWT	JSON Web Token – authentication mechanism
OTP	One-Time Password
MFA	Multi-Factor Authentication
LKR	Sri Lankan Rupee
API	Application Programming Interface
TLS	Transport Layer Security
AES	Advanced Encryption Standard
MVP	Minimum Viable Product
MoSCoW	Must-have, Should-have, Could-have, Won't-have (prioritisation framework)
PDPA	Personal Data Protection Act – Sri Lankan data protection legislation
SLMC	Sri Lanka Medical Council – statutory body for registering medical practitioners
SLPA	Sri Lanka Psychological Association – professional body for psychologists
CBT	Cognitive Behavioural Therapy
SithBot	The AI triage chatbot component of SithCare
Pseudonymous ID	A platform identifier that hides a user's real identity in community features
Telemedicine	Remote consultation using video/audio communication tools

Table 2.1: Definitions, Acronyms and Abbreviations

2.7.1.3. References

- Many people with mental disorders do not seek professional help, creating a considerable treatment gap. (health.gov.lk/wp-content/uploads/2025/04/Mental-Health-Service-Providers-Directory-2023.pdf)
- Stigma, poor awareness, transport, language, and financial barriers reduce help seeking.([Mental health prevalence, healthcare use and access between 2018 and 2022 in Sri Lanka: an analysis of survey data - NCBI Bookshelf](#))



- Sri Lanka's National Mental Health Policy 2020–2030 emphasizes community support, early detection, continuing care, and stigma reduction. ([National-Mental-Health-Policy-2020-2030-compressed.pdf](#))
- Sri Lanka's Digital Health Blueprint supports digital tools to improve healthcare accessibility and timely service delivery. ([health.gov.lk/wp-content/uploads/2023/11/Digital-Health-Blue-Print-Full-Book-01.11.2023-Final.pdf](#))
- Sri Lanka has only 2.5 mental health workers per 100,000 population. This indicates the need for technology-supported access, appointment coordination, and better connection between patients and available professionals. ([Mental-health-atlas-2024.pdf](#))

SENG 31242 System Design Project Course Guideline and Industrial Standards, University of Kelaniya.

2.7.1.4. Overview of Document

The remainder of this document is organised as follows:

- Chapter 2 – Overall Description: system context, user classes, operating environment, assumptions, and user personas.
- Chapter 3 – System Analysis: fact-finding summary, existing system analysis, competitive analysis, actors, use case list, use case descriptions, and activity diagram plan.
- Chapter 4 – Functional Requirements: complete numbered FR list with MoSCoW priority, source, and verification method.
- Chapter 5 – Non-Functional Requirements: performance, security, usability, reliability, scalability, and maintainability.
- Chapter 6 – Alternative Solutions and Feasibility Study.
- Appendices: glossary, fact-finding artefacts, risk register, MoSCoW scope table, and repository artefact map.

2.7.2. Overall Description

2.7.2.1. Product Perspective

SithCare operates as a standalone, centralised, role-based web application rather than a set of separate systems. A single platform manages shared services such as authentication, user roles, appointment scheduling, payments, communication,



community moderation, and emergency support. Users access different dashboards according to their role. The system interfaces with the following external services:

External System	Purpose	Integration Method
PayHere	Sri Lankan payment gateway for online appointment payments in LKR	REST API / PayHere Checkout SDK
SMS Provider (Dialog/Mobitel)	OTP delivery, appointment reminders, queue notifications	SMS Gateway API
Email Service (SMTP/SendGrid)	Booking confirmations, password resets, notifications	SMTP / REST API
Video SDK (Jitsi / Daily.co)	Telemedicine video consultation rooms	JavaScript SDK / REST API
Google Maps / OpenStreetMap	Clinic location display and directions	Maps API
Ministry of Health Helpline Data	Emergency helpline numbers (static integration)	Static configuration

Table 2.2: System Interfaces with Following External Services

2.7.2.2. Product Functions

- Authentication and role-based access for patients, clinic-admins ,system-admins.
- Patient onboarding, profile management, privacy consent, and Adult Mode for elderly users.
- Patient onboarding, profile management, privacy consent handling, and Adult Mode for elderly/low-literacy users.
- Specialist search/filtering by category, language, availability, clinic, and consultation type.
- Appointment booking, rescheduling, cancellation, confirmation, and status tracking with SMS/email notifications.
- Clinic and doctor availability management including multiple clinics/branches and emergency same-day slots.
- Telemedicine support through secure consultation links with one-tap phone fallback.
- Payments using local PayHere gateway, manual/cash payment tracking, and PDF receipt generation.
- Emergency support with helplines, urgent booking, and consent-based location sharing.
- Live patient queue tracker displaying current token number and sending proximity notifications.



- Anonymous community consultation using pseudonymous identities.

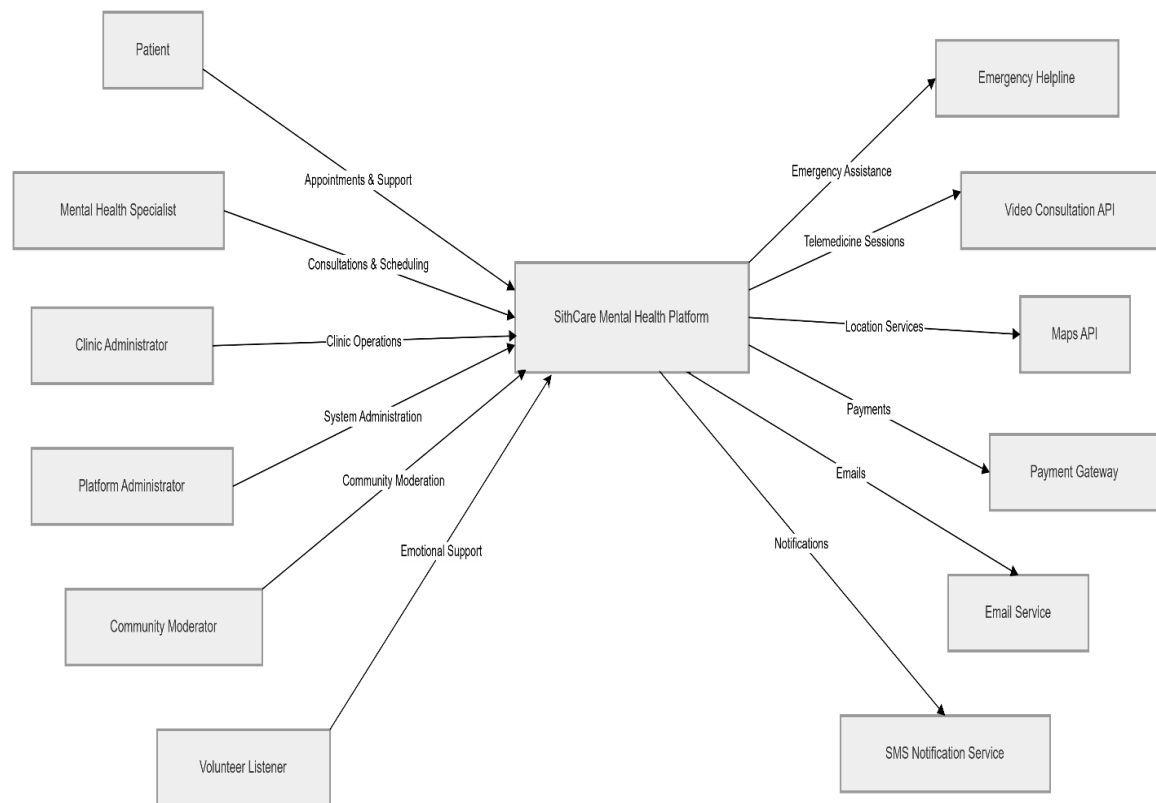


Figure 2.1: Context Diagram of the SithCare Mental Health Platform

2.7.2.3. User Classes and Characteristics

User Class	Technical Proficiency	Primary Needs
Patient (Urban)	Moderate – smartphone/web literate	Private appointment booking, telemedicine, community support, resource library
Patient (Rural/Elderly)	Low – minimal digital literacy	Simplified Adult Mode, SMS reminders, in-person booking, queue tracking
Mental Health Specialist / Doctor	Moderate – clinical background	View daily appointments, manage regular activities. (Future implementation)
Clinic Administrator	Moderate – administrative background	Schedule management, payment tracking, patient records, doctor coordination



Platform Administrator	High – technical background	Clinic onboarding, platform analytics, user management, compliance reporting
Volunteer Listener	Low to Moderate	Access volunteer portal, join listener sessions, view training materials
Community Moderator	Moderate	Moderate anonymous forum posts, flag harmful content, manage community
Emergency Helplines / Responders	External actor	Shown or contacted during emergency/crisis support flows

Table 2.3: User Classes and Characteristics

The SithCare platform serves multiple categories of users with varying technical expertise, responsibilities, and interaction requirements.

The following 5 personas represent the primary user groups identified during requirements analysis. These personas help define user expectations, accessibility considerations, operational workflows, and role-based functionalities within the system.

1. Persona 1 – Patient

Attribute	Description
Name	Kavindu Perera
Age	23
Occupation	University Student
Technical Proficiency	Intermediate
Primary Goals	Book appointments, access online consultations, receive emotional support anonymously
Responsibilities	Manage personal appointments and mental health activities
Pain Points	Long waiting times, stigma around mental health, difficulty finding trusted professionals
System Expectations	Simple interface, privacy protection, mobile accessibility, multilingual support

Figure 2.2: Persona 1 - Patient



2. Persona 2 – Clinic Administrator

Attribute	Description
Name	Nadeeka Fernando
Age	34
Occupation	Clinic Operations Administrator
Technical Proficiency	Advanced
Primary Goals	Manage appointments, clinic schedules, queues, and payment tracking
Responsibilities	Coordinate clinic operations and appointment workflows
Pain Points	Manual scheduling conflicts, patient no-shows, inefficient communication
System Expectations	Real-time updates, appointment management tools, reporting dashboards

Figure 2.3: Persona 2 – Clinic Administrator

3. Persona 3 – Mental Health Specialist

Attribute	Description
Name	Dr. Tharindu Silva
Age	41
Occupation	Consultant Psychiatrist
Technical Proficiency	Intermediate to Advanced
Primary Goals	Conduct consultations, manage patient schedules, provide secure telemedicine sessions
Responsibilities	Deliver mental health care and maintain patient consultation records
Pain Points	Schedule overload, fragmented communication, administrative delays
System Expectations	Secure telemedicine support, efficient scheduling, reliable patient communication

Figure 2.4: Persona 3 – Mental Health Specialist



4. Persona 4 – System Administrator

Attribute	Description
Name	Isuru Jayawardena
Age	29
Occupation	System Administrator
Technical Proficiency	Advanced
Primary Goals	Monitor platform performance, manage users, maintain security and system reliability
Responsibilities	User management, access control, system monitoring, report management
Pain Points	Security threats, unauthorized access attempts, server downtime
System Expectations	Administrative dashboards, security monitoring, audit logs, system analytics

Figure 2.5: Persona 4 – System Administrator

5. Persona 5 – Community Moderator

Attribute	Description
Name	Dinithi Rajapaksha
Age	27
Occupation	Community Support Moderator
Technical Proficiency	Intermediate
Primary Goals	Maintain a safe community environment and moderate harmful content
Responsibilities	Review reported posts, enforce community guidelines, assist users in discussions
Pain Points	Harmful user behavior, misinformation, emotionally sensitive discussions
System Expectations	Content moderation tools, reporting systems, user management capabilities

Figure 2.6: Persona 5 – Community Moderator



2.7.2.4. Operating Environment

Dimension	Details
Web Application	Deployed on cloud hosting (AWS / Google Cloud / DigitalOcean) with auto- scaling capability
Browser Support	Chrome 110+, Firefox 110+, Safari 16+, Edge 110+
Network	Designed for Sri Lankan 4G LTE; telemedicine degrades gracefully on 3G (>1 Mbps minimum)
Database	Mongo DB database with JSON usage
Server OS	Ubuntu 22.04 LTS
Localisation	Sinhala (Unicode), Tamil (Unicode), English
Future	Native Android/iOS mobile applications are out of scope for v1.0 but may be considered in a later phase

Table 2.4: Operating Environment

2.7.2.5. Design and Implementation Constraints

- The system must comply with Sri Lanka's data protection obligations under the proposed Personal Data Protection Act (PDPA).
- Patient mental health records must be encrypted at rest (AES-256) and in transit (TLS 1.3).
- The AI chatbot (SithBot) must not provide clinical diagnoses; all AI-generated suggestions must display a medical disclaimer.
- Payment processing must use PayHere or equivalent Sri Lankan certified payment gateway only.
- Emergency location sharing requires explicit user confirmation before activation on a per-session basis.
- Volunteers must not provide clinical or medical advice; the system must enforce this through UI constraints and a signed code of conduct.
- Telemedicine is optional; patients must always be able to use in-person appointment booking without a video component.
- The initial design must be feasible within the SENG 31242 design timeline and later development semester.



- All major requirements and design artefacts must be version-controlled through the project GitHub organization.

2.7.2.6. Assumptions and Dependencies

ID	Type	Assumption / Dependency
A1	Assumption	Users have access to a smartphone, desktop, or internet-enabled device for online services.
A2	Assumption	Clinics and mental health specialists are willing to maintain availability and appointment information digitally.
A3	Assumption	Payment gateway, SMS/email, video SDK, and hosting services remain available and affordable for the project scope.
A4	Assumption	Stakeholder questionnaires and interviews will provide enough evidence to refine the requirements before SDS.
A5	Assumption	Mental health specialists registered on the platform hold valid professional licenses verified during onboarding.
D1	Dependency	System communication depends on external SMS/email providers (Dialog, Mobitel, SendGrid).
D2	Dependency	Online payments depend on the PayHere payment gateway; PayHere sandbox is used for testing.
D3	Dependency	Telemedicine depends on video SDK/service reliability and regional latency (Southeast Asia servers required).
D4	Dependency	Emergency support depends on helpline contact information supplied by the Ministry of Health.
D5	Dependency	Emergency helpline numbers provided by MoH are assumed to remain static throughout the project lifecycle.

Table 2.5: Assumptions and Dependencies

2.7.3. System Analysis

2.7.3.1. Fact-Finding Techniques Used

Technique	Session Details	Key Findings
Structured Interview	Conducted 5 separate interviews with mental patients to identify the pain points	Stigma is primary barrier; patients prefer anonymous search; clinics need payment tracking; elderly need simplified UX.
Online Questionnaire / Survey	Distributed to 150 respondents via Google Forms (2026-04-15 to 2026-04-22).	Analysis of the findings has been uploaded to the research repository accordingly



Survey for the mental health specialists	Conducted 6 interviews for mental health specialists and counsellors	Identified the pain points, actual requirements of the specialists for conducting the activities in a better way.
Document Analysis	WHO Mental Health Atlas 2024, MoH Directory 2023, National Mental Health Policy 2020–2030, Digital Health Blueprint 2023.	2.5 mental health workers per 100,000; significant treatment gap; government digital health push creates market opportunity.

Table 2.6: Fact-Finding Techniques Used

2.7.3.2. Existing System Analysis

2.7.3.2.1. Current State and Pain Points

Mental health care access in Sri Lanka currently relies on outpatient department (OPD) queues at government hospitals, manual telephone bookings at private psychiatric clinics, and informal WhatsApp groups for community support. Key pain points identified from interviews and research:

- No centralised directory of licensed mental health professionals accessible to the public.
- Telephone-only booking creates stigma – fear of being overheard by family members or colleagues.
- No digital queue tracking; patients wait at clinic premises for 1–4 hours with no information about wait times.
- Payment records kept in paper ledgers; no digital receipts or payment history for patients.
- No telemedicine option for rural patients or those with mobility issues.
- Crisis support helplines are unknown to most patients; numbers are not widely publicised.

2.7.3.2.2. Competitive Analysis Summary

Feature	BetterHelp	Talkspace	Wysa	Nalanda Care	SithCare
Multi-clinic management	✗	✗	✗	✗	✓
Sri Lanka localization	✗	✗	Partial (AI)	Partial	✓ Full
Live queue tracker	✗	✗	✗	✗	✓
AI triage chatbot	✗	✗	✓	✗	✓
Emergency support	✗	✗	Basic	✗	✓ Full



Telemedicine	✓	✓	✗	✗	✓
Anonymous community	✗	✗	✓	✗	✓
PayHere / LKR payment	✗	✗	✗	✓	✓
Adult/Elderly mode	✗	✗	✗	✗	✓
Volunteer listener	✗	✗	✗	✗	✓

Table 2.7: Competitive Analysis Summary

2.7.3.3. System Boundary and Actors

The system boundary of SithCare encompasses all interactions between the following actors and the platform:

Actor	Type	Role / Interaction
Patient	Primary	Registers, searches specialists, books appointments, joins consultations, uses emergency/community/support features.
Elderly Patient / Adult Mode User	Primary	Uses a simplified interface for essential functions: booking, guidance, and emergency access.
Mental Health Specialist / Doctor	Primary	Manages profile, availability, appointments, telemedicine link, session notes, and consultation status.
Clinic Administrator	Primary	Manages clinic branches, schedules, queues, payments, and appointment approvals.
Platform Administrator	Primary	Manages users, roles, clinics, content, reports, fundraising approvals, and system settings.
Volunteer Listener	Primary	Provides supportive listening sessions following the code of conduct and confidentiality rules.
Community Moderator	Primary	Reviews reported community content, volunteer sessions, and recovery stories.
AI Chatbot (SithBot)	System Actor	Automated triage interacts with patients to suggest specialist categories.
Emergency Helpline / Responder	External	Shown or contacted during emergency/crisis support flows (Sumithrayo 1926, CCcline).
Payment Gateway (PayHere)	External	Processes online payments and returns transaction status to SithCare.
SMS/Email Notification Service	External	Delivers appointment reminders, OTPs, and booking confirmations.



Video Consultation Service	External	Hosts and manages telemedicine video rooms.
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Table 2.8: System Boundary and Actors

2.7.3.4. Use Case List

ID	Use Case Name	Primary Actor(s)	Description
UC-01	Register and Login	Patient / Doctor / Admin / Volunteer	User creates an account with OTP verification, logs in securely, and is routed to their role-based dashboard.
UC-02	Manage User Profile and Consent	Patient	Patient updates personal information, sets language preferences, manages privacy consent, and adds emergency contacts.
UC-03	Search for Mental Health Specialist	Patient	Patient searches and filters mental health professionals by category, language, consultation mode, location, and availability.
UC-04	Book Appointment	Patient / Clinic Admin / Doctor	Patients select a slot and book an appointment; the system confirms the booking and notifies all relevant parties.
UC-05	Manage Doctor Availability	Doctor / Clinic Admin	The doctor or clinic admin defines updates or blocks available time slots across one or more clinic locations.
UC-06	Approve / Reschedule Appointment	Clinic Admin / Doctor	Clinic staff reviews pending appointment requests and approvals, rejects, or reschedules as required.
UC-07	Join Telemedicine Consultation	Patient / Doctor	Patient joins a secure video consultation via a doctor-generated room link; fallback phone call provided if video fails.
UC-08	Process Payment and Generate Receipt	Patient / Payment Gateway / Clinic Admin	Patient pays online (PayHere) or via cash; system updates payment status and generates a PDF receipt.
UC-09	Track Live Patient Queue	Patient / Clinic Admin	System displays the current token being served and the patient's position; clinic admin advances the queue.
UC-10	Request Emergency Support	Patient / Doctor / Emergency Helpline	Patient activates emergency button; system displays crisis options, initiates contact, and shares location if consented.



UC-11	Use Anonymous Community Consultation	Patient / Community Moderator	Patient posts general mental health concerns anonymously; community responds, and moderator oversees safety.
UC-12	Complete AI Triage Questionnaire	Patient / AI Triage Module (SithBot)	Patient answers guided questions; SithBot recommends a specialist category with a mandatory non-diagnosis disclaimer.

Table 2.9: Use Case List

2.7.3.5. Use Case Descriptions and Diagrams

2.7.3.5.1. UC-01 – Register and Login

Use Case ID	UC-01
Use Case Name	Register and Login
Actor(s)	Patient, Doctor, Clinic Admin, Volunteer, Platform Admin
Preconditions	User is not logged in. User has a valid email address or phone number.
Postconditions – Success	Authenticated session created; user routed to correct role-based dashboard.
Postconditions – Failure	Account not created or login rejected; appropriate error message displayed.
Main Flow	1. User opens SithCare and navigates to /register or /login. 2. For registration: user selects role, enters full name, email, phone, date of birth, preferred language, and password. 3. System validates input (email format, password strength, phone format). 4. System checks for duplicate email/phone. 5. System sends OTP to registered email/phone. 6. User enters OTP, System verifies OTP, activates account, and redirects to profile completion wizard. 7. For login: user enters credentials, system verifies role and account status, creates a secure JWT session, and redirects to dashboard.
Alternative Flow 1	At step 2 (Registration): If password is weak, system highlights requirements (min 8 chars, 1 uppercase, 1 digit, 1 special character). User corrects and resubmits.
Alternative Flow 2	At step 5: If email is already registered, system displays a generic message to prevent user enumeration. OTP resend is limited to 3 attempts per hour.
Alternative Flow 3	At step 8 (Login): Invalid credentials or suspended account – system displays error. After 5 failed attempts, account is temporarily locked for 15 minutes.
Business Rules	• BR-01: OTPs expire after 10 minutes. • BR-02: Max 3 OTP resend attempts per hour. • BR-03: Min password: 8 chars, 1 uppercase, 1 digit, 1 special character. • BR-04: JWT access tokens expire after 15 minutes; refresh tokens expire after 7 days.

Table 2.10: UC-01 – Register and Login

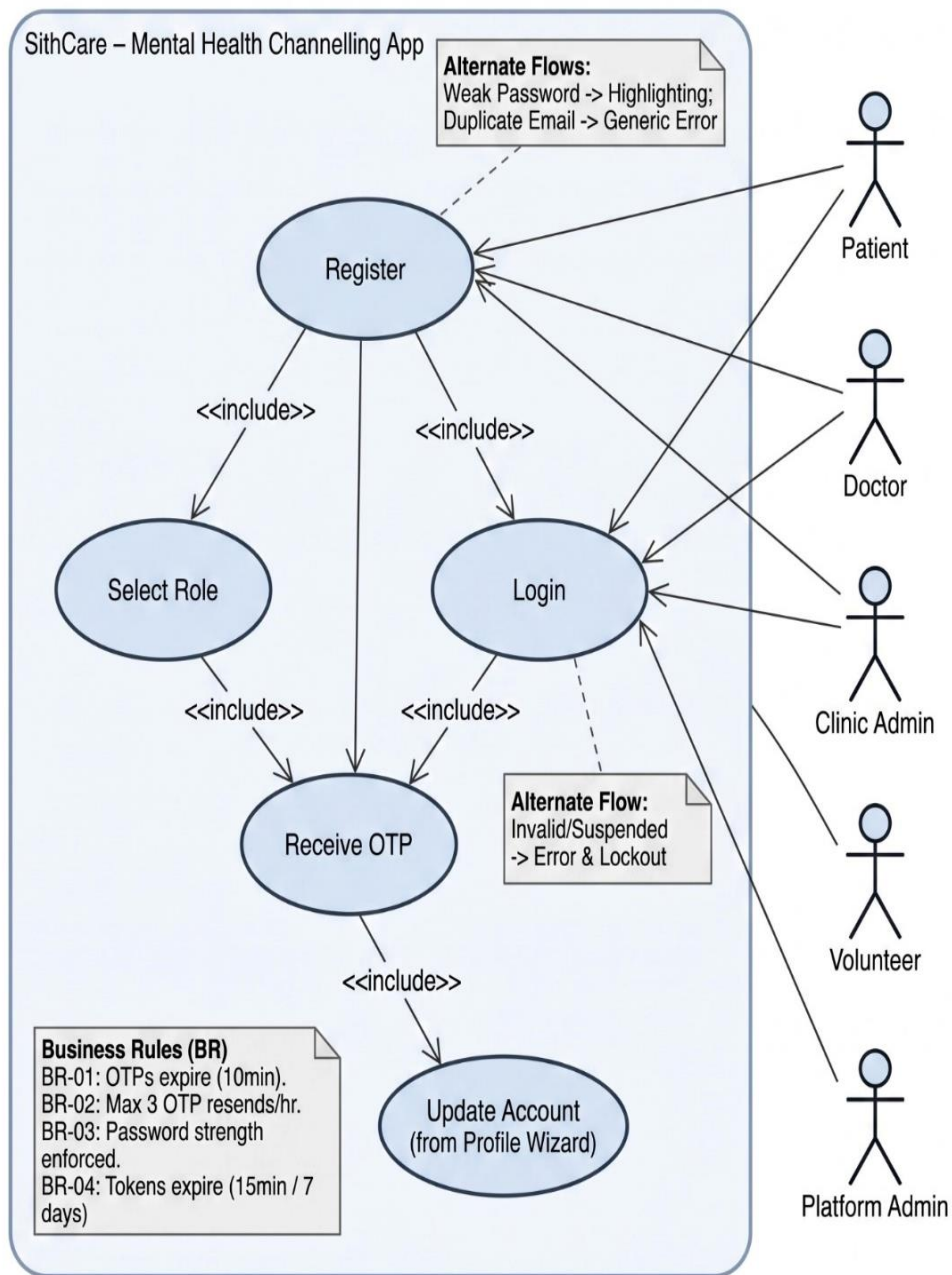


Figure 2.7: UC-01 – Register and Login



2.7.3.5.2. UC-02 – Manage User Profile and Consent

Use Case ID	UC-02
Use Case Name	Manage User Profile and Consent
Actor(s)	Patient
Preconditions	Patient is registered and logged in to their dashboard.
Postconditions – Success	Patient profile is updated with the latest personal information, consent preferences, and emergency contact details.
Postconditions – Failure	Profile changes not saved; validation error message displayed; existing data remains unchanged.
Main Flow	1. Patient navigates to 'My Profile' from the dashboard. 2. Patient views current profile information: name, contact details, date of birth, preferred language, and profile photo. 3. Patient edits desired fields and clicks 'Save Changes'. 4. System validates inputs (e.g., phone number format, valid date of birth). 5. System saves updated profiles and displays a success confirmation. 6. Patient navigates to 'Privacy & Consent' settings. 7. System displays privacy policy and current consent status. 8. Patient reviews and updates consent preferences (e.g., data sharing for anonymous analytics). 9. Patient adds or updates emergency contact name and phone number. 10. System saves preferences and logs the consent update with a timestamp.
Alternative Flow 1	At step 4: If validation fails (e.g., invalid phone format), system highlights the specific field error and prevents saving until corrected.
Alternative Flow 2	At step 8: If patient withdraws previously given consent, system updates settings and queues data minimisation actions where applicable.
Alternative Flow 3	Patient enables Adult Mode from profile settings: system switches to simplified navigation, larger font size (minimum 18px), and reduced feature set for the session.
Business Rules	• BR-05: Consent withdrawal is recorded with timestamp and user ID for audit purposes. • BR-06: Emergency contact information is used only during emergency support flow (UC-10) and is not shared publicly. • BR-07: Patients can upload a profile photo; file size must not exceed 2 MB; accepted formats are JPG and PNG.

Table 2.11: UC-02 – Manage User Profile and Consent

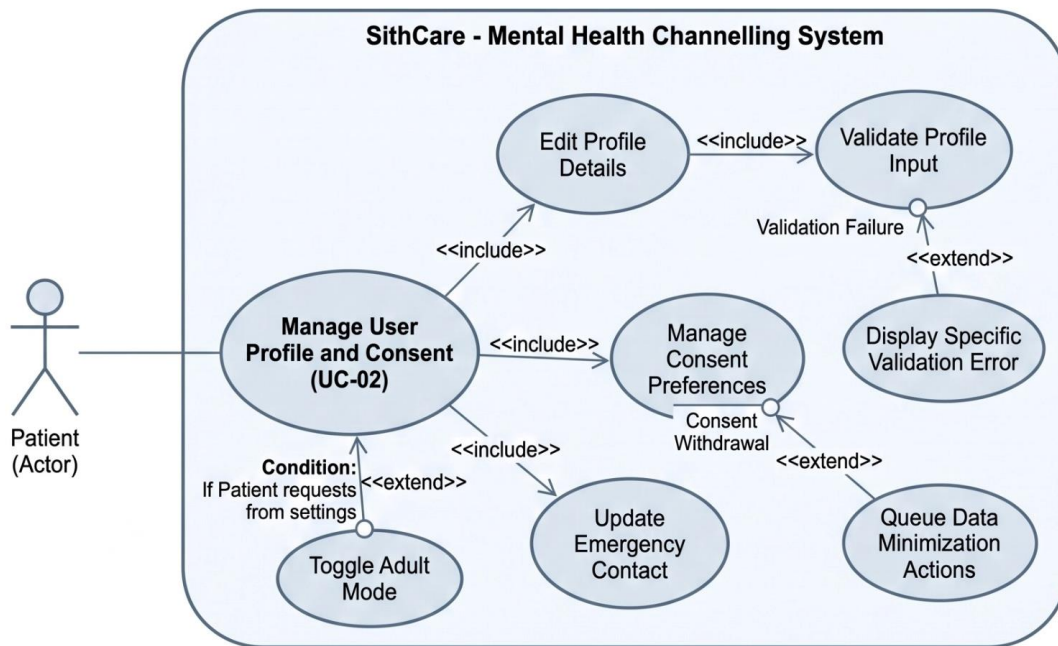


Figure 2.8: UC-02 – Manage User Profile and Consent

2.7.3.5.3. UC-03 – Search Mental Health Specialist

Use Case ID	UC-03
Use Case Name	Search Mental Health Specialist
Actor(s)	Patient
Preconditions	Patients are logged in. At least one verified specialist is listed on the platform.
Postconditions – Success	A filtered list of matching specialists is displayed with profile summaries and available slots.
Postconditions – Failure	No matching specialists found; system displays 'No results found' with suggestions to broaden the search criteria.
Main Flow	1. Patient navigates to 'Find a Specialist' from the dashboard or the AI chatbot recommendation. 2. Patient enters search criteria using available filters: specialist category (e.g., CBT Therapist, Psychiatrist, Counsellor), preferred language (Sinhala, Tamil, English), consultation type (online/in-person), location/district, and availability date. 3. System queries the database and returns a ranked list of matching specialists. 4. System displays each specialist with: name, photo, category, SLMC/SLPA registration number, consultation fee in LKR, languages spoken, average rating, and next available slot. 5. Patient clicks a specialist card to view the full profile. 6. Patient reviews full profile and clicks 'Book Appointment' to proceed to UC-04.



Alternative Flow 1	At step 3: If no specialists match all filters, system relaxes lower-priority filters (e.g., consultation type) and displays results with a note: 'Showing nearest matches – some filters were broadened.'
Alternative Flow 2	At step 4: If the patient was referred via the AI chatbot (UC-14), the recommended specialist category is pre-filled in the search filter.
Alternative Flow 3	Patient selects 'View on Map' to see clinic locations pinned on an interactive map (Google Maps / OpenStreetMap integration).
Business Rules	- BR-08: Only verified specialists with 'Active' status appear in search results. - BR- 09: Search results are sorted by availability (soonest slot first) by default. - BR-10: Specialist SLMC/SLPA registration number must be verified by the Platform Admin before the profile is publicly listed (UC-15).

Table 2.12: UC-03 – Search Mental Health Specialist

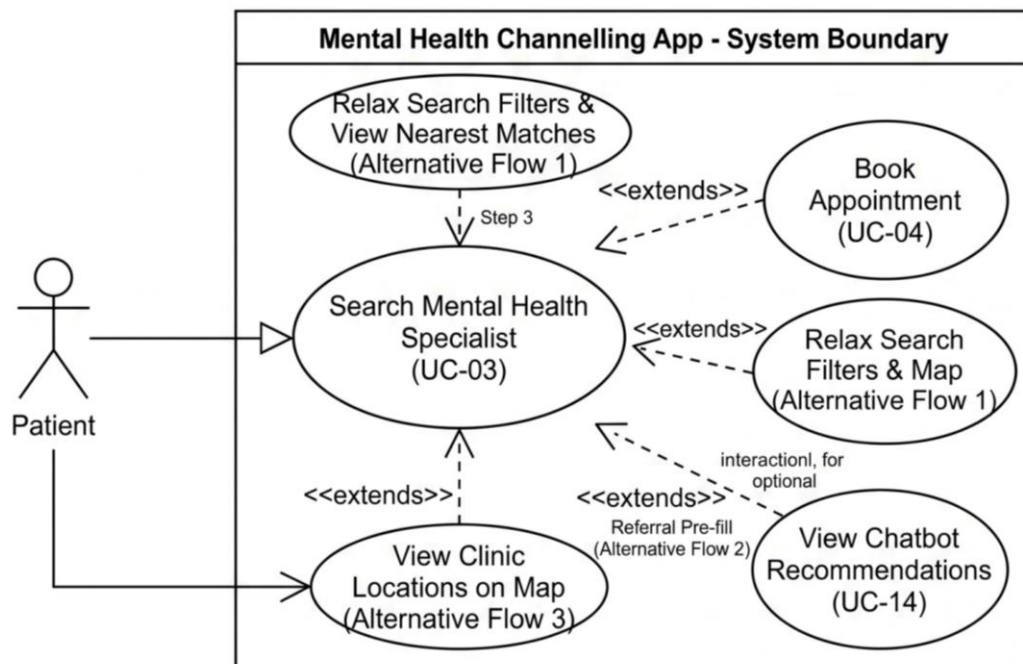


Figure 2.9: UC-03 – Search Mental Health Specialist



2.7.3.5.4. UC-04 – Book Appointment

Use Case ID	UC-04
Use Case Name	Book Appointment
Actor(s)	Patient, Clinic Appointment System, Payment Gateway (PayHere), SMS/Email Service
Preconditions	Patients are registered and logged in. Patients have selected a specialist from the search results (UC-03).
Postconditions – Success	Appointment confirmed; booking reference generated; SMS and email confirmation sent; appointments appear on patient and doctor dashboards.
Postconditions – Failure	Appointment not booked; selected slot remains available; patient notified with reason.
Main Flow	1. Patient selects a specialist and clicks 'Book Appointment'. 2. System displays available time slots for the selected clinic/doctor. 3. Patient selects a date and time slot and chooses consultation type (in-person or video). 4. Patient reviews booking summary: doctor name, clinic, slot, consultation type, and fee in LKR. 5. Patient selects payment method: online (PayHere) or cash at clinic. 6. If online: patient completes PayHere checkout; system receives payment confirmation. 7. System creates appointment record with status 'Confirmed'. 8. System generates a unique booking reference number. 9. System sends SMS and email confirmation to the patient. 10. Appointment appears on both patient and doctor dashboards.
Alternative Flow 1	At step 6: If PayHere payment fails, system retains the slot for 5 minutes and prompts a retry. After 2 failures, slot is released, and patient is redirected to search.
Alternative Flow 2	At step 2: If no regular slots are available, system offers 'Emergency Same-Day Booking' option and displays the earliest next available slot, subject to clinic approval.
Alternative Flow 3	Patient cancels before step 7: system releases the slot and no record is created.
Business Rules	• BR-11: Appointment slots are 50 minutes by default. • BR-12: Cancellation is free if more than 24 hours before the appointment; a cancellation fee policy applies within 24 hours. • BR-13: Online payment receipt is generated as a PDF automatically upon payment confirmation. • BR-14: Slot is reserved for 5 minutes during payment processing.

Table 2.13: UC-04 – Book Appointment

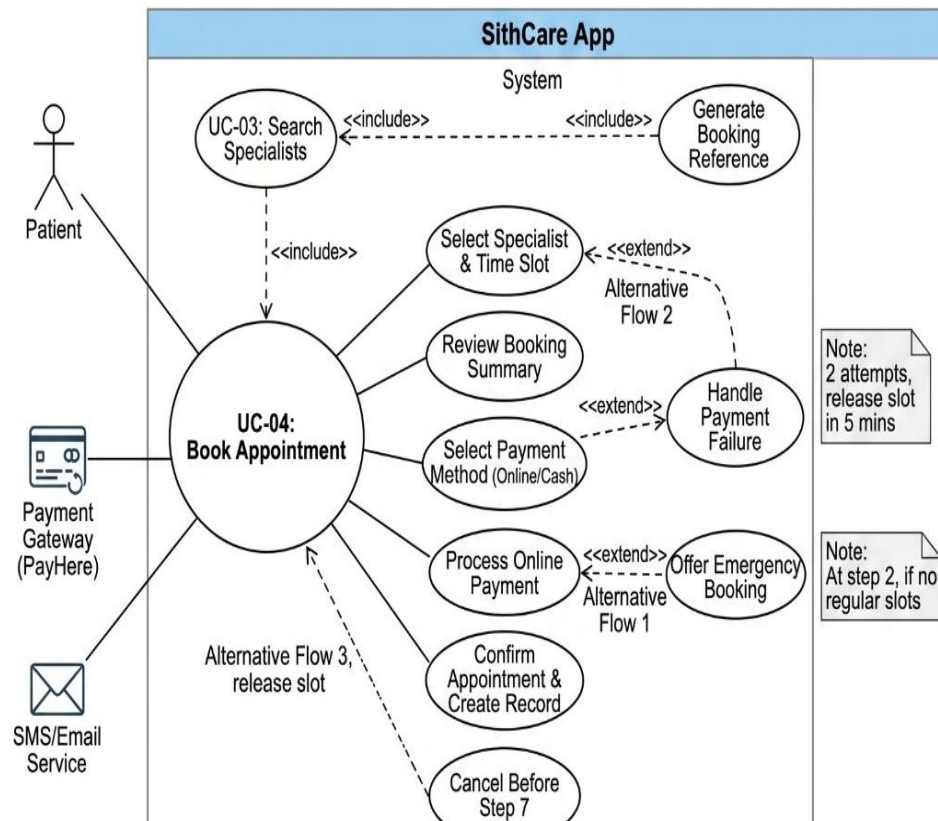


Figure 2.10: UC-04 – Book Appointment

2.7.3.5.5. UC-05 – Manage Doctor Availability

Use Case ID	UC-05
Use Case Name	Manage Doctor Availability
Actor(s)	Doctor, Clinic Administrator
Preconditions	Doctor or clinic admin is authenticated and has the appropriate role permissions.
Postconditions – Success	Availability calendar is updated and reflects correctly in patient-facing specialist search.
Postconditions – Failure	Schedule conflict detected; changes not saved; user informed of the conflict.
Main Flow	1. Doctor or admin opens the availability management panel and selects a clinic/branch. 2. System displays a weekly calendar view of current availability. 3. User selects a date and defines available time slots (start time, end time, slot duration). 4. User can block specific dates/times for holidays or personal leave. 5. System validates for conflicts with existing confirmed appointments. 6. System saves the updated availability schedule. 7. Updated slots immediately become available in patient-facing search.



Alternative Flow 1	At step 5: If a conflict exists with an already-confirmed appointment, system highlights the conflict and prevents saving until resolved.
Alternative Flow 2	Doctor is linked to multiple clinics: system allows setting separate availability schedules per clinic branch.
Alternative Flow 3	Clinic admin sets availability on behalf of a doctor: system logs the change with the admin's account for audit purposes.
Business Rules	- BR-15: A doctor may be associated with up to 5 clinic branches simultaneously. - BR-16: Availability changes affecting already-confirmed appointments require explicit admin confirmation and trigger a patient notification.

Table 2.14: UC-05 – Manage Doctor Availability

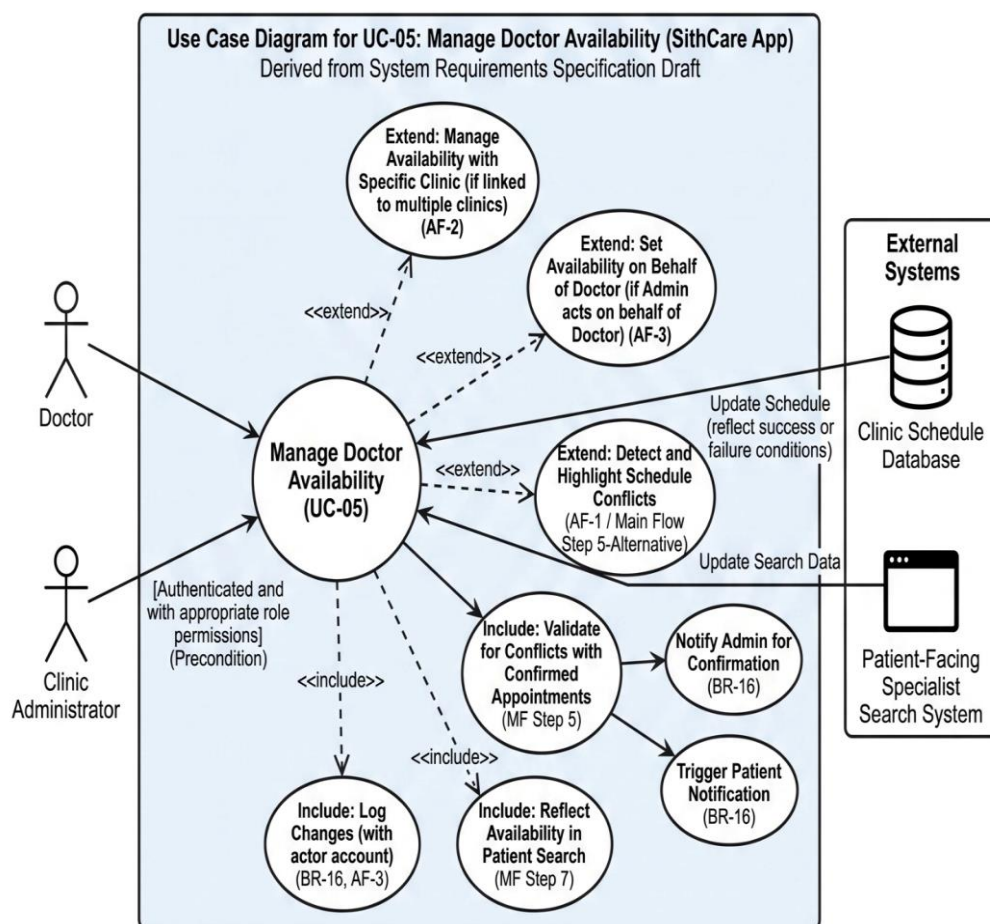


Figure 2.11: UC-05 – Manage Doctor Availability

**2.7.3.5.6. UC-06 – Approve / Reschedule Appointment**

Use Case ID	UC-06
Use Case Name	Approve / Reschedule Appointment
Actor(s)	Clinic Administrator, Doctor, SMS/Email Notification Service
Preconditions	One or more appointment requests with status 'Pending Approval' exist in the clinic admin dashboard.
Postconditions – Success	Appointment status updated (Confirmed/Rejected/Rescheduled); patient notified via SMS and email.
Postconditions – Failure	Action not taken; appointment remains in pending state; error displayed to admin.
Main Flow	1. Clinic admin logs in and navigates to 'Pending Appointments'. 2. System displays a list of pending appointment requests sorted by date. 3. Admin selects a request and reviews patient details, requested slot, and consultation type. 4. Admin chooses one of: Approve, Reject, or Reschedule. 5a. (Approve): System updates appointment status to Confirmed and sends confirmation to patient. 5b. (Reject): Admin enters a rejection reason. System updates status to Rejected and notifies patient with the reason and suggestion to re-book.
Alternative Flow 1	At step 5c: If patient does not confirm the rescheduled slot within 24 hours, the appointment is automatically cancelled and the patient is notified.
Alternative Flow 2	Emergency Same-Day Booking (from UC-04): admin is presented with a priority flag and must approve or reject within 30 minutes; a reminder notification is sent if not acted on within 15 minutes.
Business Rules	• BR-17: Manual approval is required only for clinics that have enabled the 'Approval Required' setting. • BR-18: Rejection reason is recorded for audit and reported to the Platform Administrator monthly.

Table 2.15: UC-06 – Approve / Reschedule Appointment

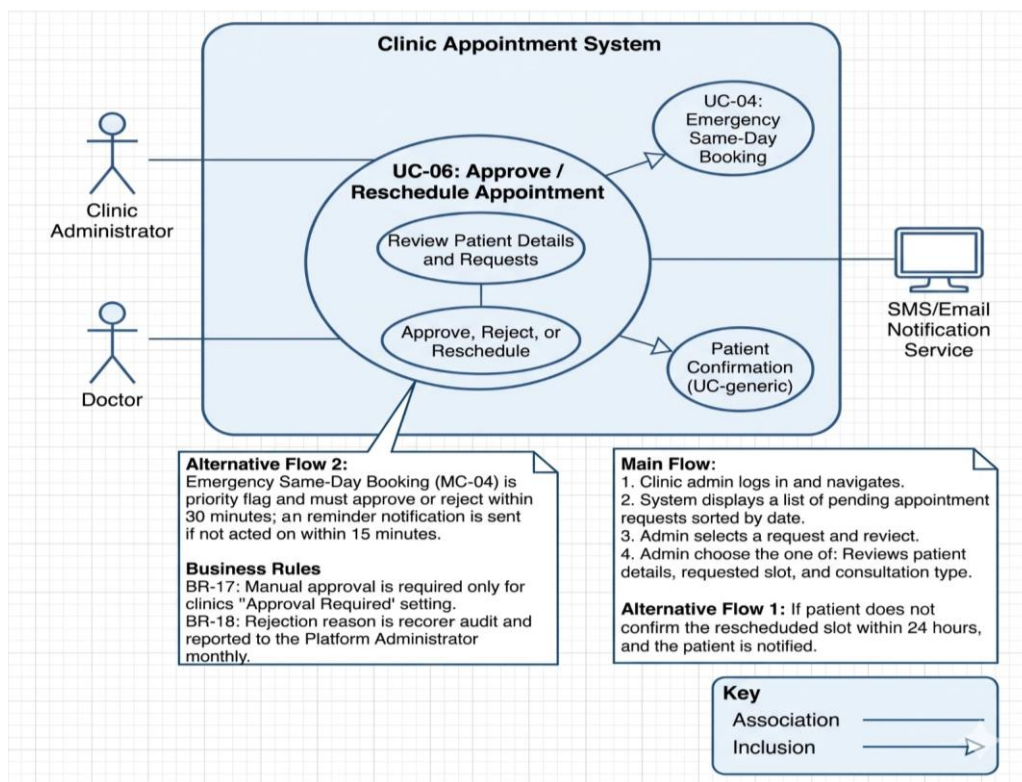


Figure 2.12: UC-06 – Approve / Reschedule Appointment

2.7.3.5.7. UC-07 – Join Telemedicine Consultation

Use Case ID	UC-07
Use Case Name	Join Telemedicine Consultation
Actor(s)	Patient, Doctor, Video Consultation Service (Jitsi/Daily.co)
Preconditions	Appointment is confirmed with consultation type 'Online/Video'. Appointment time is within 15 minutes.
Postconditions – Success	Consultation session is started; both parties connected in the video room.
Postconditions – Failure	Video session fails; fallback phone call option is displayed to both parties.
Main Flow	1. System generates a unique, time-limited video room link 15 minutes before the appointment. 2. Patient opens the appointment in 'My Appointments' dashboard and clicks 'Join Session'. 3. System verifies appointment status and patient identity. 4. Patients are connected to the video room. 5. Doctor opens the appointment from the doctor's dashboard and clicks 'Start Session'. 6. Doctors are connected; session status changes to 'In Progress'. 7. Upon completion, doctor clicks 'End Session'; appointment status updates to 'Completed'.

Alternative Flow 1	At step 2: Video links are expired or network failure occurs. System displays 'Unable to connect' with a one-tap phone call fallback button to call the doctor's registered phone number.
Alternative Flow 2	At step 4: Patient's device lacks camera/microphone permissions. System displays instructions for enabling permission; audio-only fallback is offered.
Business Rules	- BR-19: Video room links expire 30 minutes after the scheduled appointment end time. - BR-20: Sessions are not recorded by the SithCare platform; recording is the doctor's responsibility if clinically required. - BR-21: One-tap phone fallback must always be available if video fails.

Table 2.16: UC-07 – Join Telemedicine Consultation

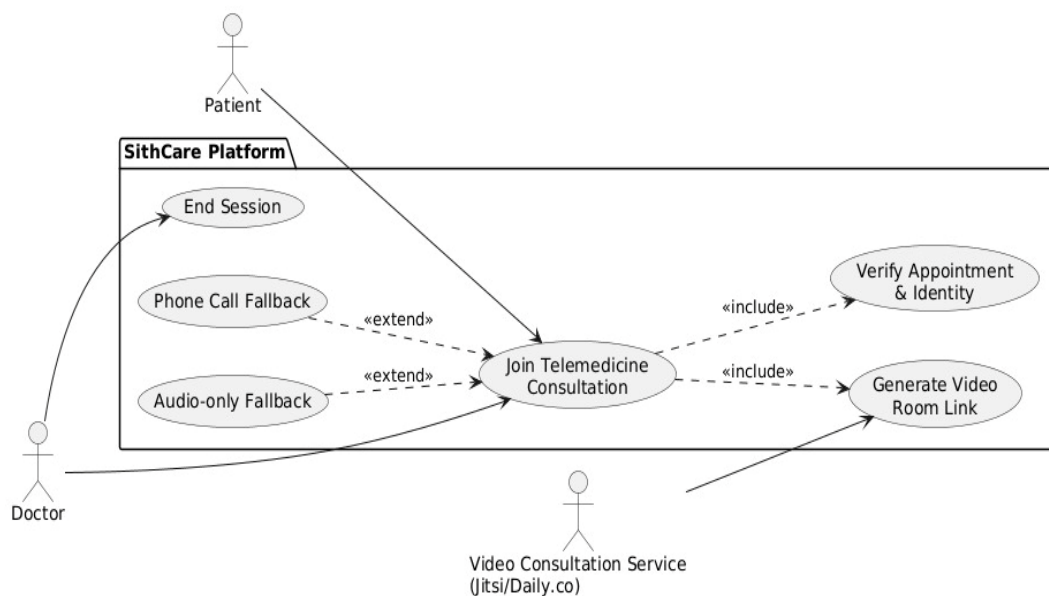


Figure 2.13: UC-07 – Join Telemedicine Consultation

2.7.3.5.8. UC-08- Process Payment and Generate Receipt

Use Case ID	UC-08
Use Case Name	Process Payment and Generate Receipt
Actor(s)	Patient, Payment Gateway (PayHere), Clinic Administrator
Preconditions	Appointment requires payment. Patients have selected a payment method (online or cash).
Postconditions – Success	Payment status updated to Paid; PDF receipt generated and available to patient and clinic admin.
Postconditions – Failure	Payment not completed; appointment status remains Pending Payment; patient notified.

Main Flow	1. Patient views payment due on the appointment summary screen. 2. Patient selects payment method: Online (PayHere) or Cash at Clinic. 3a. (Online): System redirects to PayHere checkout page with appointment details and fee in LKR. PayHere processes payment and return status to SithCare via webhook. 3b. (Cash): Clinic admin marks the appointment as Cash Paid after receiving payment at the clinic. 4. System updates appointment/payment record to 'Paid'. 5. System generates a PDF receipt with appointment details, fee, payment method, date, and booking reference. 6. Receipt is available for download from patient dashboard and clinic admin panel.
Alternative Flow 1	At step 3a: PayHere payment fails. System retains appointment slot for 5 minutes and prompts the patient to retry. After 2 failures, slot is released.
Alternative Flow 2	Refund required (appointment cancelled within refund window): System initiates refund via PayHere API; refund status tracked and patient notified of completion.
Business Rules	• BR-22: Payment card data is never stored on SithCare servers; all card handling is delegated to PayHere's PCI-DSS compliant infrastructure. • BR-23: All amounts are displayed and processed in Sri Lankan Rupees (LKR). • BR-24: PDF receipts are retained for 12 months and accessible to both patient and clinic admin.

Table 2.17: UC-08-Process Payment and Generate Receipt

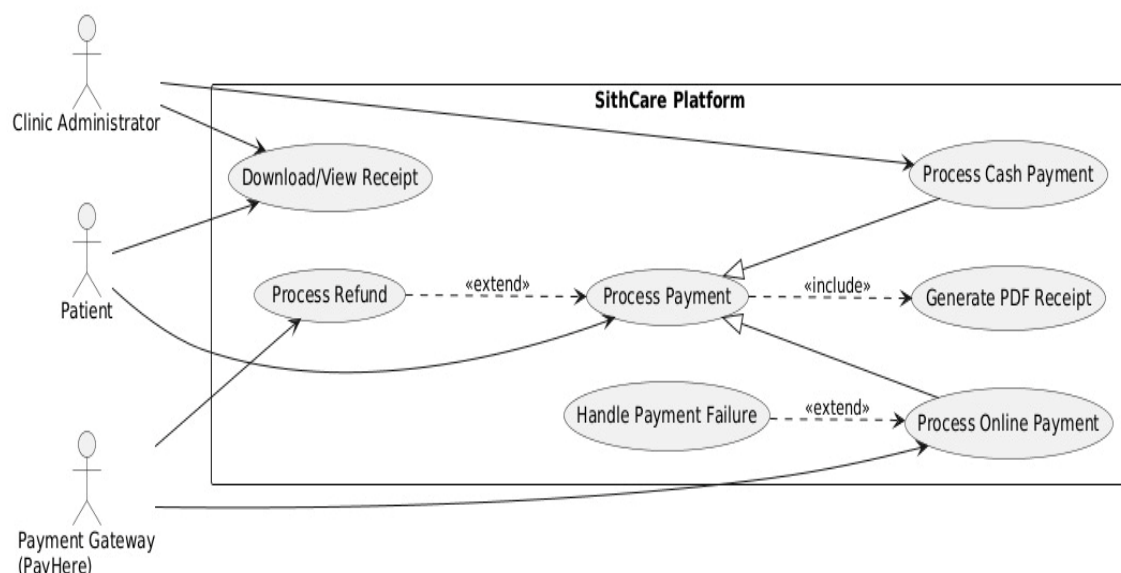


Figure 2.14: UC-08-Process Payment and Generate Receipt



2.7.3.5.9. UC-09 – Track Live Patient Queue

Use Case ID	UC-09
Use Case Name	Track Live Patient Queue
Actor(s)	Patient, Clinic Administrator
Preconditions	Patients have a confirmed in-person appointment for today with an assigned queue token. Clinic has queue tracking enabled.
Postconditions – Success	Patient can view their real-time queue position without being physically present at the clinic.
Postconditions – Failure	Queue data unavailable; static message displayed: 'Queue data temporarily unavailable. Please contact the clinic.'
Main Flow	1. Clinic admin starts the daily queue from the clinic dashboard. 2. System assigns each confirmed in-person patient a queue token number. 3. Patient logs in, navigates to 'My Appointments', and selects today's appointment. 4. System displays: current token being served, patient's token number, estimated wait time. 5. Queue tracker refreshes automatically every 60 seconds. 6. Clinic admin clicks 'Next Patient' to advance the queue. 7. System updates the display on all connected patient views in real time. 8. When patient is 3 tokens away, system sends a push/SMS notification: 'You are next soon – please head to the clinic.'
Alternative Flow 1	At step 5: If the clinic administrator pauses the queue (e.g., lunch break), a banner is shown: 'Queue is currently paused. Expected to resume at [time].'
Alternative Flow 2	Emergency slot is inserted by admin: queue is renumbered; affected patients receive an SMS notification of the delay.
Business Rules	• BR-25: Queue position is updated only by the clinic administrator. • BR-26: Patients are notified automatically when within 3 tokens of their turn. • BR-27: Queue resets automatically at the end of each clinic day.

Table 2.18: UC-09 – Track Live Patient Queue

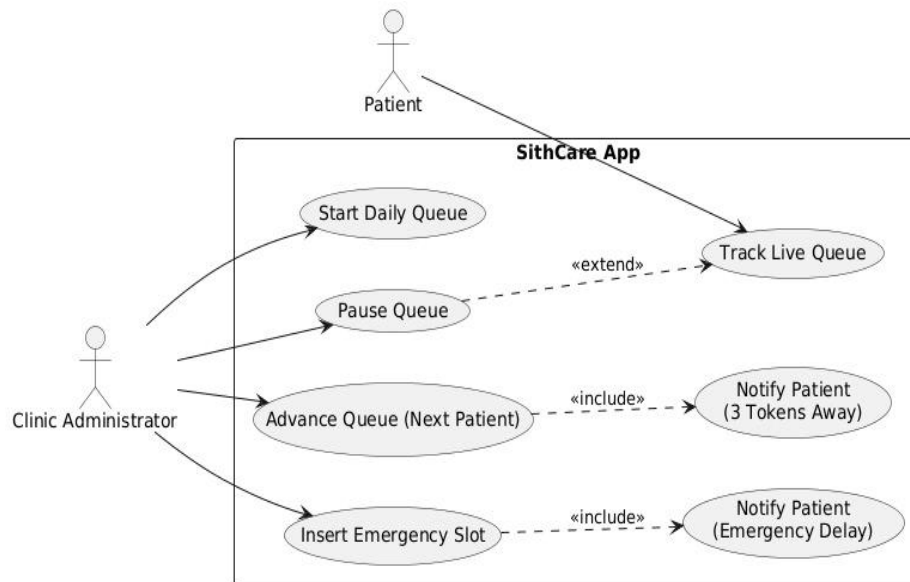


Figure 2.15: UC-09 – Track Live Patient Queue

2.7.3.5.10. UC-10 – Request Emergency Support

Use Case ID	UC-10
Use Case Name	Request Emergency Support
Actor(s)	Patient, Doctor, Emergency Helpline
Preconditions	Patient is logged in and in distress. Emergency support button is visible on all screens.
Postconditions – Success	Patient is connected to assigned doctors or helplines; location shared (if consented); crisis event logged with timestamp.
Postconditions – Failure	Connection fails; patients are shown static helpline numbers with tap-to-call functionality.
Main Flow	1. Patient taps the emergency button (persistent red floating action button visible on all screens). 2. System displays: 'You are about to contact your assigned mental health professional. Your location will be shared. Confirm?' along with the disclaimer: SithCare does not replace emergency services. Dial 119 (police) or 1990 (ambulance) for immediate danger. 3. Patient taps Confirm. 4. System request's location permission from device (if not already granted). 5. System attempts to call the assigned doctor. 6. System logs the crisis event with timestamp and anonymised patient ID. 7. If doctor answers patient is connected; consented location is shared with doctor.



Alternative Flow 1	At step 3: Patient taps Cancel. System returns to previous screen. No call placed; no location shared. No crisis event logged.
Alternative Flow 2	At step 4: Location services are disabled or permission denied. System skips location sharing, proceeds with call only, and informs the doctor that location was unavailable.
Business Rules	• BR-28: Location sharing requires explicit per-session user consent. • BR-29: Crisis events are logged for safeguarding review by the Platform Administrator. • BR-30: SithCare must always display: 'SithCare is not a substitute for emergency services. Dial 119 for police or 1990 for ambulance.'

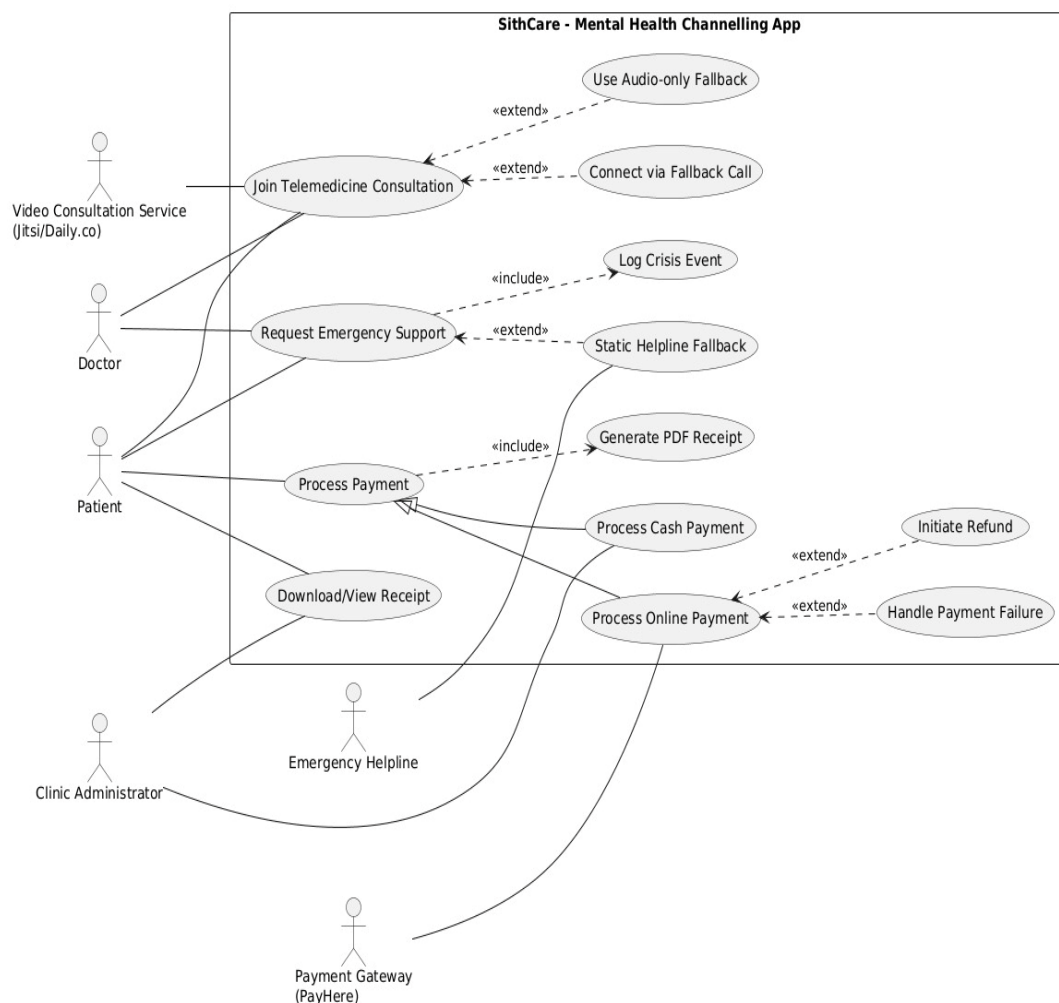


Figure 2.16: UC-10 – Request Emergency Support

**2.7.3.5.11. UC-11 – Use Anonymous Community Consultation**

Use Case ID	UC-11
Use Case Name	Use Anonymous Community Consultation
Actor(s)	Patient, Community Moderator
Preconditions	Patient is logged in and has accepted community guidelines.
Postconditions – Success	General questions are posted pseudonymously; community responses are visible; moderator can review if reported.
Postconditions – Failure	Post flagged and hidden; user informed of guideline violation.
Main Flow	1. Patient navigates to 'Community Forum' from the dashboard. 2. System displays community safety guidelines. Patients accept to proceed. 3. Patient submits a general mental health question using their pseudonymous platform identity. 4. System hides the real user identity & displays only the pseudonymous username. 5. Other community members (patients) can respond under their own pseudonymous identities. 6. Moderator monitors the forum and can flag, hide, or remove posts violating community guidelines.
Alternative Flow 1	Post contains private identifying information, clinical details, or emergency-level distress keywords: system warns user and recommends contacting a professional or using the emergency support feature.
Alternative Flow 2	User reports a post: moderator receives a review notification and must action within 24 hours.
Business Rules	• BR-31: Real user identity is never publicly linked to community posts. • BR-32: Community posts are not clinical advice; a disclaimer is permanently visible in the forum. • BR-33: Moderators can escalate concerning posts to Platform Admin for further review.

Table 2.20: UC-11 – Use Anonymous Community Consultation

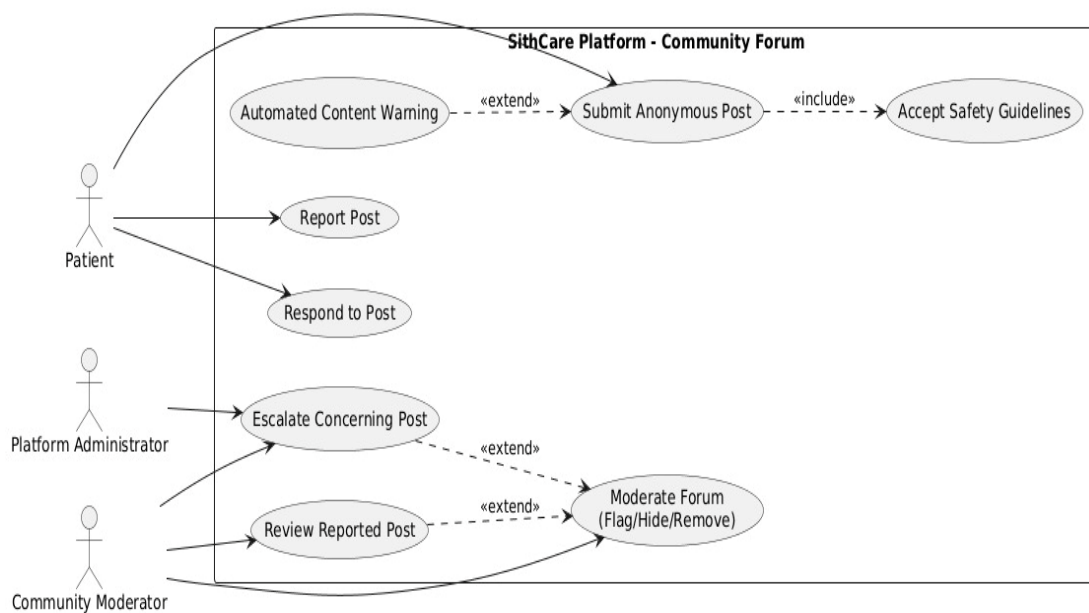


Figure 2.18: UC-11 – Use Anonymous Community Consultation

2.7.3.5.12. UC-12 – Complete AI Triage Questionnaire (SithBot)

Use Case ID	UC-12
Use Case Name	Complete AI Triage Questionnaire (SithBot)
Actor(s)	Patient (or Unregistered Visitor), AI Triage Module (SithBot)
Preconditions	User has accessed the SithCare platform (login is not required for basic triage access).
Postconditions – Success	User receives a specialist category recommendation and is guided to book an appointment, access resources, or contact emergency support.
Postconditions – Failure	Chatbot cannot determine a recommendation; users are guided to contact the platform support team or browse resources manually.
Main Flow	1. User clicks 'Talk to SithBot' on the home or dashboard screen. 2. SithBot greets user and asks: 'How are you feeling today?' with quick-reply options (Anxious, Depressed, Stressed, Struggling, Not Sure). 3. User describes feelings or selects options. 4. SithBot asks 2–4 follow-up questions based on the initial response. 5. SithBot analyses responses and suggests a specialist 6. SithBot presents options: Book Appointment Browse Resources Talk to a Volunteer Emergency Support. 7. Medical disclaimer is displayed: 'SithBot does not provide medical diagnoses. Please consult a qualified mental health professional.'



Alternative Flow 1	At step 3: User types crisis keywords (e.g., 'suicidal', 'harm myself', 'end my life'). SithBot immediately surfaces emergency support options and helpline numbers, skipping the triage flow entirely.
Alternative Flow 2	At step 5: Chatbot confidence is below threshold. SithBot responds: 'I want to make sure you get the right support. Let me connect with a volunteer listener or a platform support contact.'
Business Rules	- BR-34: SithBot must never use the word 'diagnose' or imply clinical assessment. - BR-35: All chatbot interactions are logged in anonymized form for safety review by Platform Admin. - BR-36: Medical disclaimer must appear on every SithBot response screen, not just the final screen.

Table 2.21: UC-12 – Complete AI Triage Questionnaire (SithBot)

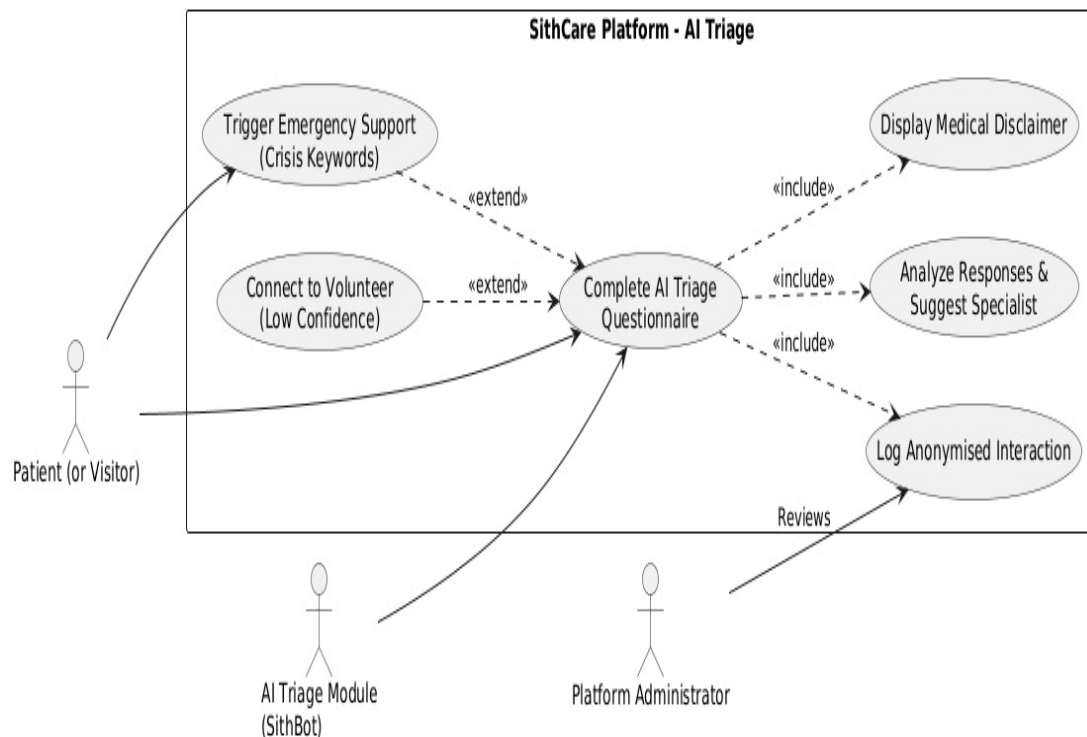


Figure 2.19: UC-12 – Complete AI Triage Questionnaire (SithBot)



2.7.3.6. Activity Diagram Plan

2.7.3.6.1. Activity Diagram Descriptions

The following activity diagrams shall be created using Draw.io (UML 2.x notation) and committed to the repository under documents/diagrams/activity/ with both source (.drawio) and exported SVG/PDF files.

Diagram ID	Process / Use Case	Scope Description
AD-01	User Authentication (UC-01)	Role-based login and registration flow, OTP delivery, input validation, session creation, and dashboard routing.
AD-02	Appointment Booking (UC-03, UC-04)	Specialist search filters, slot selection, payment flow, booking confirmation, and SMS/email notification.
AD-03	Emergency Support (UC-10)	Emergency button activation, disclaimer display, consent confirmation, doctor call attempt, helpline fallback, and crisis log.
AD-04	Telemedicine Consultation (UC-07)	Video room link generation, patient join flow, secure consultation session, and phone fallback handling.
AD-05	Community Posting (UC-11)	Anonymous post creation, pseudonymous identity masking, moderation rules, community responses, and report handling.

Table 2.22: Activity Diagram Descriptions

2.7.3.6.2. Activity Diagrams

Below are the key activity diagrams drawn to depict the behavior of the SithCare Application.

Source codes and diagrams have been uploaded to the GitHub “SithCare” Organization.

1. User Authentication

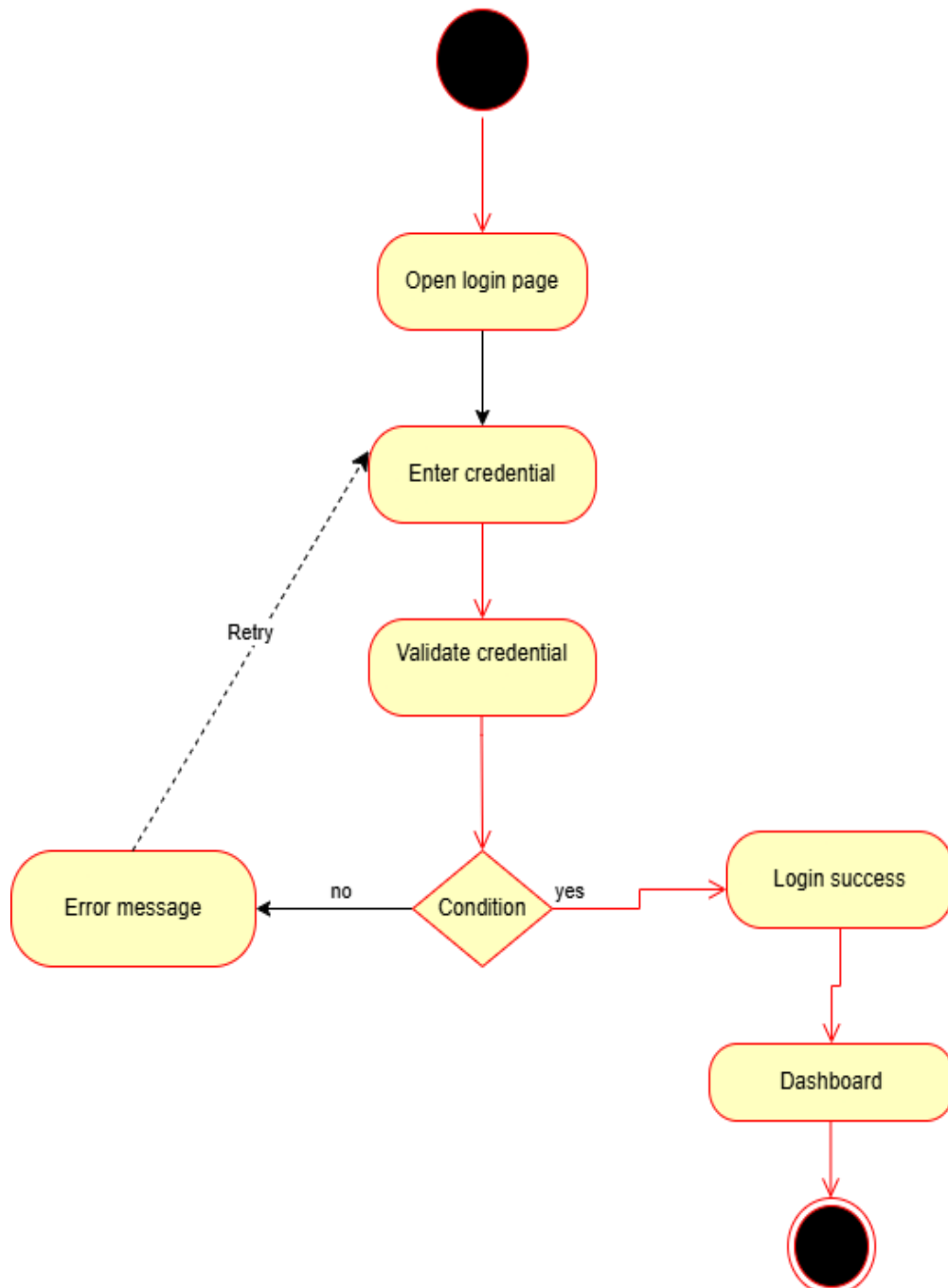


Figure 2.20: Activity Diagram - User Authentication



2. Booking an Appointment

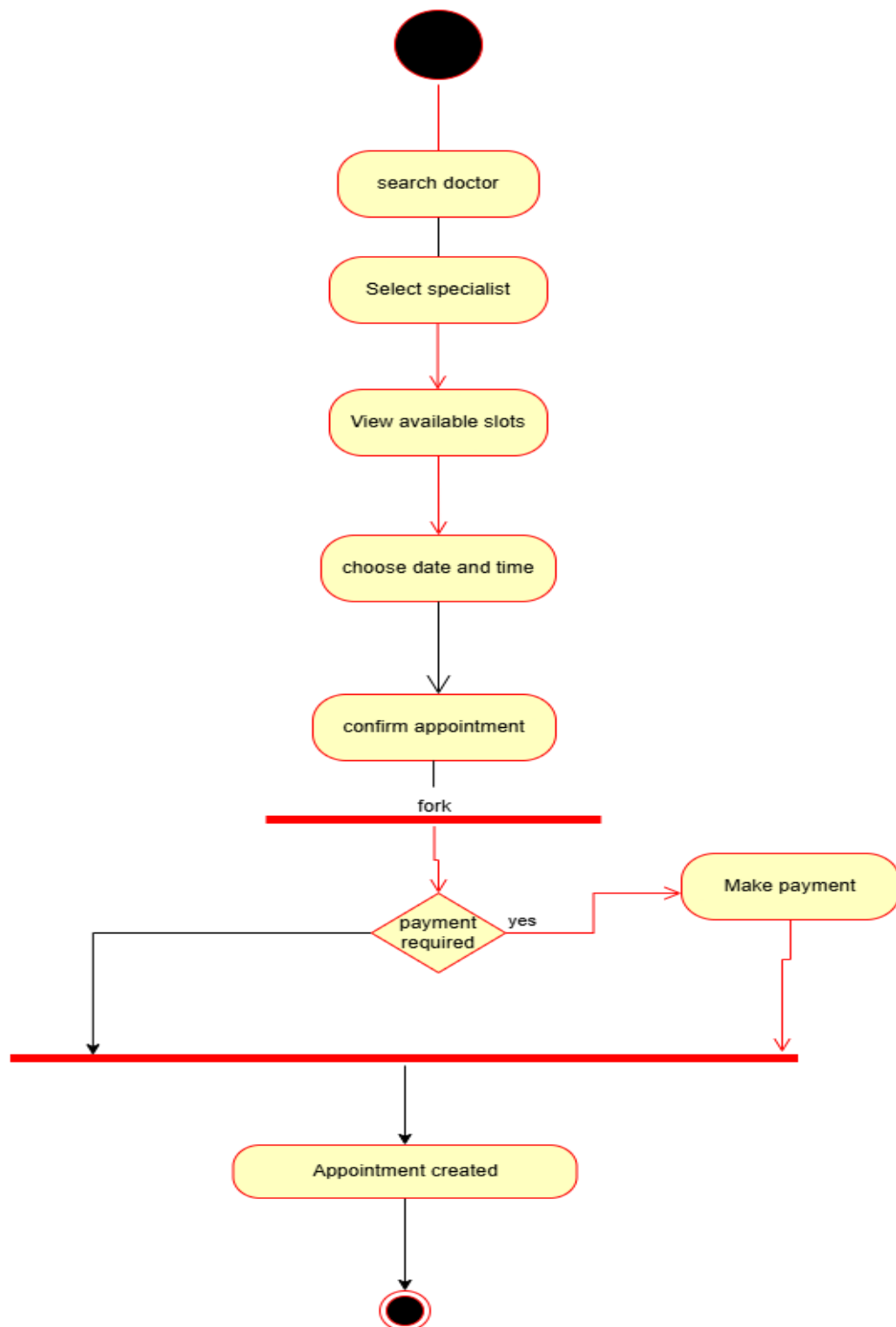


Figure 2.21: Activity Diagram - Booking an Appointment



3. Emergency Support

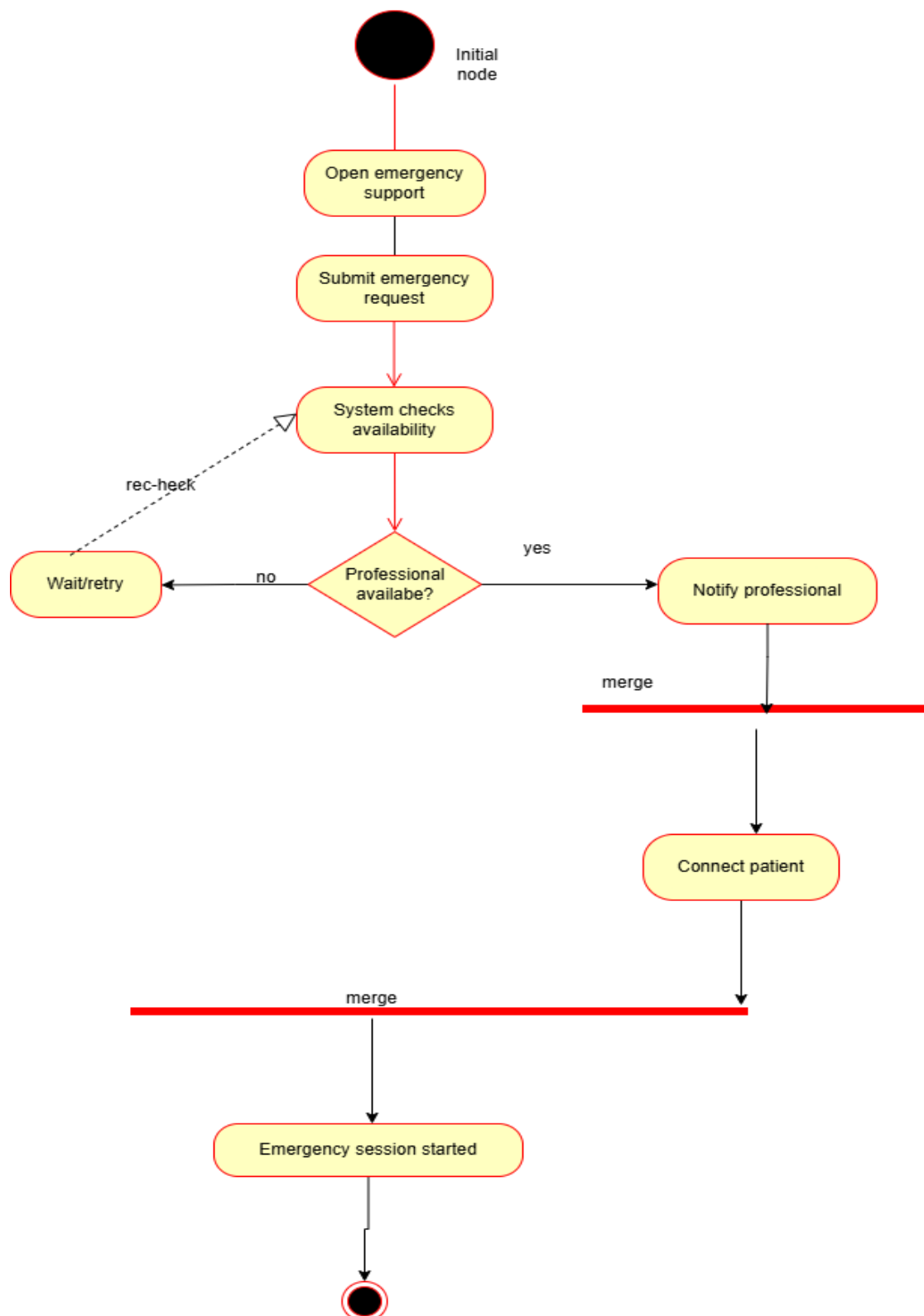


Figure 2.22: Activity Diagram - Emergency Support



4. Telemedicine Consultation

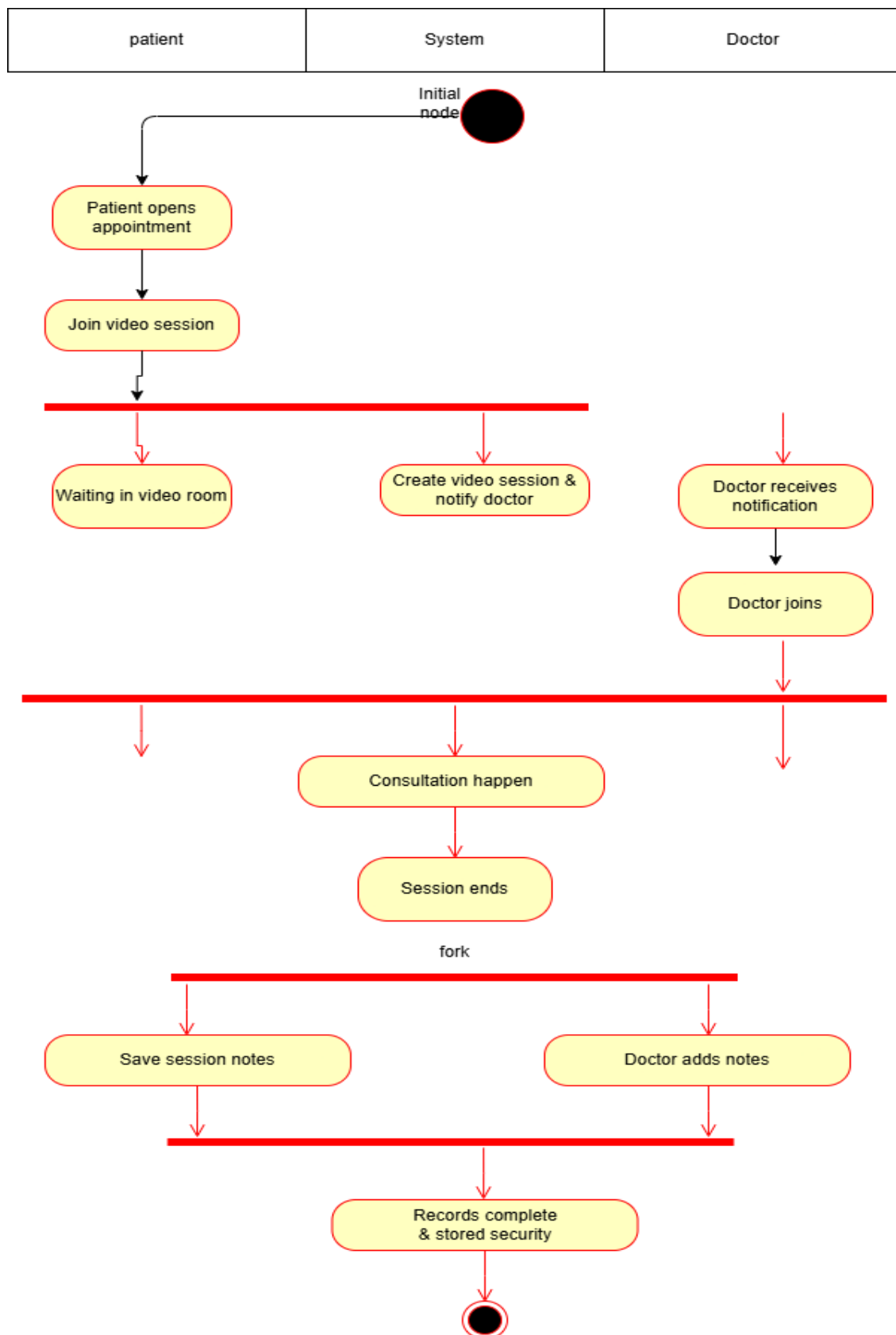


Figure 2.23: Activity Diagram - Telemedicine Consultation



5. Community Posting

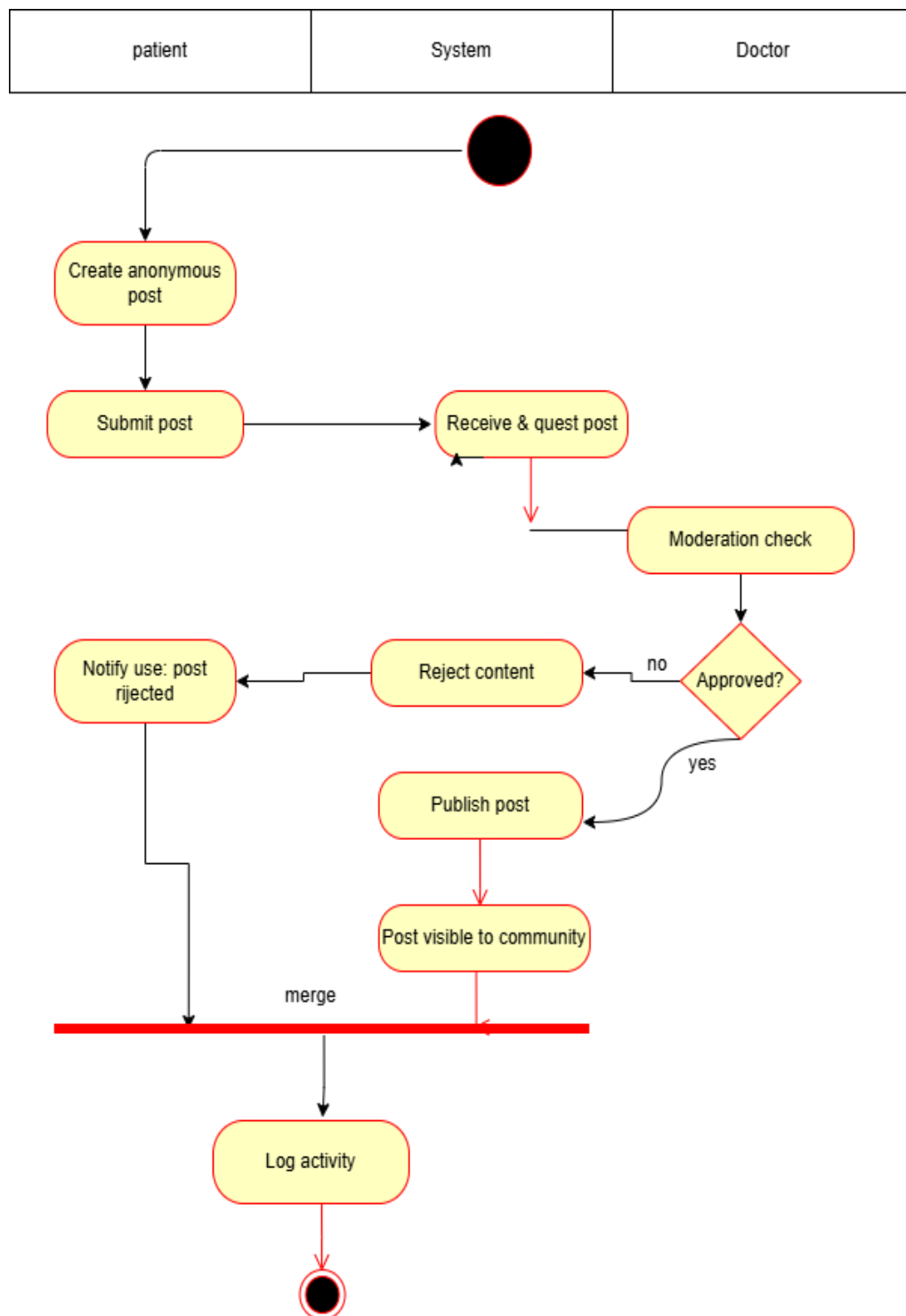


Figure 2.24: Activity Diagram - Community Posting



2.7.4. Functional Requirements

Functional requirements are prioritised using the MoSCoW framework: M = Must have (MVP critical), S= Should have (high value), C = Could have (desirable), W = Won't have in first release. Each requirement references the source use case and includes a verification test case ID.

2.7.4.1. Authentication and Access Control

ID	Description	Priority	Source	Verification
FR-01	The system shall allow unregistered users to create a patient account using email or phone number with OTP verification.	M – Must	UC-01	TC-FR-01: Register with valid email; verify OTP; confirm account activated.
FR-02	The system shall enforce password complexity: minimum 8 characters, 1 uppercase, 1 digit, 1 special character.	M – Must	UC-01; NFR-S	TC-FR-02: Attempt registration with weak passwords; confirm rejection and error message.
FR-03	The system shall support role-based access control (RBAC) with roles: Patient, Doctor, Clinic Admin, Platform Admin, Volunteer, Moderator.	M – Must	UC-01	TC-FR-03: Login as each role; verify different dashboards and menus are displayed.
FR-04	The system shall provide a secure password reset flow via email OTP expiring in 10 minutes.	M – Must	UC-01	TC-FR-04: Request reset; verify OTP expiry at 10 min; confirm password changed successfully.
FR-05	The system shall automatically log out users after 30 minutes of inactivity.	S – Should	NFR-S	TC-FR-05: Leave session idle for 31 minutes; confirm redirect to login page.
FR-06	The system shall support optional Multi- Factor Authentication (MFA) for Clinic and Platform Administrators.	S – Should	NFR-S	TC-FR-06: Enable MFA; login from new device; confirm OTP prompt is displayed.

Table 2.23: Authentication and Access Control



2.7.4.2. Profile and Consent Management

ID	Description	Priority	Source	Verification
FR-07	The system shall allow patients to update personal profile information including name, contact details, language preference, and profile photo.	M – Must	UC-02	TC-FR-07: Update profile fields; save; confirm changes persist on re-login.
FR-08	The system shall display privacy consent options and record consent acceptance and withdrawal with a timestamp.	M – Must	UC-02; PDPA	TC-FR-08: Withdraw consent; confirm timestamp recorded and data minimisation triggered.
FR-09	The system shall allow patients to add or update emergency contact name and phone number on their profile.	M – Must	UC-02; UC-10	TC-FR-09: Add emergency contact; trigger emergency flow; confirm contact details are shown.
FR-10	The system shall provide an Adult Mode with minimum 18px font, simplified navigation, and reduced feature set.	S – Should	UC-02; Persona 2	TC-FR-10: Enable Adult Mode; confirm menu reduced and text enlarged to 18px minimum.

Table 2.24: Profile and Consent Management

2.7.4.3. Specialist Search and Discovery

ID	Description	Priority	Source	Verification
FR-11	The system shall allow patients to search for mental health specialists using filters: category, location, consultation type, language, and availability date.	M – Must	UC-03; Survey Q3	TC-FR-11: Apply all filters; confirm returned results match all applied criteria exactly.
FR-12	The system shall display specialist profiles with: name, photo, qualifications, SLMC/SLPA registration number, languages, consultation fee in LKR, rating, and available slots.	M – Must	UC-03	TC-FR-12: View specialist profile; confirm all specified fields are present and accurate.
FR-13	The system shall display clinic locations on an interactive map (Google Maps / OpenStreetMap).	S – Should	UC-03; Survey Q6	TC-FR-13: View clinic; confirm correct map pin and directions link.



FR-14	The system shall allow patients to read and submit star ratings and text reviews for specialists, subject to moderation.	C – Could	UC-03; Competitive Analysis	TC-FR-14: Submit review; confirm displayed on profile after moderation approval.
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Table 2.25: Specialist Search and Discovery

2.7.4.4. Appointment Management

ID	Description	Priority	Source	Verification
FR-15	The system shall allow patients to book in-person or video appointments with available specialists.	M – Must	UC-04; Interview #1	TC-FR-15: Book appointment; confirm slot blocked and confirmation sent.
FR-16	The system shall allow patients to cancel appointments and trigger a refund if cancellation is more than 24 hours before the scheduled time.	M – Must	UC-04	TC-FR-16: Cancel 25+ hrs in advance; confirm refund initiated via PayHere.
FR-17	The system shall allow clinic administrators to approve, reject, or reschedule pending appointment requests.	M – Must	UC-06; Interview #2	TC-FR-17: Admin rejects booking; confirm patient receives notification with reason.
FR-18	The system shall provide an Emergency Same-Day Booking option when regular slots are full, subject to clinic approval.	S – Should	UC-04; UC-06	TC-FR-18: Attempt booking when full; confirm emergency option appears and admin notified.
FR-19	The system shall send SMS and email reminders to patients 24 hours and 1 hour before their appointment.	M – Must	UC-04; Survey Q7	TC-FR-19: Book appointment; verify two reminders are sent at the correct scheduled times.
FR-20	The system shall allow doctors to manage their weekly availability schedule (add, edit, and block slots).	M – Must	UC-05; Interview #2	TC-FR-20: Doctor blocks a slot; confirm slot is unavailable in-patient search immediately.

Table 2.26: Appointment Management



2.7.4.5. Telemedicine

ID	Description	Priority	Source	Verification
FR-21	The system shall generate a secure, time-limited video consultation room link for confirmed online appointments, available 15 minutes before the scheduled time.	M – Must	UC-07	TC-FR-21: Confirm online appointment; verify video link appears in dashboard 15 min before slot.
FR-22	The system shall allow the doctor to initiate the video session from the doctor dashboard.	M – Must	UC-07	TC-FR-22: Doctor clicks 'Start Session'; confirm video room opens and status updates.
FR-23	The system shall provide a one-tap phone call fallback option if the video connection fails.	S – Should	UC-07; Risk	TC-FR-23: Simulate video failure; confirm phone call fallback button appears.

Table 2.27: Telemedicine

2.7.4.6. Queue Management and Clinic Operations

ID	Description	Priority	Source	Verification
FR-24	The system shall assign a queue token to each confirmed in-person appointment and display the live queue position to patients.	M – Must	UC-09	TC-FR-24: Admin advances queue; confirm patient dashboard updates within 60 seconds.
FR-25	The system shall allow clinic administrators to advance, pause, and reset the daily patient queue.	M – Must	UC-09; Interview #2	TC-FR-25: Admin pauses queue; confirm patients see pause banner with expected resume time.
FR-26	The system shall send a push/SMS proximity notification when patient is within 3 tokens of their turn.	S – Should	UC-09	TC-FR-26: Advance queue to 3 tokens before patient; confirm notification is delivered.
FR-27	The system shall provide clinic administrators with a daily appointment list sortable by time, doctor, and status.	M – Must	UC-06; Interview #2	TC-FR-27: Login as clinic admin; verify list sorts correctly by all three criteria.

Table 2.28: Queue Management and Clinic Operations



2.7.4.7. Payment Processing

ID	Description	Priority	Source	Verification
FR-28	The system shall integrate PayHere payment gateway for online appointment payments in LKR.	M – Must	UC-08	TC-FR-28: Complete online payment; confirm PayHere redirects and payment confirmed via webhook.
FR-29	The system shall allow clinic administrators to record and track cash payment status.	M – Must	UC-08; Interview #2	TC-FR-29: Mark appointment as cash paid; confirm payment record status updated to Paid.
FR-30	The system shall generate a PDF payment receipt for all confirmed transactions (online and cash).	M – Must	UC-08	TC-FR-30: Generate receipt; confirm PDF opens with correct appointment details, fee, and reference.
FR-31	The system shall process refunds for cancelled appointments via the original PayHere payment method.	S – Should	UC-08; FR-16	TC-FR-31: Cancel eligible appointment; confirm refund initiated and patient notified of status.

Table 2.29: Payment Processing

2.7.4.8. Emergency Support

ID	Description	Priority	Source	Verification
FR-32	The system shall provide a persistent emergency support button visible on all screens when a patient is logged in.	M – Must	UC-10	TC-FR-32: Login as patient; navigate multiple screens; confirm button is visible on all.
FR-33	The system shall require explicit user confirmation before initiating a crisis call or sharing location data.	M – Must	UC-10; PDP A	TC-FR-33: Press emergency button; confirm confirmation dialog appears with location consent prompt.
FR-34	The system shall display Sri Lankan emergency helplines if the assigned doctor is unreachable within 20 seconds.	M – Must	UC-10	TC-FR-34: Simulate doctor unavailability; confirm helplines (1926, 0112 696 666, 0115 679 900) appear at 20s mark.
FR-35	The system shall display a non-substitution disclaimer: 'SithCare is not a substitute for emergency services. Dial 119 (Police) or 1990 (Ambulance).'	M – Must	UC-10; Risk	TC-FR-35: Open emergency screen; confirm disclaimer text is displayed prominently.



FR-36	The system shall log all crisis events with timestamp, anonymised patient ID, action taken, and outcome.	M – Must	UC-10; Risk	TC-FR-36: Trigger emergency; confirm log entry created and visible in Platform Admin panel.
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Table 2.30: Emergency Support

2.7.4.9. AI Chatbot (SithBot)

ID	Description	Priority	Source	Verification
FR-37	The system shall provide SithBot, an AI triage chatbot accessible without login for basic mental health triage.	M – Must	UC-12	TC-FR-37: Access chatbot without login; confirm full triage flow completes and recommendation displayed.
FR-38	The system shall detect crisis keywords in chatbot input and immediately surface emergency support options, bypassing the triage flow.	M – Must	UC-12, Risk	TC-FR-38: Enter crisis keyword; confirm emergency options displayed within one chatbot response.
FR-39	The system shall display a non- diagnosis disclaimer on every SithBot response screen.	M – Must	UC-12, Risk	TC-FR-39: Complete triage; confirm disclaimer present on every screen, not only the final.

Table 2.31: AI Chatbot (SithBot)

2.7.4.10. Community Features

ID	Description	Priority	Source	Verification
FR-40	The system shall provide an anonymous community forum where users post using pseudonymous platform identities.	S – Should	UC-11	TC-FR-40: Post anonymously; confirm real name is hidden from all other users.
FR-41	The system shall allow community moderators to flag, hide, and delete posts that violate community guidelines.	S – Should	UC-11	TC-FR-41: Moderator hides post; confirm removed from public view within 60 seconds.

Table 2.32: Community Features

2.7.4.11. Localization and Communication

ID	Description	Priority	Source	Verification
FR-49	The system shall support English, Sinhala, and Tamil language switching accessible from any screen.	M – Must	Survey Q4; 1.2 Scope	TC-FR-49: Switch to Sinhala; confirm all key UI strings render correctly in Unicode Sinhala.



FR-50	The system shall display all monetary amounts in Sri Lankan Rupees (LKR).	M – Must	1.2 Scope	TC-FR-50: Complete booking flow; confirm all fees displayed as LKR values throughout.
FR-51	The system shall display local emergency helpline numbers (Sumithrayo 1926, CCCLine 0112 696 666, Sahanaya 0115 679 900) in the emergency support section.	M – Must	UC-10; 1.2 Scope	TC-FR-51: Access emergency section; confirm all three numbers are displayed and each is tap-to-call.
FR-52	The system shall send appointment reminders and notifications via both SMS and email in the patient's preferred language.	M – Must	UC-04; Interview #1	TC-FR-52: Set language to Sinhala; book appointment; confirm both SMS and email arrive in Sinhala.

Table 2.33: Localization and Communication

2.7.5. Non-Functional Requirements

2.7.5.1. Performance

ID	Requirement	Target / Threshold	Verification
NFR-P-01	Page load time on 4G LTE (>10 Mbps).	≤ 3 seconds for 95% of requests.	Performance test with JMeter / Lighthouse.
NFR-P-02	Appointment booking transaction (excluding PayHere redirect).	End-to-end ≤ 5 seconds.	Integration performance test.
NFR-P-03	Concurrent user support without degradation.	Minimum 500 concurrent users.	Load test with JMeter.
NFR-P-04	Live queue update propagation after admin action.	All connected patients updated within 60 seconds.	Automated queue advancement test.
NFR-P-05	AI chatbot (SithBot) response time.	≤ 2 seconds for 90% of queries.	Chatbot response timing test.

Table 2.34: Performance

2.7.5.2. Security

ID	Requirement	Verification
NFR-S-01	All data in transit shall be encrypted using TLS 1.3.	SSL/TLS configuration review; automated SSL scan.
NFR-S-02	All patient health data at rest shall be encrypted using AES-256.	Database encryption configuration review.



NFR-S-03	JWT access tokens shall expire after 15 minutes; refresh tokens after 7 days.	Token expiry test: confirm expired token rejected.
NFR-S-04	The system shall implement OWASP Top 10 mitigations including input sanitisation, CSRF protection, and rate limiting.	OWASP security checklist review; penetration test.
NFR-S-05	Patient health/session records shall be accessible only to the authoring doctor and Platform Administrator.	RBAC access control test for each role.
NFR-S-06	Community forum posts shall use pseudonymous IDs; real user identity shall not be publicly linked.	Data exposure test; inspect forum post metadata.
NFR-S-07	Payment card data shall never be stored on SithCare servers; all payment data handled by PayHere (PCI- DSS).	Architecture review; confirm no card data in SithCare DB.
NFR-S-08	All authentication events (login, logout, failed attempts) shall be logged with timestamp and IP address.	Log audit review; confirm log entries for all auth events.

Table 2.35: Security

2.7.5.3. Usability

ID	Requirement	Target / Threshold	Verification
NFR-U-01	System Usability Scale (SUS) score in usability testing.	≥ 70 (acceptable) in user testing sessions.	SUS questionnaire with test participants.
NFR-U-02	Core task completion time for new users (register, find doctor, book appointment).	Completable in under 5 minutes.	Timed usability test with 5 participants.
NFR-U-03	Adult Mode font size and navigation.	Minimum 18px font; single-level navigation menus.	Visual inspection and accessibility audit.
NFR-U-04	Screen reader / accessibility compliance.	WCAG 2.1 Level AA.	Accessibility audit using axe / WAVE tools.
NFR-U-05	Error messages across all user-facing screens.	Plain language in the user's selected language; no technical jargon.	Content review with native language reviewers.

Table 2.36: Usability



2.7.5.4. Reliability

ID	Requirement	Target	Verification
NFR-R-01	System availability (excluding scheduled maintenance).	$\geq 99.5\%$ uptime.	Uptime monitoring (UptimeRobot or equivalent).
NFR-R-02	Scheduled maintenance windows.	Not during 08:00–22:00 Sri Lanka Standard Time (SLST).	Maintenance schedule review; advance notice to users.
NFR-R-03	Automated database backup frequency and recovery point.	Every 6 hours; RPO = 6 hours.	Backup log review; test restore from backup.
NFR-R-04	Emergency support subsystem availability during main application failure.	Emergency call/helpline feature must remain accessible even if non-critical modules fail.	Failover test: disable non-critical modules; confirm emergency feature operational.

Table 2.37: Reliability

2.7.5.5. Scalability

ID	Requirement	Verification
NFR-SC-01	Application layer shall support horizontal scaling without code changes.	Architecture review; container/deployment configuration review.
NFR-SC-02	Database schema-per-tenant design shall support up to 200 clinics without schema migration.	Tenant isolation test; load test with 200 simulated clinic schemas.
NFR-SC-03	System shall handle a $5\times$ traffic spike during public health campaigns.	Stress test: simulate $5\times$ baseline load; confirm no downtime or degraded core function.

Table 2.38: Scalability

2.7.5.6. Maintainability

ID	Requirement	Verification
NFR-M-01	All code shall follow consistent coding standards documented in CONTRIBUTING.md.	Code review; linting pass in CI pipeline.
NFR-M-02	Unit test coverage for all backend service modules.	Minimum 70% line coverage reported by test runner.



NFR-M-03	All API endpoints shall be documented using OpenAPI 3.0 specification.	OpenAPI spec review; confirm all endpoints documented.
NFR-M-04	No hardcoded secrets in source code; environment-based configuration.	Static analysis scan; confirm secrets in .env files excluded from repository.
NFR-M-05	All code changes via pull requests with at least one peer review before merge.	GitHub PR history review; confirm branch protection rules active.

Table 2.39: Maintainability

2.7.6. Alternative Solutions and Feasibility Study

2.7.6.1. Alternative Solutions Considered

2.7.6.1.1. Alternative A – Manual Phone / Walk-in Booking (Current State)

- Patients contact clinics manually or visit physically. This is simple but causes delays, limited transparency, no central guidance, no digital receipts, no queue tracking, poor support for remote/elderly users, and significant stigma barriers from phone-based booking overheard by family members. This does not address the identified problem.

2.7.6.1.2. Alternative B – Adapt an Existing Open-Source Platform (e.g., OpenMRS)

- OpenMRS is an open-source electronic medical records system used in developing countries. It could be extended to support mental health appointments. However, it is primarily designed for in-person clinical workflows in hospital settings, lacks patient-facing consumer UX, has no native community or volunteer features, and would require substantial customisation to meet Sri Lankan mental health context requirements. The learning curve and configuration overhead were assessed as too high for an 8-week design phase.

2.7.6.1.3. Alternative C – Build on a Healthcare SaaS Platform (e.g., Medplum)

- Medplum is a FHIR-compliant healthcare developer platform providing authentication, patient records, and scheduling APIs. It is optimised for the US healthcare market (HL7 FHIR, US insurance workflows). Sri Lankan context requirements (PayHere, Sinhala/Tamil localisation, local emergency helplines, LKR pricing) would require significant custom development on top of a complex proprietary stack. Vendor dependency and licensing costs are additional concerns for a university project.



2.7.6.1.4. Alternative D – Social Media / Informal Support Groups (e.g., WhatsApp Groups)

- Users currently use public or private groups for community support, but privacy, misinformation, moderation, crisis handling, and professional connection are poor. This does not provide appointment management, verified specialist discovery, payment handling, or emergency support.

2.7.6.1.5. Recommended Solution – Custom-Built Web Application (SithCare)

- A custom-built solution using established open-source technologies (React, Node.js/FastAPI, PostgreSQL) provides full control over localisation, UX, data privacy, and compliance. It directly addresses the Sri Lankan context without the constraints of an existing platform and is achievable within the team's skill set and the SENG 31242 timeline.

2.7.6.2. Feasibility Assessment

Dimension	Assessment	Justification
Technical	Feasible	Team skills include React, Node.js, SQL, and REST APIs from prior coursework. Telemedicine SDK (Jitsi) is free and open-source. PayHere provides a developer sandbox for testing.
Economic	Feasible (Academic)	Development cost is team effort only. Cloud hosting can use free-tier (AWS Educate / DigitalOcean student credits). PayHere has no monthly fee for sandbox testing. SMS/email usage within free-tier limits during development.
Operational	Feasible	Clinic administrators are technically capable of using web dashboards per interview findings. Patient-facing UI designed for low-literacy users (Adult Mode, multi-language, guided flows).
Schedule	Feasible with MoSCoW	8-week design phase is sufficient for core MVP requirements (Auth, Booking, Emergency, Queue). Advanced features (AI chatbot, volunteer portal, community) are Could-have and deferred to the SENG 34213 development phase.



Legal / Ethical	Feasible with Constraints	Requires strict privacy controls, PDPA-awareness checklist, RBAC, data minimisation, consent recording, and clear disclaimers that SithCare is not an emergency service or diagnostic tool.
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Table 2.40: Feasibility Assessment

2.7.6.3. MoSCoW Scope Prioritization

Priority	Features
Must Have	Authentication and privacy, role-based dashboards (Patient, Clinic Admin, Platform Admin), mental health specialist search and verified profiles, appointment booking with PayHere and cash payment, clinic/doctor availability management, admin user management, emergency support with helpline display and crisis logging, basic SMS/email notifications, multilingual support (Sinhala, Tamil, English).
Should Have	Telemedicine (video consultation with fallback), PDF receipt generation, live patient queue tracker, doctor verification workflow, content moderation tools for moderators, appointment approval/reschedule workflow, Adult Mode, multi-branch clinic management, appointment cancellation and refund.
Could Have	Anonymous community consultation forum, volunteer listener portal, AI triage chatbot (SithBot), doctor session notes, recovery stories module, specialist ratings and reviews, analytics dashboard for Platform Admin, mood and progress tracking.
Won't Have (v1.0)	AI clinical diagnosis, automatic emergency service dispatch without user confirmation, integration with government EHR systems, native Android/iOS mobile applications, insurance claim processing, complex fundraising automation.

Table 2.41: MoSCoW Scope Prioritization



3. SYSTEM DESIGN

3.1. Architectural Design

3.1.1. Architectural Pattern

SithCare employs a Client-Server Architecture utilising a Layered (N-Tier) model on the back end, tightly integrated with the Model-View-Controller (MVC) design pattern. The application is built using the MERN stack (MongoDB, Express.js, React.js, Node.js). The architectural layers are:

- Presentation Layer (Client): React-based Single Page Application handling UI, routing, and state management for Patients, Specialists, and Administrators.
- Application/Controller Layer (Server): Express.js REST APIs that intercept HTTP requests, validate payloads, and manage Role-Based Access Control via JSON Web Tokens.
- Business Logic Layer (Server): Node.js services managing core functionalities such as appointment scheduling, queue management, AI triage (SithBot), and emergency routing.
- Data Access Layer (Database): Mongoose Object Data Modeling (ODM) wrapping a NoSQL MongoDB database.

3.1.2. Quality Attribute Justification

The chosen architecture directly supports the Non-Functional Requirements:

- Scalability: Node.js non-blocking I/O supports 500 concurrent users and 5x traffic spikes. MongoDB flexible schema accommodates schema-per-tenant design for multiple clinic branches.
- Maintainability: MVC and Layered approach enforce separation of concerns. JavaScript across the entire stack reduces context switching and improves code maintainability.
- Security: Dedicated Controller layer verifies JWTs and enforces RBAC before business logic execution. MongoDB field-level encryption aids PDPA compliance.
- Reliability: React virtual DOM ensures sub-3-second page loads. Separation of external services ensures third-party failures do not crash the primary application.



Usability: React combined with Tailwind CSS provides a responsive, intuitive experience. Simplified navigation flows reduce cognitive load.

3.1.3. Component Design

The platform consists of six major components:

- 1) Patient Module handling registration, authentication, specialist search, booking, queue tracking, emergency support, and community interaction.
- 2) Mental Health Specialist Module managing appointments, schedules, telemedicine, and patient communication.
- 3) Clinic Administration Module providing clinic profile management, doctor allocation, queue management, and appointment monitoring.
- 4) SithCare Admin Module for oversight of all portals, user management, and risk control.
- 5) Community Support Module enabling anonymous consultation, recovery stories, peer support, and volunteer listener sessions.
- 6) AI Assistance Module (SithBot) providing symptom-based triage, specialist recommendations, and navigation assistance.

3.1.4. Deployment Architecture

The deployment topology utilises free-tier cloud services optimised for MERN stack applications. Client devices access the SPA via HTTPS (TLS 1.3). Static React assets are hosted on Vercel or Netlify with global CDN. The backend Node.js Express API runs on Render or Railway with SSL termination. MongoDB Atlas Free Tier provides cloud-hosted database persistence. All external integrations communicate securely over HTTPS.

3.2. Technology Stack

The platform is built using the MERN stack, selected for its unified JavaScript ecosystem, high development velocity, and alignment with team skills.

- Frontend: React + Vite (component-based architecture, fast rendering, reusable UI, scalable SPA)
- Styling: Tailwind CSS (rapid UI development, utility-first responsive design, consistent design system)



- Backend: Node.js + Express.js (lightweight, event-driven async architecture, large middleware ecosystem)
- Database: MongoDB Atlas (flexible document schema, cloud-native, high scalability)
- ODM: Mongoose (schema validation, query handling, middleware, model abstraction)
- Auth: JWT + bcrypt (stateless authentication, secure password storage, RBAC support)
- Payments: PayHere Gateway (local Sri Lankan payment support, easy MERN integration)
- SMS: Twilio SMS API (reliable delivery, appointment reminders, emergency notifications)
- Email: Nodemailer (lightweight integration, OTP verification, password recovery)
- Video: Jitsi Meet API (open-source, lightweight, minimal infrastructure overhead)
- Maps: Google Maps API (accurate clinic mapping, consent-based emergency location sharing)
- Hosting: Vercel/Netlify + Render (free-tier support, automatic SSL, seamless GitHub integration)

3.3. Detailed Design

3.3.1. Class Diagram

The class diagram defines the complete object model including all classes with full attribute and method specifications. Key entity classes include User (with subclasses Patient, Doctor, ClinicAdmin, PlatformAdmin), Appointment, Clinic, Specialization, AvailabilitySlot, Payment, QueueEntry, CommunityPost, CrisisEvent, and ConsultationSession. Service classes include AuthService, AppointmentService, QueueManager, PaymentService, NotificationService, TelemedicineService, EmergencyHandler, CommunityService, and SithBotService.

3.3.2. Sequence Diagrams

Twelve sequence diagrams were produced, one per use case, modelling interactions between actors, frontend components, backend controllers, services, and external systems. Key interactions include OTP verification (UC-01), PayHere webhook handling (UC-08), Jitsi video room lifecycle (UC-07), and emergency fallback (UC-10).



3.3.3. State Transition Diagrams

State diagrams for all twelve use cases model entity lifecycles. Notable state machines include Appointment (Pending Approval to Confirmed to In Progress to Completed/Cancelled), Queue Entry (Waiting to Called to Completed/No-show), and Payment (Pending to Processing to Paid/Failed to Refunded).

3.3.4. Entity-Relationship Diagram

The ER diagram models the MongoDB document-based schema with collections for Users, Clinics, Doctors, Appointments, Payments, QueueEntries, CommunityPosts, CrisisEvents, and ConsultationSessions. Relationships use reference fields and embedded documents with indexing for high-frequency query paths.

3.4. User Interface Design

The UI is organised into four dashboard interfaces. The Patient Dashboard features a calming design with teal accents. Key screens include landing page with SithBot access, specialist search with map integration, booking wizard, video consultation waiting room, live queue tracker, community forum, and emergency activation. Adult Mode provides simplified navigation with 18px minimum font. The Clinic Admin Dashboard provides appointment lists, availability calendars, queue control, payment tracking, and reports. The SithCare Admin Dashboard offers user management, clinic verification, crisis log review, and system analytics.

3.5. Security Design

3.5.1. Authentication and Authorization

Authentication uses JWT access tokens (15-minute expiry) and refresh tokens (7-day expiry). Passwords stored as bcrypt salted hashes. Optional MFA for Clinic and Platform Administrators. RBAC enforces six distinct roles with clearly defined permissions.

3.5.2. Data Protection

Data in transit uses HTTPS with TLS 1.3. Patient data at rest encrypted with AES-256. Payment card data never stored; PayHere handles PCI-DSS. Consent records stored with timestamp for audit.



3.5.3. OWASP Top 10 Considerations

- Broken Access Control: RBAC middleware, ownership checks, route protection, admin logging
- Cryptographic Failures: TLS 1.3, AES-256, bcrypt, no card data storage
- Injection: Backend validation, Mongoose schema validation, sanitised filters
- Insecure Design: Controller validation, service isolation, audit logging
- Security Misconfiguration: Environment variables, secure CORS, disabled debug output
- Vulnerable Components: Dependency review, package updates, security alerts
- Auth Failures: OTP verification, strong passwords, token expiry, MFA, account lockout
- Data Integrity: Pull requests, code review, CI checks, webhook verification
- Logging Failures: Auth events, role changes, payments, emergencies logged

3.6. Non-Functional Design Decisions

The following mapping shows how each NFR is addressed by design decisions:

1. **NFR-P-01 (Page load $\leq$ 3s):** React+Vite, Vercel/Netlify CDN, compressed images, Tailwind CSS
2. **NFR-P-02 (Booking $\leq$ 5s):** Dedicated Appointment Service, simple DB operations, async notifications
3. **NFR-P-03 (500 concurrent users):** Node.js non-blocking I/O, stateless REST APIs, horizontal scaling
4. **NFR-S-01 (TLS 1.3 in transit):** HTTPS for all frontend-backend and external API communication
5. **NFR-S-02 (AES-256 at rest):** MongoDB Atlas storage encryption for sensitive data
6. **NFR-U-01 (SUS score $\geq$ 70):** Simple role-based dashboards, clear navigation, guided workflows
7. **NFR-U-03 (Adult Mode 18px font):** Simplified navigation, larger text, reduced feature set
8. **NFR-R-01 (99.5% uptime):** Cloud hosting with uptime monitoring, critical error logging
9. **NFR-R-04 (Emergency if non-critical fails):** Static helpline configuration, independent emergency UI
10. **NFR-SC-01 (Horizontal scaling):** Stateless REST APIs, JWT session state



11. **NFR-SC-02 (200 clinics no migration):** MongoDB flexible schema, clinic-based tenant separation
12. **NFR-M-01 (Consistent coding standards):** GitHub, PRs, code reviews, linting, CONTRIBUTING.md
13. **NFR-M-02 (70%-unit test coverage):** Modular backend services for independent testing

3.7. Software Design Specifications Document

3.7.1. Introduction

3.7.1.1. Purpose

The purpose of this Software Design Specification (SDS) document is to describe the detailed system design and technical architecture of the SithCare Mental Health Care Platform. This document translates the functional and non-functional requirements specified in the Software Requirements Specification (SRS) into a structured technical design that guides the development, implementation, testing, and deployment phases of the project.

The SDS defines:

- The architectural design of the system
- System components and interactions
- Database and deployment structures
- User interface design considerations
- Security mechanisms
- Technology stack selections
- and infrastructure configurations.

This document serves as the primary technical reference for developers, testers, supervisors, and stakeholders throughout the system development lifecycle.

3.7.1.2. Scope

SithCare is a web-based mental health care platform designed to improve accessibility, privacy, and efficiency of mental health support services in Sri Lanka. The platform provides secure digital healthcare services for patients, mental health specialists, clinic administrators, volunteers, moderators, and system administrators.

The system includes functionalities such as:



- User authentication and role-based access control
- Appointment booking and queue management
- Telemedicine consultations
- Emergency mental health assistance
- Anonymous community support
- AI-assisted specialist recommendations
- Clinic administration and scheduling
- Payment processing

The SDS focuses on the internal technical design and implementation structure required to realize these functionalities using the MERN technology stack and a layered MVC-based architecture.

3.7.1.3. References

The following references were used during the preparation of this document:

- SithCare Software Requirements Specification (SRS) Document
- Project Guidelines and Assessment Criteria
- IEEE 1016 – Software Design Description Standard
- MongoDB Documentation
- React.js Documentation
- Node.js and Express.js Documentation
- Tailwind CSS Documentation
- JWT Authentication Standards
- PayHere API Documentation
- Jitsi Meet API Documentation

3.7.1.4. Document Overview

This Software Design Specification document is organized into several major sections describing the overall system architecture, design components, deployment strategies, and implementation details of the SithCare platform.

The document includes:

- System overview and relationship to the SRS
- Architectural and deployment design
- Technology stack justifications
- Component and package diagrams



- Database and class design
- User interface wireframes and prototypes
- Sequence diagrams
- Security and infrastructure considerations

The SDS acts as the technical blueprint for implementing the SithCare system according to the requirements identified during the analysis phase.

3.7.2. System Overview

SithCare is a centralized digital mental health care platform developed to connect patients, mental health specialists, clinics, volunteers, and administrators through a secure and scalable web-based system.

The platform addresses several challenges currently faced in mental healthcare services, including:

- Limited accessibility to mental health specialists,
- Long waiting times and inefficient appointment management,
- Lack of anonymous support systems,
- Emergency response delays,
- and social stigma associated with seeking mental health assistance.

The system enables users to securely access mental health services through multiple integrated modules such as appointment scheduling, telemedicine consultations, AI-assisted specialist recommendations, emergency support systems, anonymous community interactions, and peer support services.

3.7.2.1. Relationship with the SRS

This SDS document is directly derived from the Software Requirements Specification (SRS) document prepared during the requirements engineering phase.

The SRS defines:

- Functional requirements
- Non-functional requirements
- User roles



- Use cases
- System constraints
- and business objectives

The SDS transforms those requirements into:

- Detailed architectural structures
- System components
- Database schemas
- Interface designs
- Deployment configurations
- and implementation-level technical decisions

All design decisions described within this document are aligned with the requirements, constraints, and quality attributes defined in the SRS document.

3.7.2.2. High-Level System Architecture

SithCare follows a Layered MVC Architecture enhanced with modular service-oriented design principles.

The system consists of the following major architectural layers:

Layer	Responsibility
 Presentation Layer	User interfaces and frontend interactions
 Application/Controller Layer	API routing, validation, and RBAC
 Business Logic Layer	Core healthcare workflows and services
 Data Access Layer	Database communication and persistence

Figure 3.1: Major Architectural Layers

3.7.2.3. Technology Overview

SithCare is implemented using the MERN technology stack:

- MongoDB Atlas - Cloud database
- Express.js - Backend API framework



- React.js - Frontend SPA development
- Node.js - Backend runtime environment

Additional technologies include:

- Tailwind CSS
- JWT Authentication
- GitHub Actions
- PayHere API
- Twilio SMS API
- Jitsi Meet API
- and Google Maps API

3.7.2.4. Infrastructure Overview

The system uses cloud-based infrastructure for scalability and availability.

Infrastructure Component	Platform
Frontend Hosting	Vercel / Netlify
Backend Hosting	Render / Railway
Database Hosting	MongoDB Atlas
Version Control	GitHub
CI/CD Pipeline	GitHub Actions

Figure 3.2: Infrastructure Components

The deployment workflow includes automated testing, linting, pull request validation, and continuous deployment pipelines to improve maintainability and development efficiency.

3.7.3. Architectural Design

3.7.3.1. Introduction

SithCare is a web-based mental health care platform developed to improve accessibility, privacy, and efficiency of mental health services in Sri Lanka. The platform provides appointment booking, telemedicine consultations, emergency mental health assistance,



anonymous community consultation, AI-assisted specialist recommendation, volunteer listener support, and clinic administration services.

The system is designed as a scalable and modular healthcare platform capable of supporting multiple stakeholders, including patients, mental health specialists, clinic administrators, volunteers, moderators, and system administrators.

This document describes the architectural design, component structure, deployment strategy, scalability considerations, and system-level technical decisions adopted for the SithCare platform.

3.7.3.2. Architectural Pattern Selected

SithCare will employ a Client-Server Architecture utilizing a Layered (N-Tier) model on the back end, tightly integrated with the Model-View-Controller (MVC) design pattern. The application is built using the MERN stack (MongoDB, Express.js, React.js, Node.js).

The layers are defined as follows:

- **Presentation Layer (Client):** A React-based Single Page Application (SPA) handling UI, routing, and state management for Patients, Specialists, and Admins.
- **Application/Controller Layer (Server):** Express.js REST APIs that intercept HTTP requests, validate payloads, and manage Role-Based Access Control (RBAC) via JSON Web Tokens (JWT).
- **Business Logic Layer (Server):** Node.js services managing core functionalities such as appointment scheduling, queue management, AI triage (SithBot), and emergency routing.
- **Data Access Layer (Database):** Mongoose Object Data Modeling (ODM) wrapping a NoSQL MongoDB database.

The architecture is designed to:

- Separate system responsibilities clearly,
- Simplify maintenance and debugging,
- Improve scalability,
- Support secure healthcare workflows,
- Enable future migration toward microservice-based deployment.



- Instead of implementing a fully distributed microservices architecture during the academic project phase, the platform uses a modular service-oriented structure where each major functionality is isolated into independent service modules.

This approach provides:

- Enterprise-level scalability planning,
- Easier future service decomposition,
- Better maintainability,
- Reduced development complexity.

3.7.3.3. Justification Referencing Quality Attributes

The chosen architecture directly supports the Non-Functional Requirements (NFRs) defined in the SRS:

3.7.3.3.1. Scalability (NFR-SC-01 to 03, NFR-P-03)

Node.js utilizes a non-blocking, event-driven I/O model which makes it exceptionally well-suited to handle the target metric of 500 concurrent users and 5x traffic spikes (NFR-SC-03). The stateless REST API design allows for horizontal scaling. MongoDB's flexible schema easily accommodates the schema-per-tenant design required for managing multiple clinic branches (NFR-SC-02).

3.7.3.3.2. Maintainability (NFR-M-01 to NFR-M-05)

The MVC and Layered approach enforces a strict separation of concerns. Frontend UI modifications will not impact backend business logic. This structure aligns perfectly with the development team's expertise and allows for parallel development. Using a single language (JavaScript) across the entire stack reduces context switching and improves code maintainability.

3.7.3.3.3. Security & Privacy (NFR-S-01 to NFR-S-08)

A dedicated Controller layer acts as a gatekeeper, verifying JWTs and enforcing RBAC (Patient, Doctor, Admin) before any business logic is executed. MongoDB allows for field-level encryption, aiding in PDPA compliance (NFR-S-02), and the isolated backend layer prevents direct database exposure to the internet.

3.7.3.3.4. Reliability & Performance (NFR-R-01, NFR-P-01)

React's virtual DOM ensures fast UI rendering, meeting the sub-3-second page load time requirement (NFR-P-01). The separation of external services (PayHere, Jitsi) from the core



backend logic ensures that failures in third-party integrations do not crash the primary application, satisfying the 99.5% uptime goal.

3.7.3.3.5. Usability & User Experience (NFR-U-01 to NFR-U-05)

The React-based frontend combined with Tailwind CSS provides clean, responsive, and intuitive user experience. Simplified navigation flows, queue tracking, emergency shortcuts, and low-complexity interfaces reduce cognitive load for mental health patients and improve overall usability.

3.7.3.4. High-Level System Components

The SithCare platform consists of the following major components:

1. Patient Module

Handles:

- Registration and authentication
- Therapist searching
- Appointment booking
- Queue tracking
- Emergency support
- Community interaction

2. Mental Health Specialist Module

Handles:

- Appointment management
- Schedule management
- Telemedicine consultations
- Patient communication

3. Clinic Administration Module

Handles:

- Clinic profile management
- Doctor allocation
- Queue management



- Appointment monitoring

4. SithCare Admin Module

Handles:

- Tracking of all the 3 portals
- Actions related to users, clinics, doctors
- Risk controlling and maintaining the progress of the application

5. Community Support Module

Handles:

- Anonymous consultation
- Recovery stories
- Peer support
- Volunteer listener sessions

6. AI Assistance Module (SithBot)

Handles:

- Symptom-based triage
- Specialist recommendations
- Navigation assistance

3.7.3.5. Component / Package Diagram Specifications

Below is the descriptive specification for the Component Diagram. (Please generate the visual diagram in Draw.io or UML tool of choice using these exact components):

- Frontend Components (React): AuthModule, Patient-Dashboard, Doctor-Portal, Clinic-AdminPanel, SithCare-AdminPanel, SithBotWidget, VideoConsultationRoom.
- Backend API Packages (Node/Express): AuthController, AppointmentService, QueueManager, NotificationService, PaymentGatewayAdapter (PayHere), EmergencyHandler.
- External Interfaces: PayHere Checkout API, Jitsi Meet SDK, Twilio SMS API, Nodemailer Email Service, Google Maps API

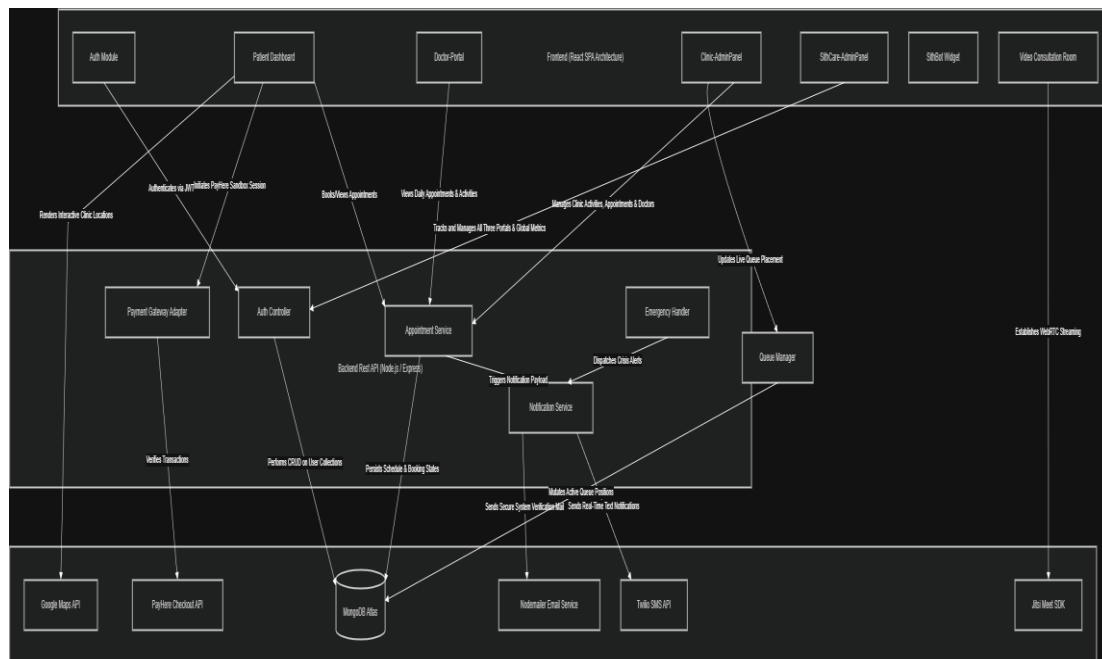


Figure 3.3: Component Diagram

Mermaid.js Diagram source code

flowchart TB

```
subgraph Frontend ["Frontend (React SPA Architecture)"]
```

AM[Auth Module]

PD[Patient Dashboard]

DP[Doctor-Portal]**CAP[Clinic-AdminPanel]**

SAP[SithCare-AdminPanel]

SB[SithBot Widget]

VC[Video Consultation Room]

end

```
subgraph Backend ["Backend Rest API (Node.js / Express)"]
```

AC[Auth Controller]

AS[Appointment Service]

QM[Queue Manager]



NS[Notification Service]

PGA[Payment Gateway Adapter]

EH[Emergency Handler]

end

subgraph External ["External Interfaces"]

PH[PayHere Checkout API]

JM[Jitsi Meet SDK]

TW[Twilio SMS API]

NM[Nodemailer Email Service]

GM[Google Maps API]

DB[(MongoDB Atlas)]

end

%% Frontend Interactions to Backend Services

AM -->|Authenticates via JWT| AC

PD -->|Books/Views Appointments| AS

PD -->|Initiates PayHere Sandbox Session| PGA

DP -->|Views Daily Appointments & Activities| AS

CAP -->|Manages Clinic Activities, Appointments & Doctors| AS

CAP -->|Updates Live Queue Placement| QM

SAP -->|Tracks and Manages All Three Portals & Global Metrics| AC

%% Direct Front-End Script Integrations

VC -->|Establishes WebRTC Streaming| JM

PD -->|Renders Interactive Clinic Locations| GM

%% Backend Service to External Systems



PGA -->|Verifies Transactions| PH

EH -->|Dispatches Crisis Alerts| NS

AS -->|Triggers Notification Payload| NS

NS -->|Sends Real-Time Text Notifications| TW

NS -->|Sends Secure System Verification Mail| NM

%% Backend to NoSQL Persistence Layer

AC -->|Performs CRUD on User Collections| DB

AS -->|Persists Schedule & Booking States| DB

QM -->|Mutates Active Queue Positions| DB

3.7.3.6. Deployment Diagram Specifications

The deployment topology utilizes free-tier cloud services optimized for MERN stack applications, adhering to the project's zero-cost infrastructure constraint.

- Client Devices: Web Browsers (Chrome, Firefox, Safari) accessing the SPA via HTTPS (TLS 1.3).
- Frontend Hosting (e.g., Vercel / Netlify): Hosts static React assets and provides a global CDN for fast initial load times.
- Backend Web Server (e.g., Render / Railway): Node.js instance running the Express API, exposed via a reverse proxy handling SSL termination.
- Database Cluster (MongoDB Atlas Free Tier): Cloud-hosted MongoDB cluster handling all data persistence.

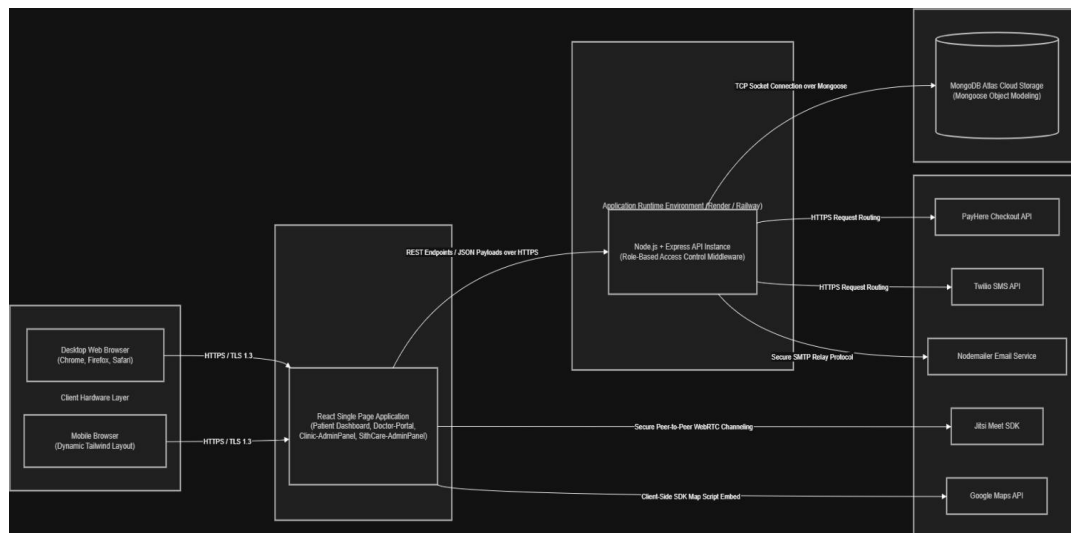


Figure 3.4: Deployment Diagram

Mermaid.js Diagram source code

flowchart LR

```
subgraph Clients ["Client Hardware Layer"]
```

```
    Browser["Desktop Web Browser\n(Chrome, Firefox, Safari)"]
```

```
    MobileBrowser["Mobile Browser\n(Dynamic Tailwind Layout)"]
```

```
end
```

```
subgraph CDN ["Static Distribution Layer (Vercel / Netlify)"]
```

```
    SPA["React Single Page Application\n(Patient Dashboard, Doctor-Portal,\nClinic-AdminPanel, SithCare-AdminPanel)"]
```

```
end
```

```
subgraph AppServer ["Application Runtime Environment (Render / Railway)"]
```

```
    NodeAPI["Node.js + Express API Instance\n(Role-Based Access Control\nMiddleware)"]
```

```
end
```



subgraph DBServer ["Database Cluster Layer"]

MongoCluster[("MongoDB Atlas Cloud Storage\n(Mongoose Object Modeling)"]

End

subgraph External ["Third-Party External Interfaces"]

PH["PayHere Checkout API"]

JM["Jitsi Meet SDK"]

TW["Twilio SMS API"]

NM["Nodemailer Email Service"]

GM["Google Maps API"]

end

%% Execution Network Vectors

Browser -->|HTTPS / TLS 1.3| SPA

MobileBrowser -->|HTTPS / TLS 1.3| SPA

SPA -->|REST Endpoints / JSON Payloads over HTTPS| NodeAPI

SPA -->|Secure Peer-to-Peer WebRTC Channeling| JM

SPA -->|Client-Side SDK Map Script Embed| GM

NodeAPI -->|TCP Socket Connection over Mongoose| MongoCluster

NodeAPI -->|HTTPS Request Routing| PH

NodeAPI -->|HTTPS Request Routing| TW

NodeAPI -->|Secure SMTP Relay Protocol|



3.7.3.7. Core Technology Stack: MERN

The platform will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js). This full-stack JavaScript environment allows the team to maintain high development velocity using a unified language ecosystem, fulfilling the 8-week MVP design and development timeline.

3.7.3.7.1. Frontend Technologies

1. React + Vite

- Purpose: Frontend web application development.
- Justification:
 - Component-based architecture improves maintainability.
 - Fast rendering and reusable UI components.
 - Efficient state management.
 - Suitable for scalable single-page applications.
 - Large community and ecosystem support.

2. Tailwind CSS

- Purpose: Responsive UI styling.
- Justification:
 - Rapid UI development.
 - Utility-first responsive design.
 - Consistent design system.
 - Simplifies glass-morphism UI implementation.
 - Lightweight production builds.

3.7.3.7.2. Backend Technologies

1. Node.js + Express.js

- Purpose: REST API backend development.
- Justification:
 - Lightweight and fast.
 - Event-driven asynchronous architecture.
 - Suitable for real-time features.



- Large middleware ecosystem.
- Easy integration with frontend applications.

3.7.3.7.3. Database Technology

1. MongoDB Atlas

- Purpose: Cloud-hosted NoSQL database system.

- Justification:

- Flexible document-based schema
- Suitable for dynamic healthcare records
- High scalability
- Cloud-native deployment support
- Faster iterative development

2. Mongoose ODM

- Purpose: Object Data Modeling (ODM) library for MongoDB.

- Justification:

- Simplifies MongoDB database operations
- Provides schema validation
- Improves query handling
- Supports middleware and model abstraction
- Improves maintainability of backend services

3.7.3.7.4. Authentication and Security Technologies

1. JWT (JSON Web Tokens)

- Purpose: Session authentication and authorization.

- Justification:

- Stateless authentication
- Secure API authorization
- Scalable session handling
- Suitable for role-based access control



2. bcrypt

- Purpose: Password hashing and encryption.

- Justification:

- Secure password storage
- Industry-standard password protection
- Reduces credential exposure risks

3. TLS 1.3

- Purpose: Secure client-server communication.

- Justification:

- Encrypts data transmission
- Prevents interception attacks
- Secures REST API communication

3.7.3.7.5. Third-Party Service Selection

1. PayHere Gateway

- Purpose: Online payment processing.

- Justification:

- Local Sri Lankan payment support
- Easy integration for MERN applications
- Supports multiple payment methods

2. Twilio SMS API

- Purpose: SMS notifications and alerts.

- Justification:

- Reliable SMS delivery
- Appointment reminders
- Emergency notifications



3. Nodemailer

- Purpose:

- Justification:

- Lightweight email integration
- Appointment confirmations
- OTP verification emails
- Password recovery support

4. Jitsi Meet API

- Purpose: Telemedicine video consultation support.

- Justification:

- Open-source video conferencing
- Lightweight integration
- Minimal infrastructure overhead
- Suitable for MVP telemedicine implementation

5. Google Maps API

- Purpose: Location and emergency mapping services.

- Justification

- Accurate clinic location mapping
- Consent-based emergency location sharing
- Easy frontend integration

3.7.3.7.6. Infrastructure and CI/CD

1. Frontend Hosting

- Selected Platforms:

- Vercel
- Netlify



- Justification:

- Robust free-tier support
- Automatic SSL provisioning using TLS 1.3
- Seamless GitHub repository integration
- Continuous deployment pipelines
- Fast global CDN delivery
- Simplified React SPA deployment

2. Backend Hosting

- Selected Platforms:

- Render
- Railway

- Justification:

- Free-tier Node.js hosting support
- Simplified cloud deployment
- Easy GitHub integration
- Automatic deployment workflows
- Environment variable management
- Scalable backend hosting capabilities

3. Version Control

- Selected Tools:

- GitHub
- GitHub Actions

- Purpose:

- Source code management
- Branch protection
- Pull request workflows
- Sprint issue tracking
- Team collaboration



- Justification:

- Code consistency
- Deployment reliability
- Team collaboration
- Faster issue detection
- Maintainability

3.7.3.8. Architectural Style Justification

SithCare adopts a Layered MVC Architecture with modular service-oriented design principles.

The architecture consists of:

1. Presentation Layer
2. Application / Controller Layer
3. Business Logic Layer
4. Data Access Layer

This architectural style was selected due to the following reasons:

Architectural Benefit	Justification
Maintainability	Independent layers simplify debugging and updates
Scalability	Modular services support future system expansion
Security	Centralized authentication and validation
Flexibility	External APIs can be independently managed
Testability	Layers can be independently tested
Extensibility	Easier future migration toward microservices

Figure 3.5: Reasons for the Architecture Style

3.7.3.9. Conclusion

The selected MERN technology stack and architectural decisions provide a scalable, secure, maintainable, and practical foundation for the SithCare platform. Architecture balances academic feasibility with enterprise-level software engineering principles while supporting future system evolution and healthcare scalability.



3.7.4. Detailed Design

3.7.4.1. Class Diagram

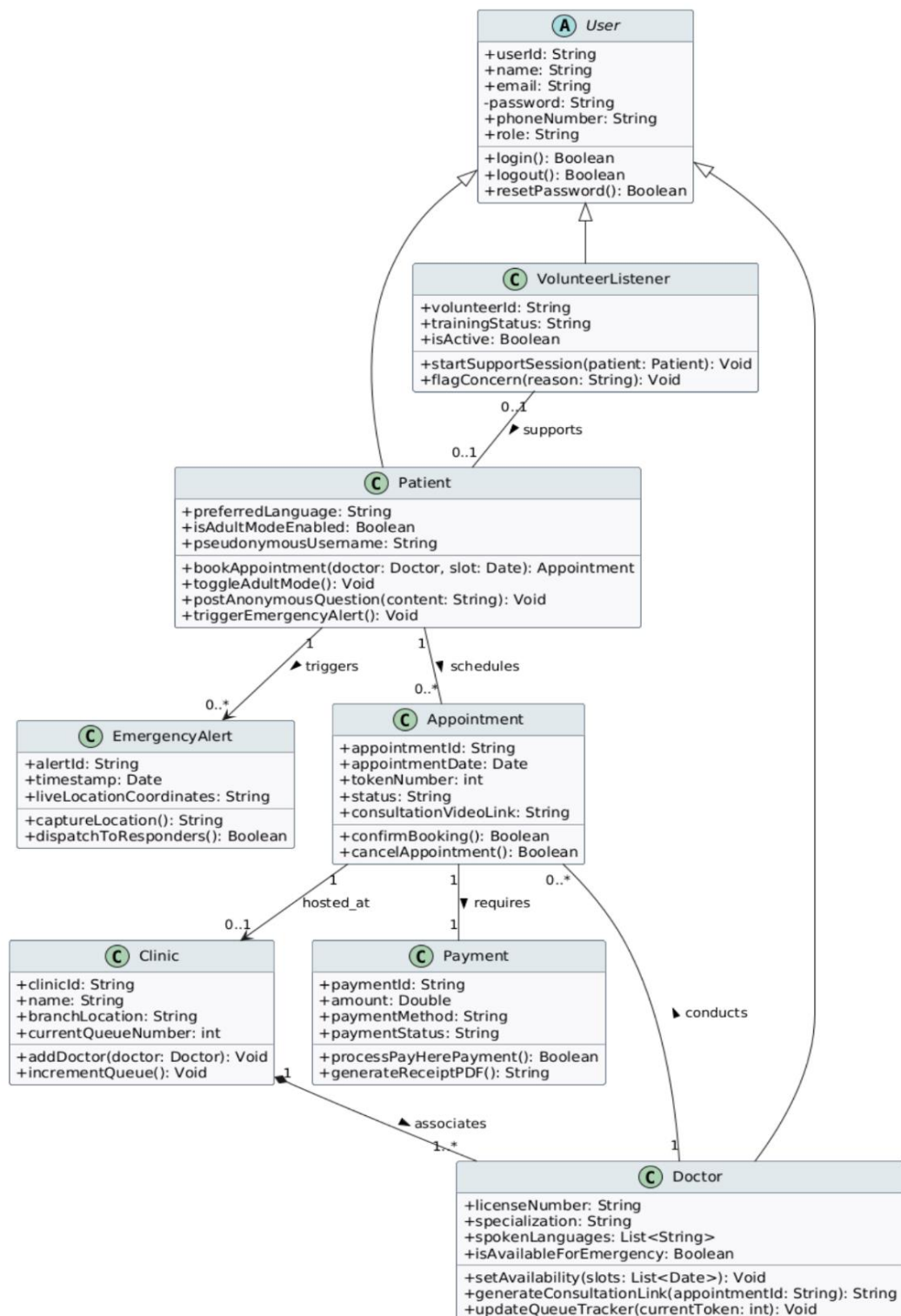


Figure 3.6: Class Diagram



3.7.4.2. High-Level System Sequence Descriptions

3.7.4.2.1. Sequence Diagram Descriptions

SSD ID	Name	Sequence Summary
SSD-01	Register and Login (UC-01)	Patient submits credentials. System sends OTP. Patient verifies OTP. System creates an account and routes the patient to role-based dashboard.
SSD-02	Manage User Profile and Consent (UC-02)	Patient updates profile details and language preferences. System saves changes to the DB. Patient toggles privacy consent. System confirms and stores the updated consent record.
SSD-03	Search Mental Health Specialist (UC-03)	Patient applies search filters (category, language, location). System queries specialist DB. System retrieves clinic locations via Maps API. System returns filtered specialist list to patient.
SSD-04	Book Appointment (UC-04)	Patient selects available slot. System verifies slot availability in DB. System initiates PayHere payment. PayHere confirms transaction via webhook. System sends booking confirmation via SMS/email.
SSD-05	Manage Doctor Availability (UC-05)	Doctor or clinic admin submits availability schedule. System validates and stores slots in DB. System confirms updated schedule to doctor/admin.
SSD-06	Approve / Reschedule Appointment (UC-06)	Clinic admin reviews pending appointment requests. System fetches appointments from DB. Admin approves or reschedules the appointment. System updates status in DB. System notifies patient via SMS/email.
SSD-07	Join Telemedicine Consultation (UC-07)	Doctor generates secure video rooms via Video SDK. Patient joins consultation through room link. Video SDK connects both parties. System provides one-tap phone fallback if video connection fails.
SSD-08	Process Payment and Generate Receipt (UC-08)	Patients select payment methods. System redirects to PayHere gateway. PayHere returns transaction status via webhook. System updates payment record in DB. System generates and delivers PDF receipt to patient.
SSD-09	Track Live Patient Queue (UC-09)	System displays current token numbers to the patient. Clinic admin advances the queue. System updates queue position in DB. System sends proximity notification to notify the next patient.
SSD-10	Request Emergency Support (UC-10)	Patients activate emergency buttons. System displays disclaimer and crisis options. Patient confirms and consents to location sharing. System initiates emergency calls to the doctor. System displays helpline numbers as fallback.
SSD-11	Use Anonymous Community Consultation (UC-11)	Patient creates anonymous posts using pseudonymous identity. System mask's real identity in DB. Community members respond to the post. System flags harmful content and routes to community moderator for review.



SSD-12	Complete AI Triage Questionnaire - SithBot (UC-12)	Patient describes feelings to SithBot. SithBot asks follow-up questions. Recommendation engine processes responses. SithBot returns specialist category recommendation with mandatory non-diagnosis disclaimer.
--------	--	---

Table 3.1: Sequence Diagram Description

3.7.4.2.2. Sequence Diagrams

The Following sequence diagrams have been drawn for the major use cases of the application

1. UC-01 Name Register and Login

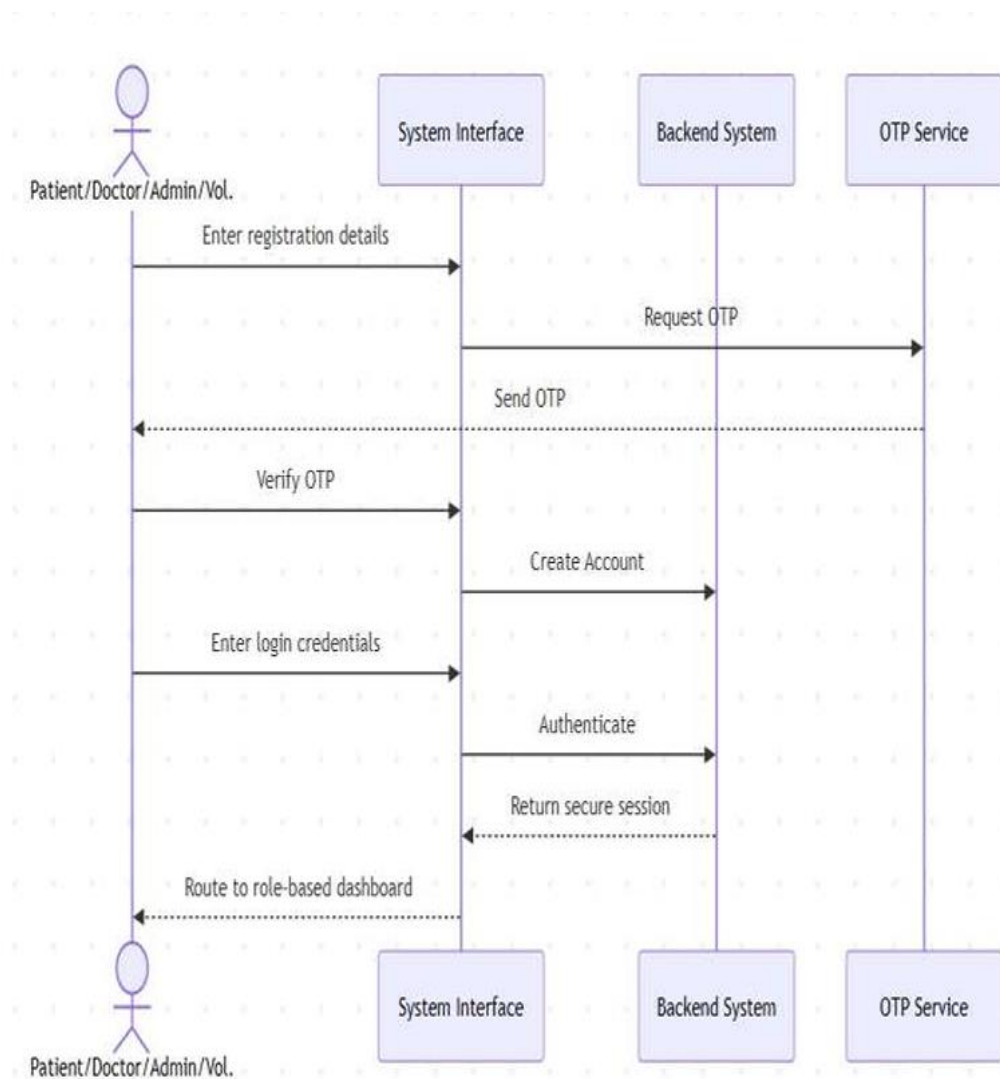


Figure 3.7: Sequence Diagram - Name Register & Login

2. UC-02 Manage User Profile and Consent

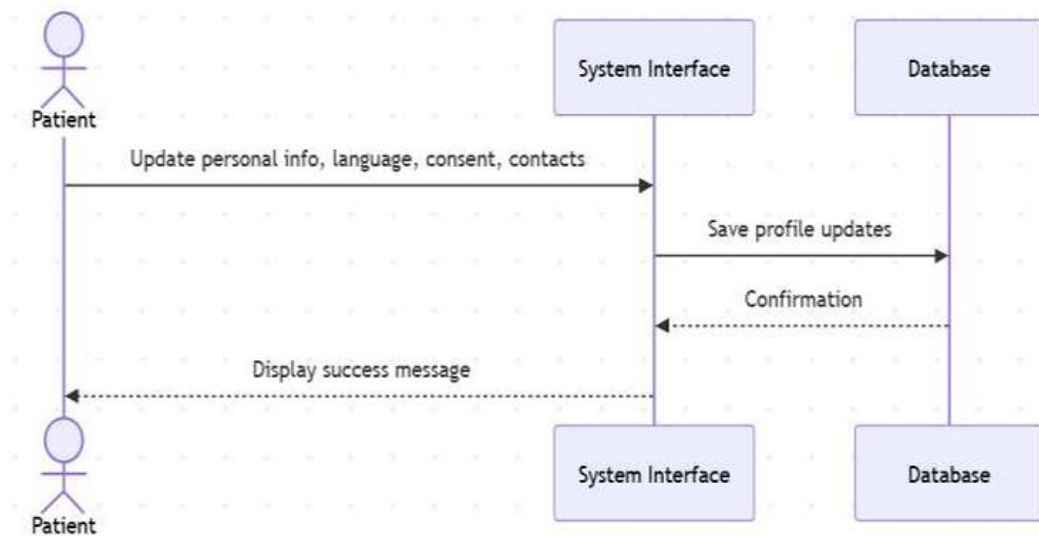


Figure 3.8: Sequence Diagram - Manage User Profile & Consent

3. UC-03 Search Mental Health Specialist

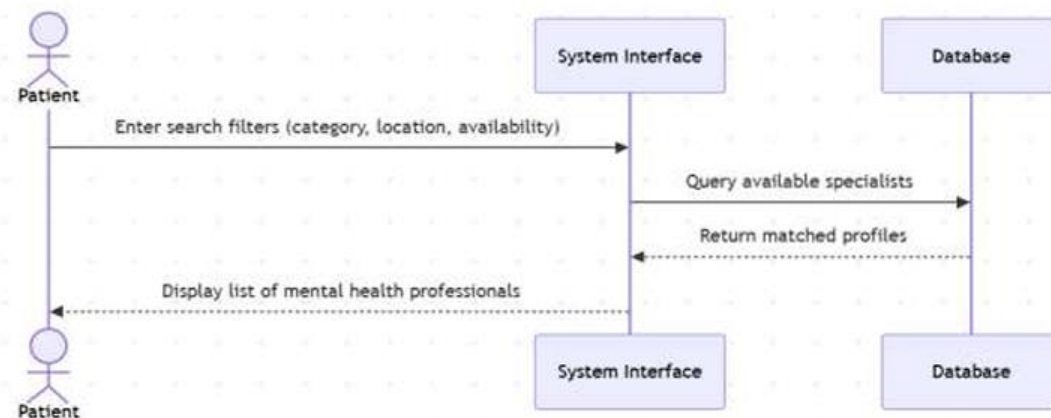


Figure 3.9: Sequence Diagram - Search Mental Health Specialist

4. UC-04 Book Appointment

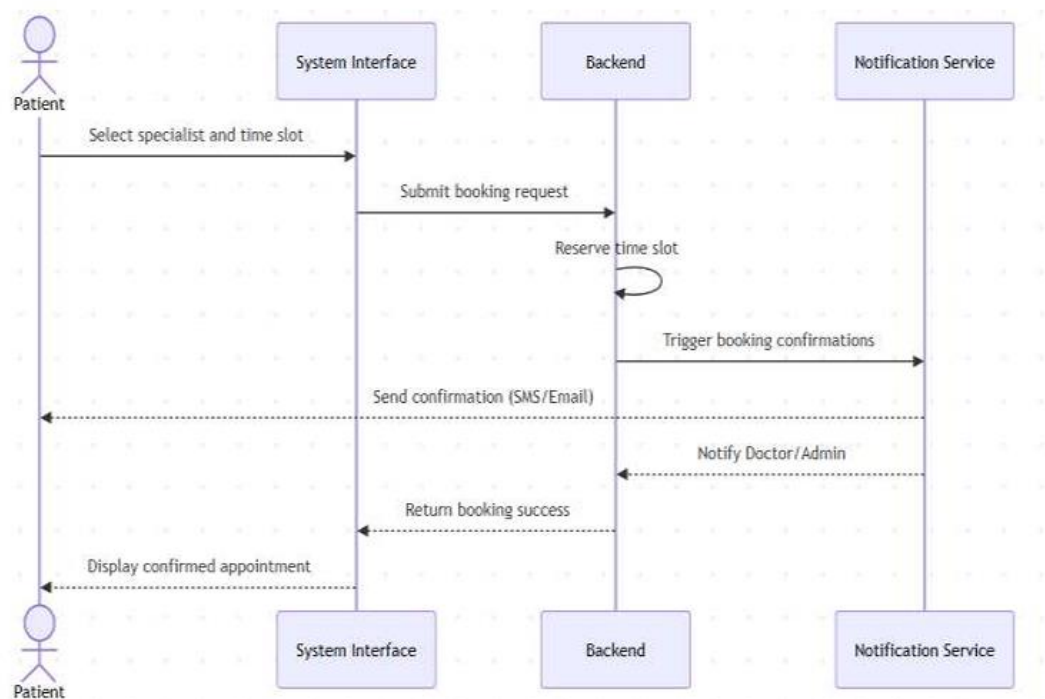


Figure 3.10: Sequence Diagram - Book Appointment

5. UC-05 Manage Doctor Availability

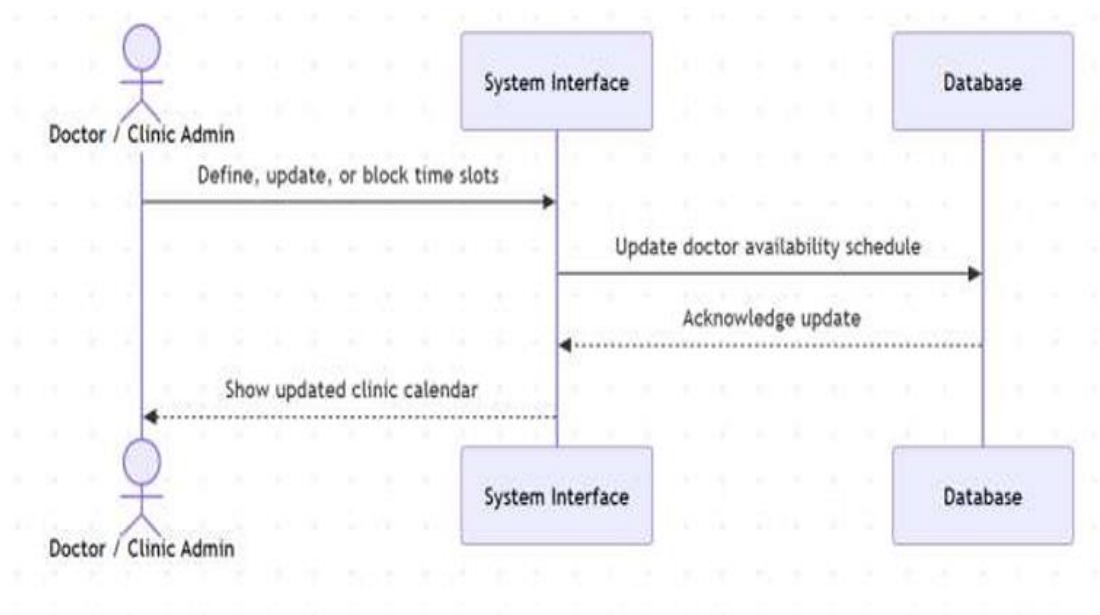


Figure 3.11: Sequence Diagram - Manage Doctor Availability

6. UC-06 Approve / Reschedule Appointment

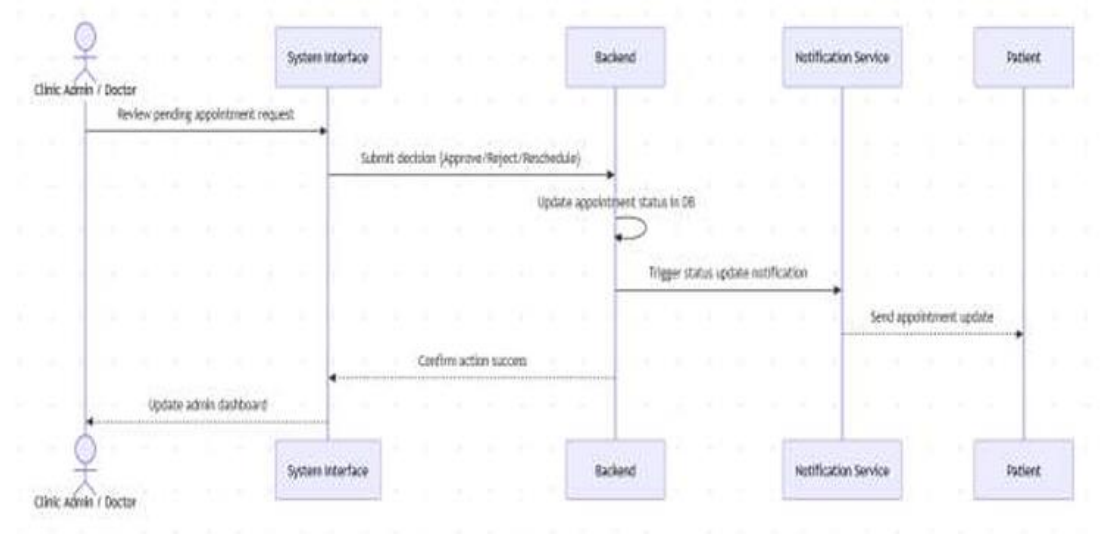


Figure 3.12: Sequence Diagram - Approve/Reschedule Appointment

7. UC-07 Join Telemedicine Consultation

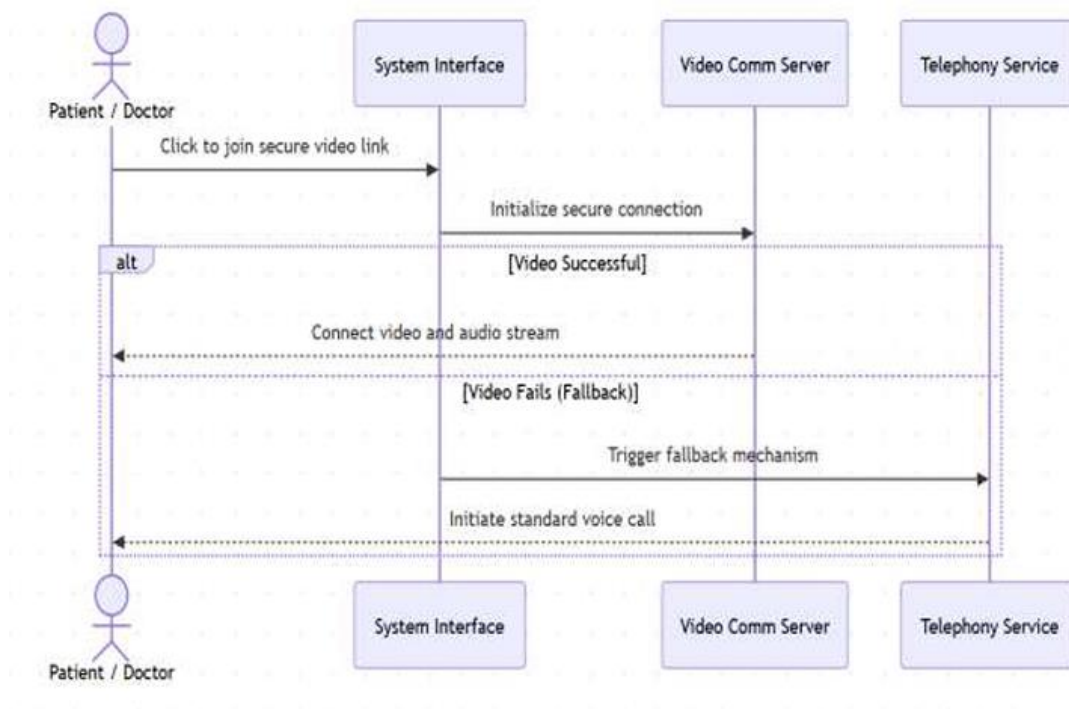


Figure 3.13: Sequence Diagram - Join Telemedicine Consultation

8. UC-08 Process Payment and Generate Receipt

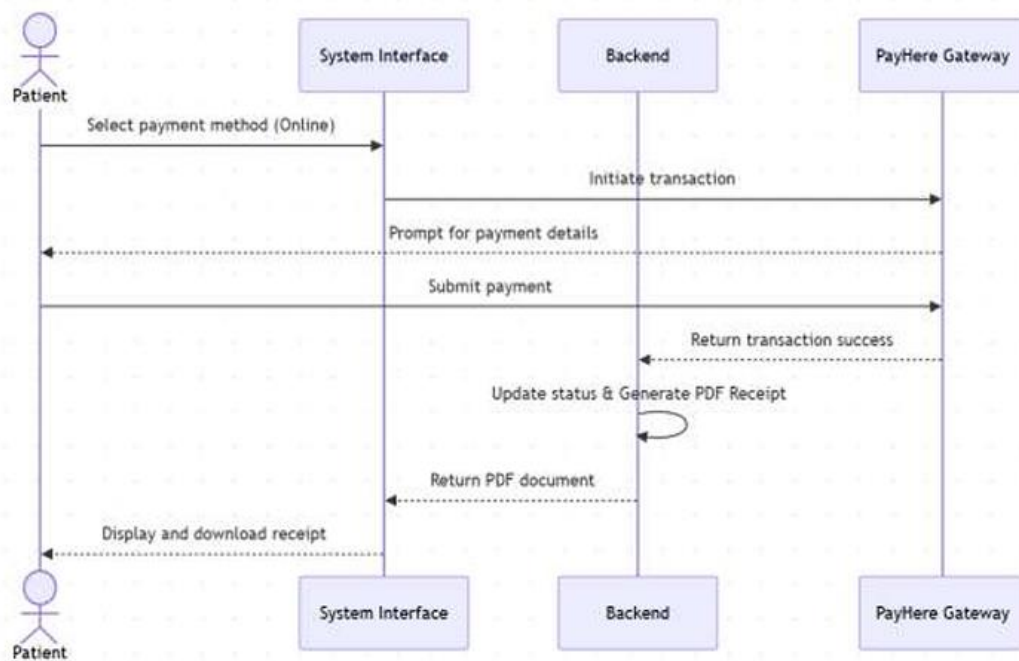


Figure 3.14: Sequence Diagram - Process Payment & Generate Receipt

9. UC-09 Track Live Patient Queue

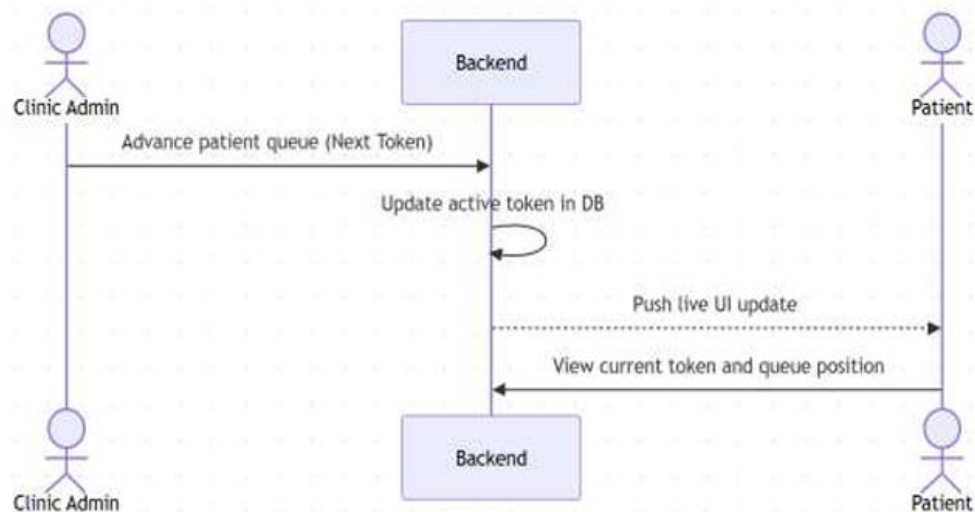


Figure 3.15: Sequence Diagram - Track Live Patient Queue

10. UC-10 Request Emergency Support

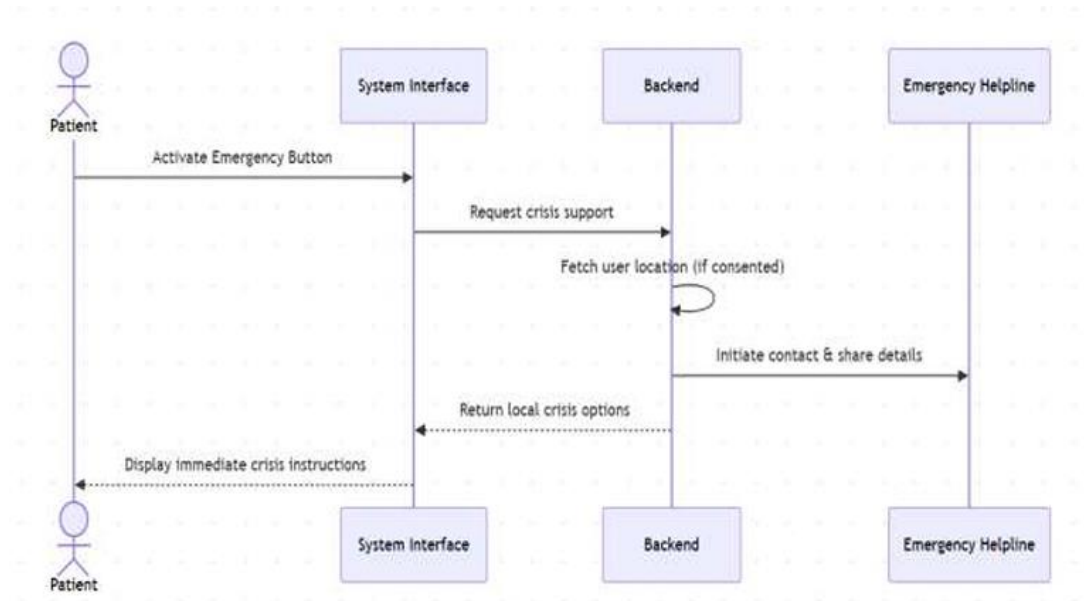


Figure 3.16: Sequence Diagram - Request Emergency Support

11. UC-11 Use Anonymous Community Consultation

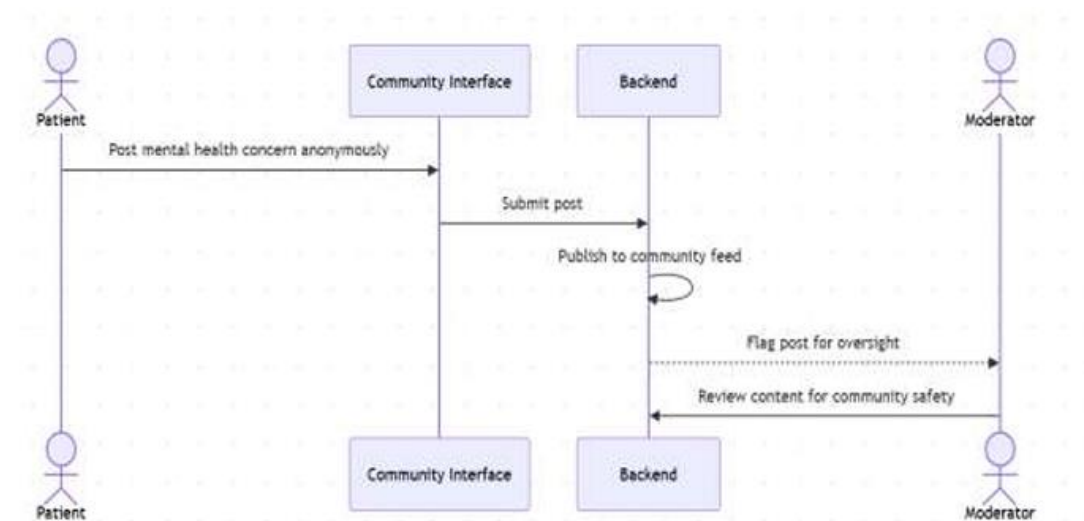


Figure 3.17: Sequence Diagram - Use Anonymous Community Consultation

12. UC-12 Complete AI Triage Questionnaire (SithBot)

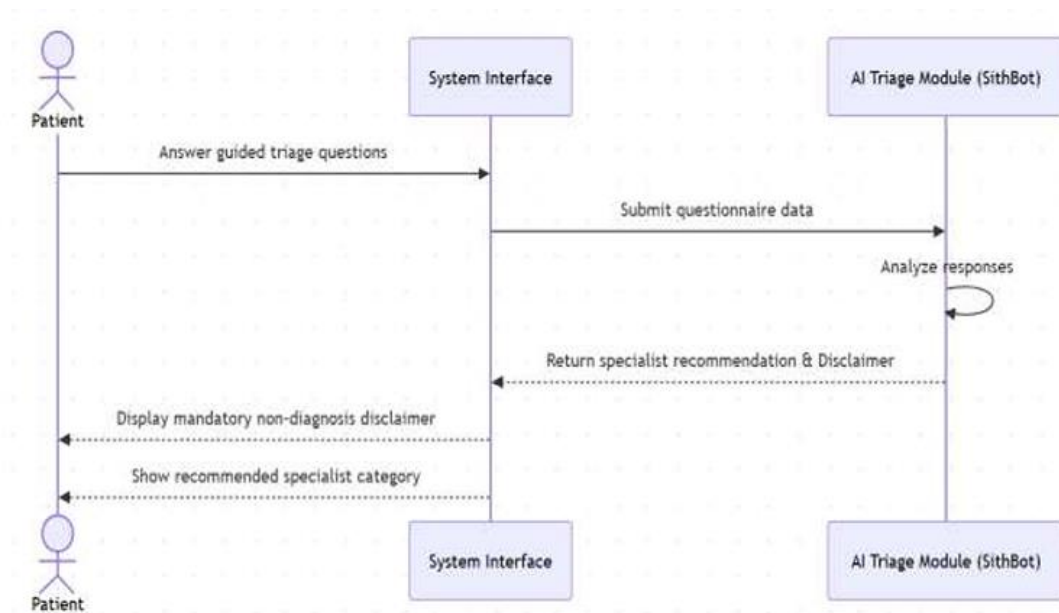


Figure 3.18: Sequence Diagram - Complete AI Triage Questionnaire (SithBot)

3.4.4.3. State Transition Diagrams

1. UC-01: Register and Login

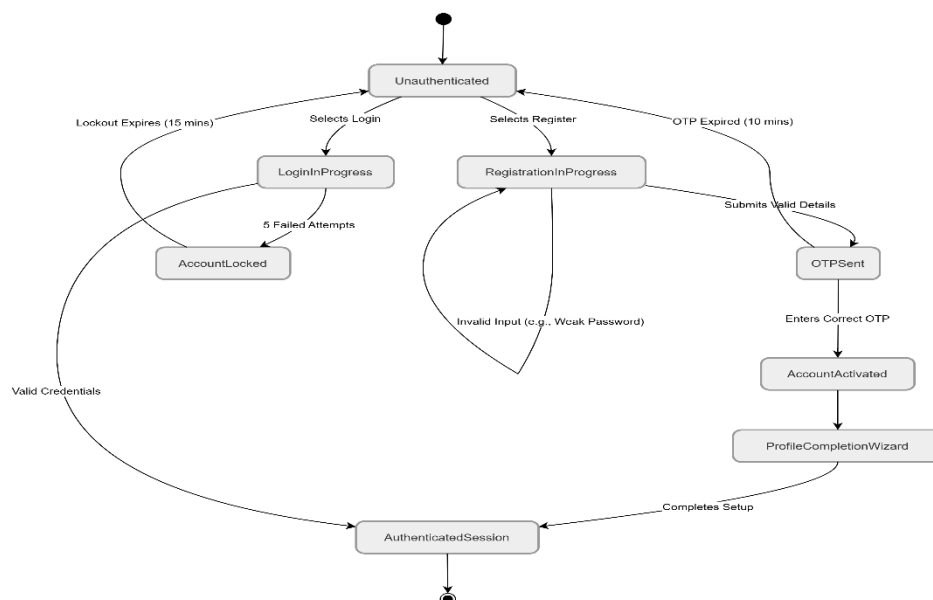


Figure 3.19: State Transition Diagram - Register & Login

2. UC-02: Manage User Profile and Consent

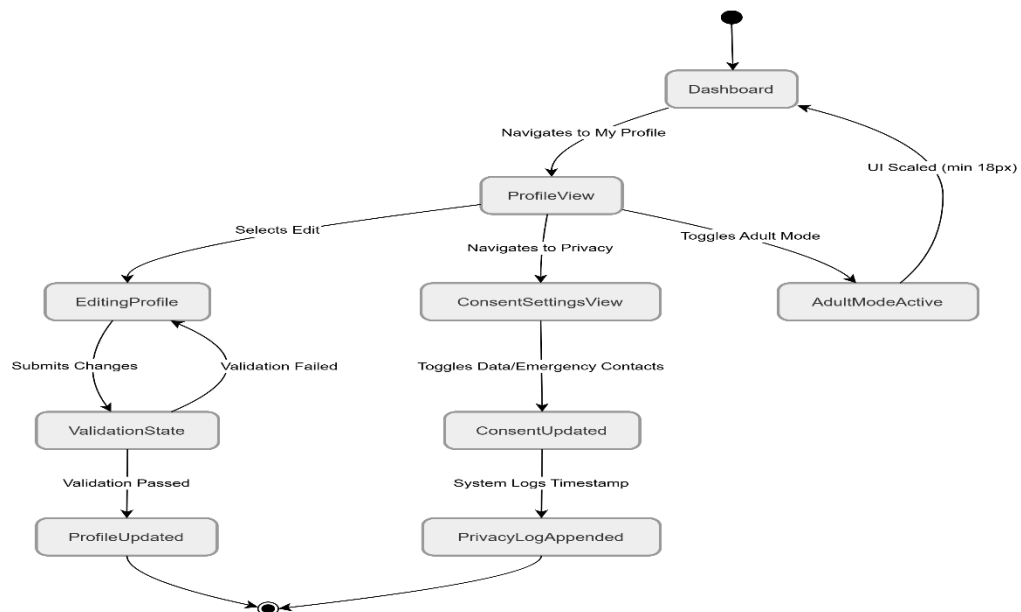


Figure 3.20: State Transition Diagram - Manage User Profile & Consent

3. UC-03: Search Mental Health Specialist

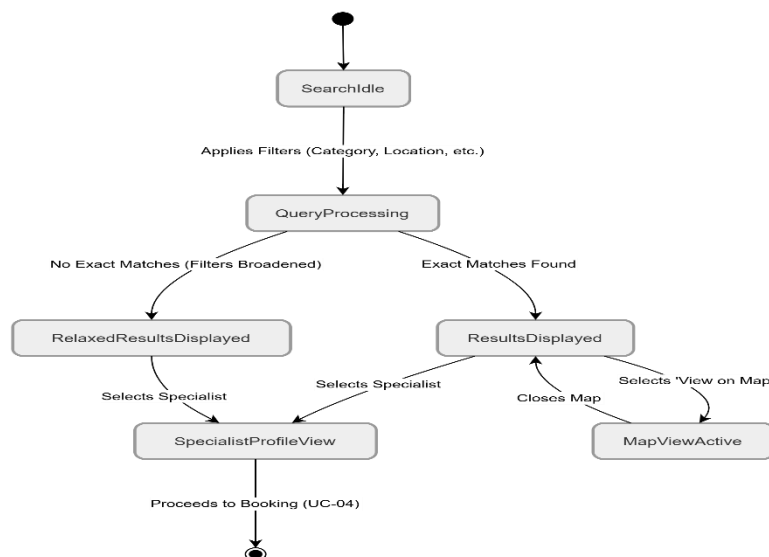


Figure 3.21: State Transition Diagram - Search Mental Health Specialist



4. UC-04: Book Appointment

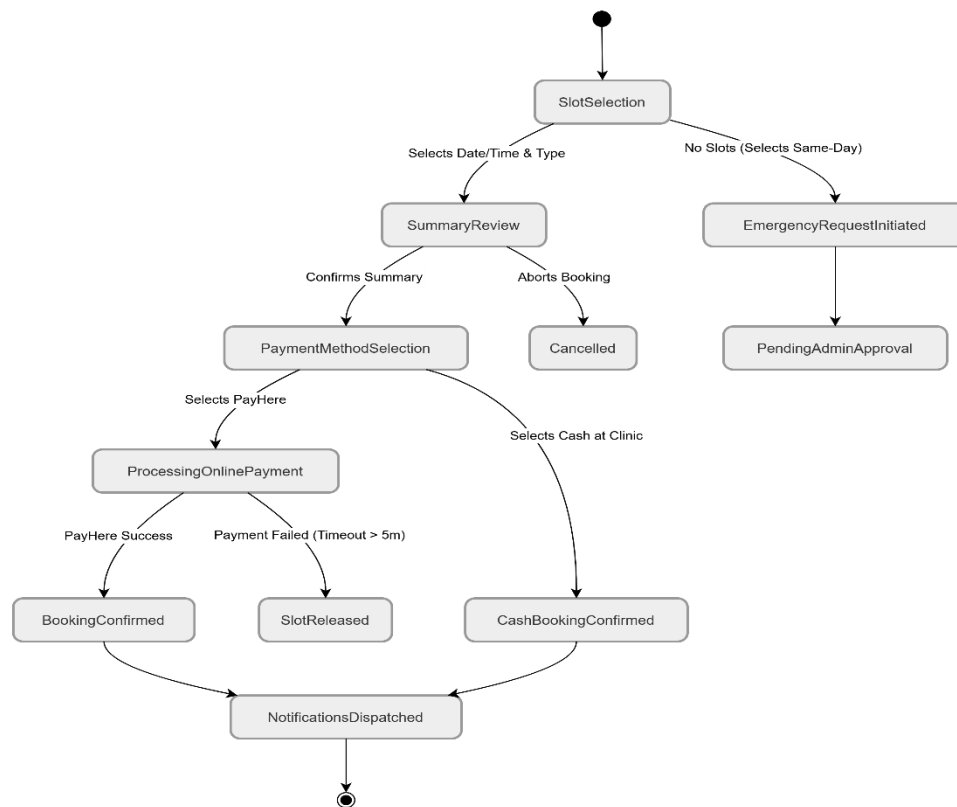


Figure 3.22: State Transition Diagram - Book Appointment

5. UC-05: Manage Doctor Availability

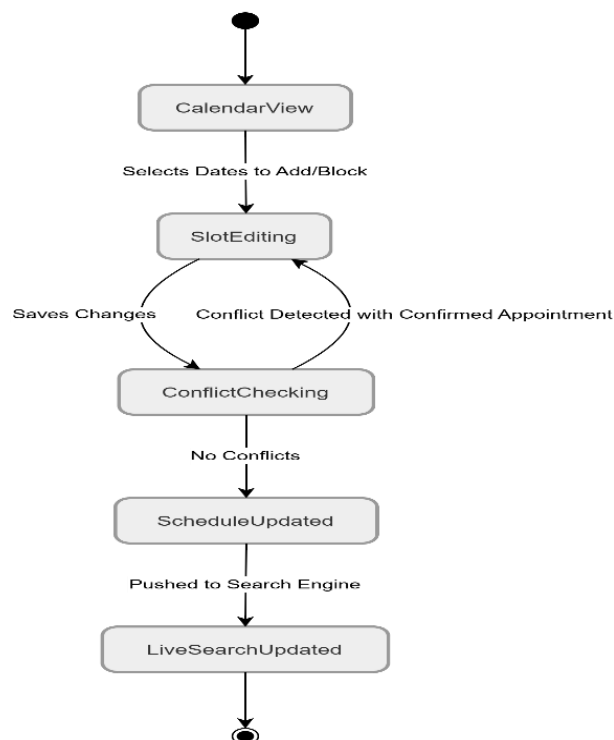


Figure 3.23: State Transition Diagram - Manage Doctor Availability

6. UC-06: Approve / Reschedule Appointment

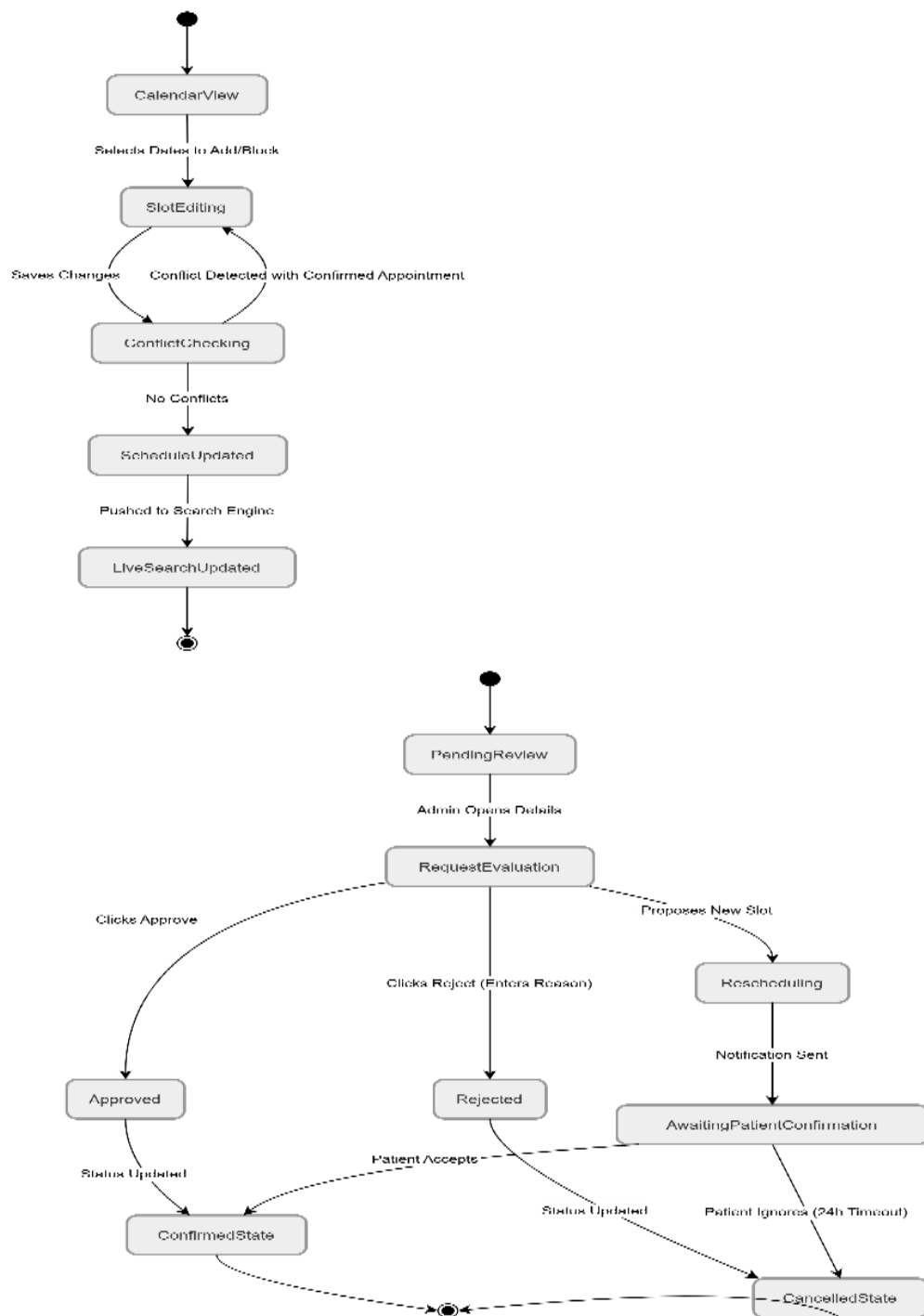


Figure 3.24: State Transition Diagram - Approve & Reschedule Appointment



7. UC-07: Join Telemedicine Consultation

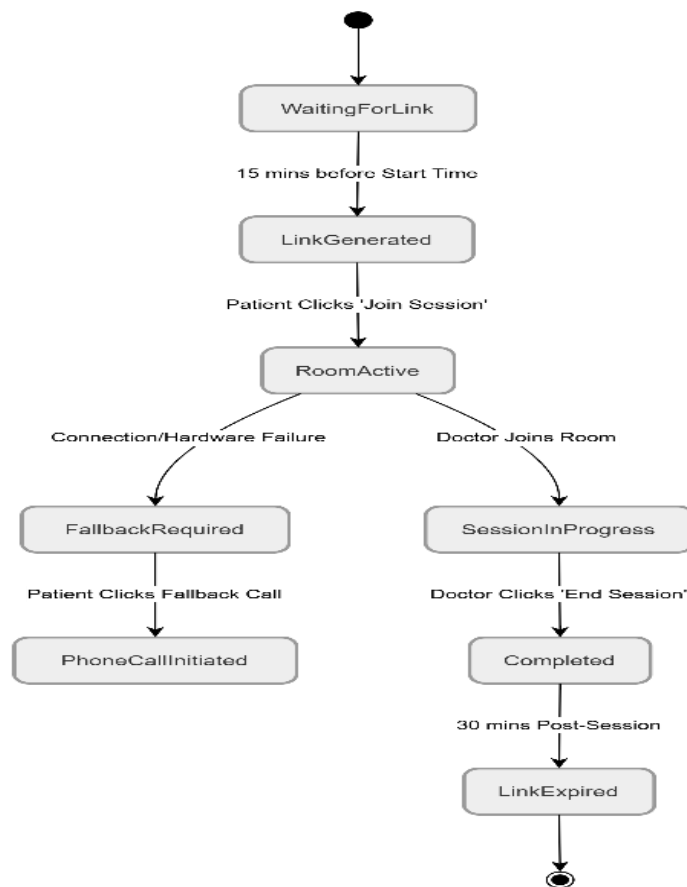


Figure 3.25: State Transition Diagram - Join Telemedicine Consultation

8. UC-08: Process Payment and Generate Receipt

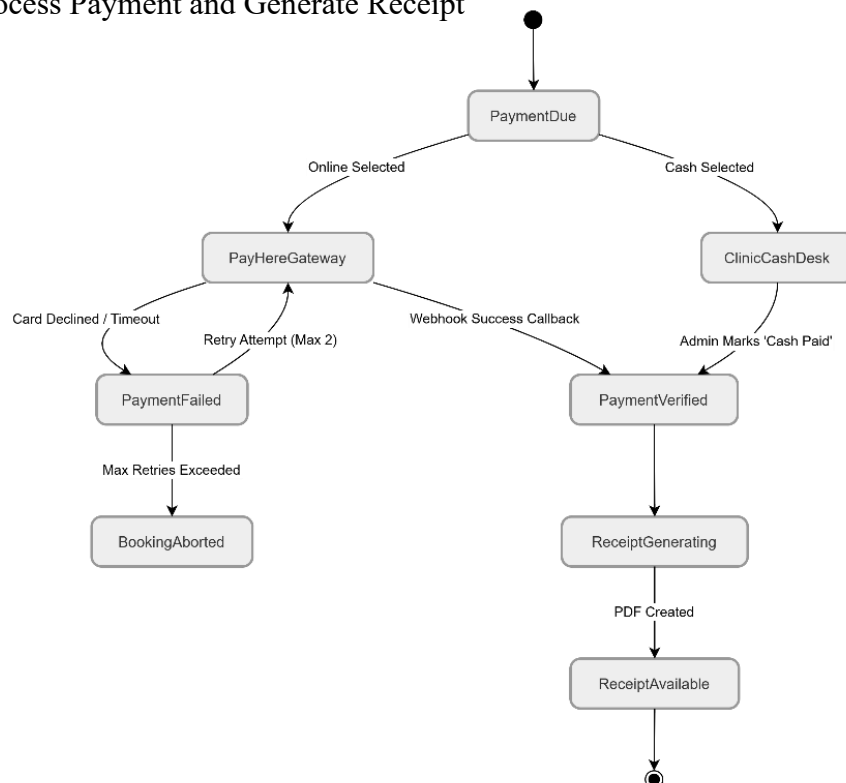


Figure 3.26: State Transition Diagram - Process Payment & Generate Receipt



9. UC-09: Track Live Patient Queue

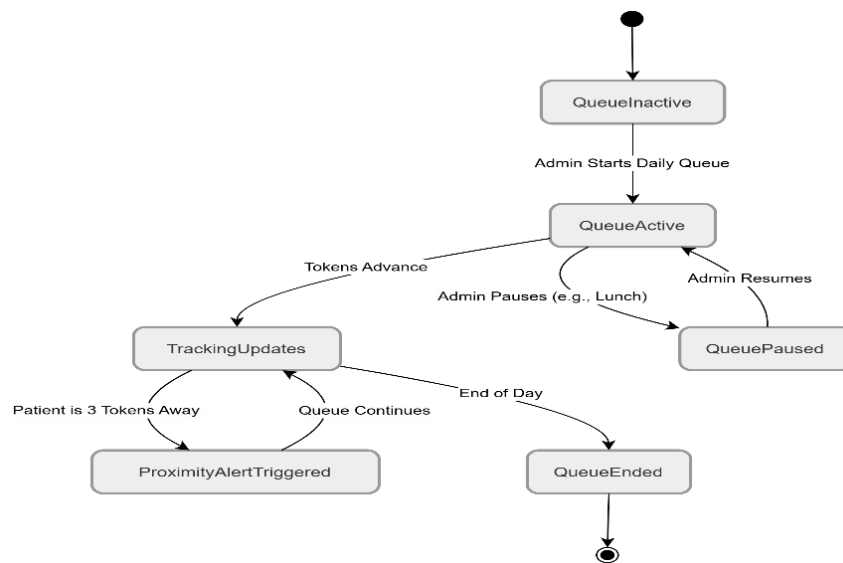


Figure 3.27: State Transition Diagram - Track Live Patient Queue

10. UC-10: Request Emergency Support

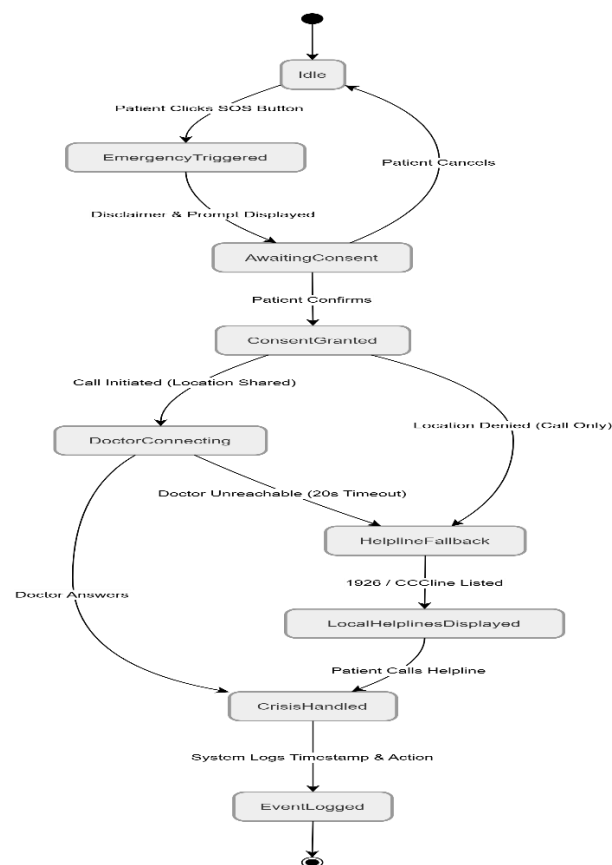


Figure 3.28: State Transition Diagram - Request Emergency Support

11. UC-11: Use Anonymous Community Consultation

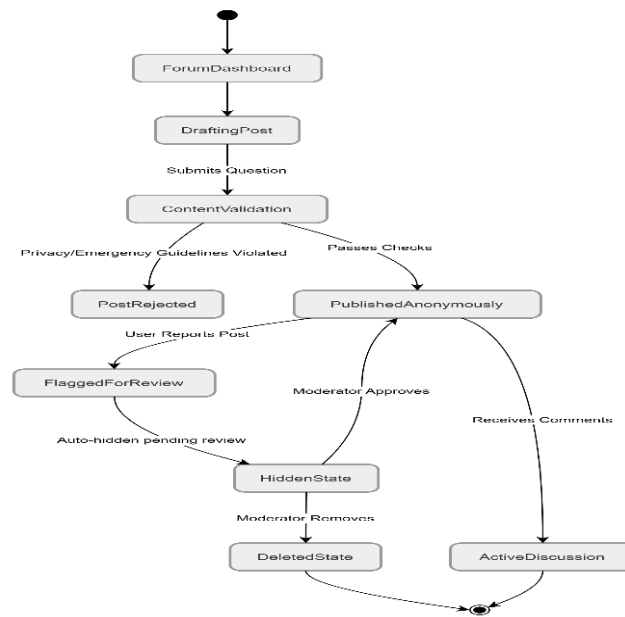


Figure 3.29: State Transition Diagram - Use Anonymous Community Consultation

12. UC-12: Complete AI Triage Questionnaire (SithBot)

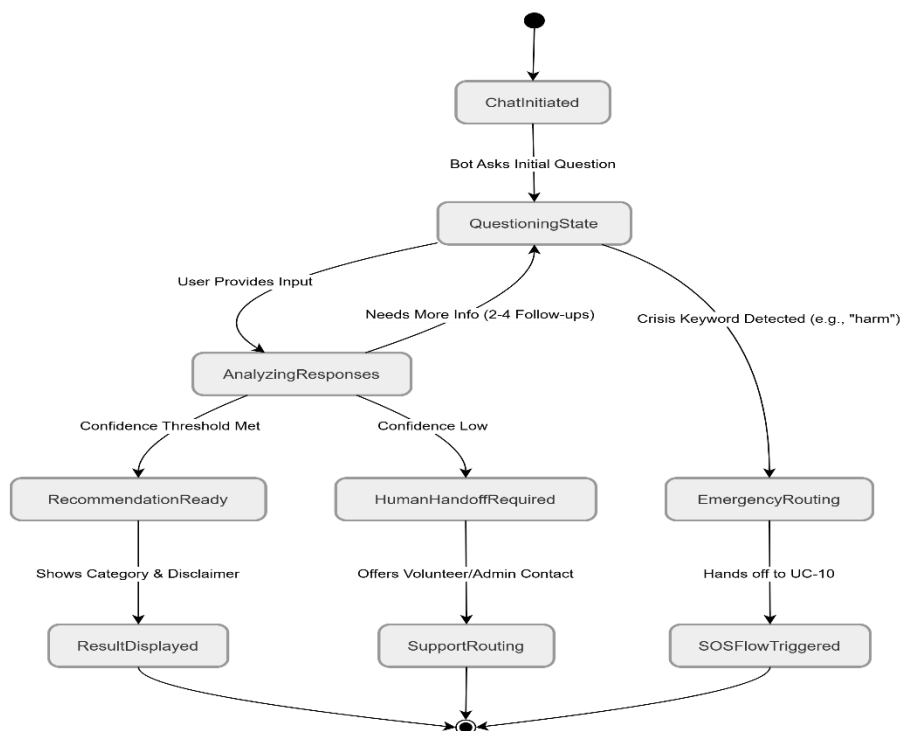


Figure 3.30: State Transition Diagram - Complete AI Triage Questionnaire



3.7.4.4. Entity-Relationship Diagram

SithCare – Entity Relationship Diagram

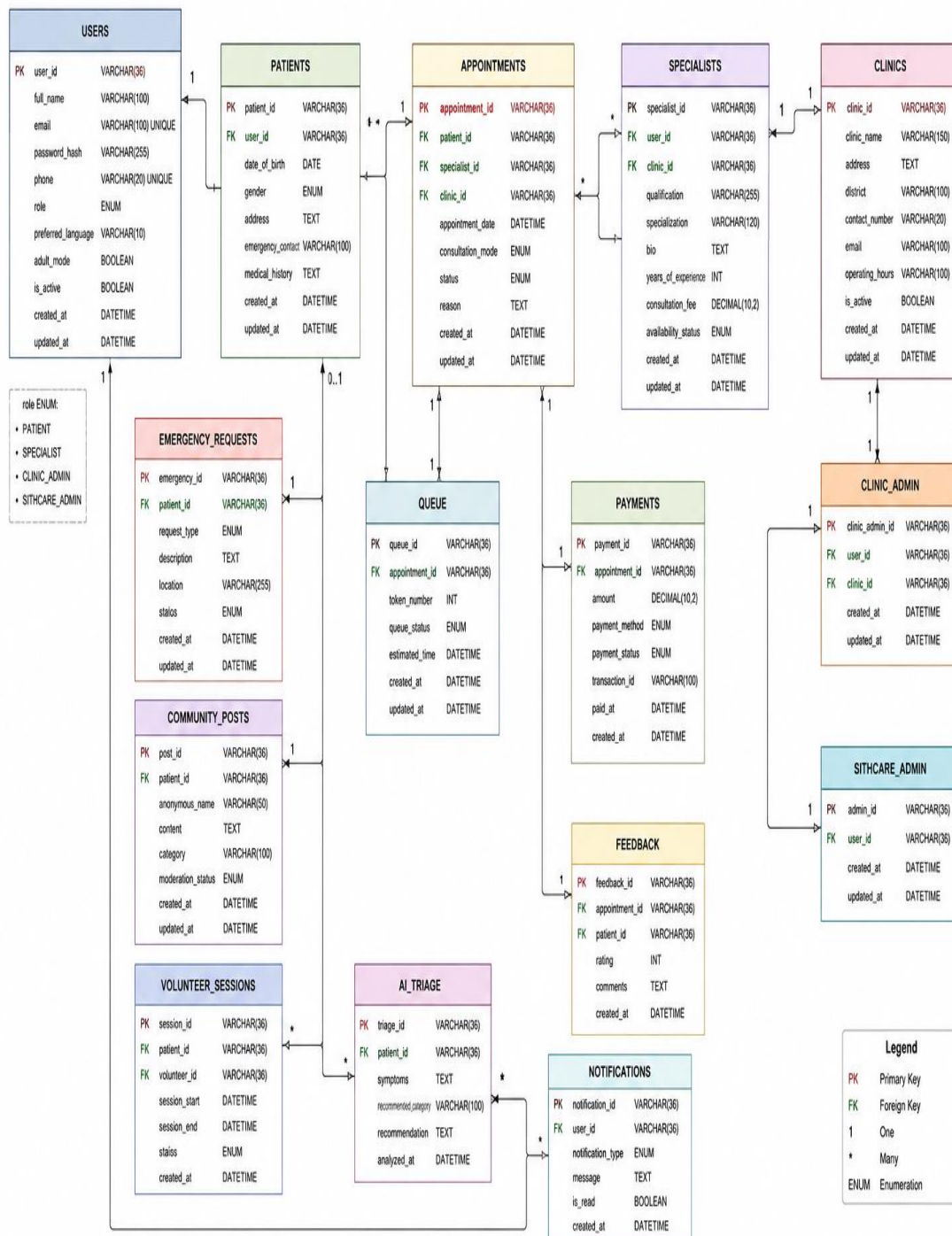


Figure 3.31: SithCare Entity Relationship Diagram



3.7.5. User Interface Design

This section details the frontend visual layout, core interface design, and layout architectures for the SithCare platform. To provide an optimal, role-specific user experience, the system's interface is divided into four main dashboard interfaces. Each interface is designed to meet the distinct functional and operational boundaries of its target users.

3.7.5.1. Clinic-Admin Dashboard

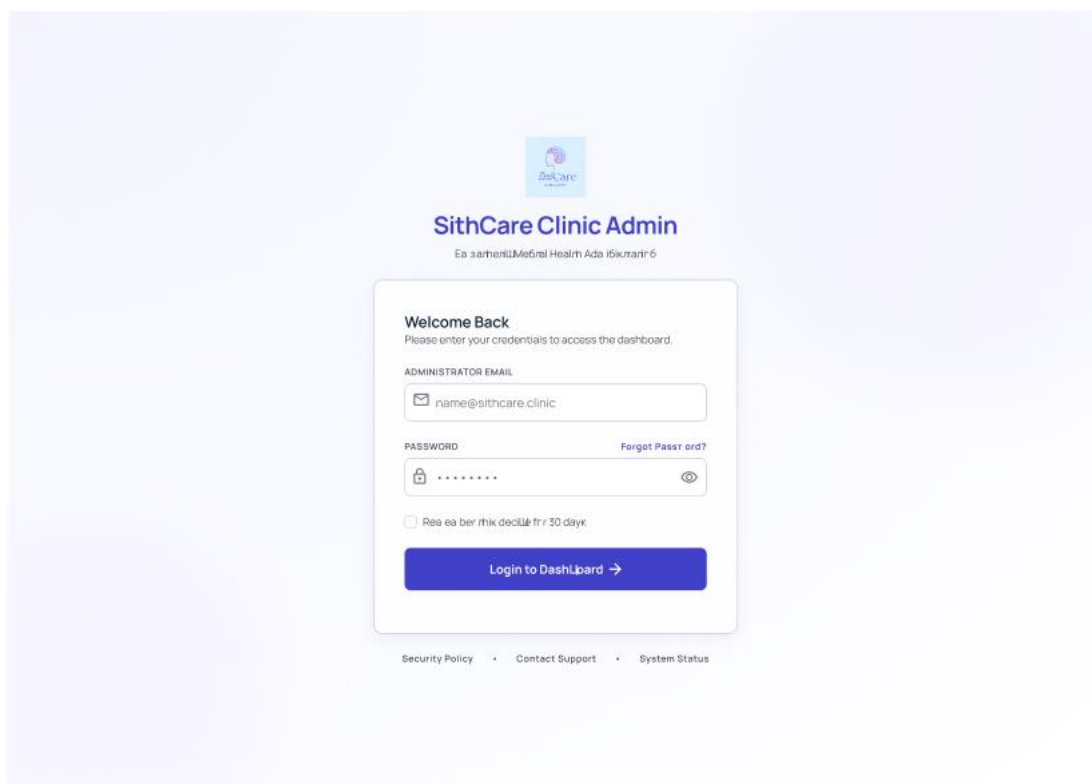


Figure 3.32: Clinic-Admin Dashboard Login Page

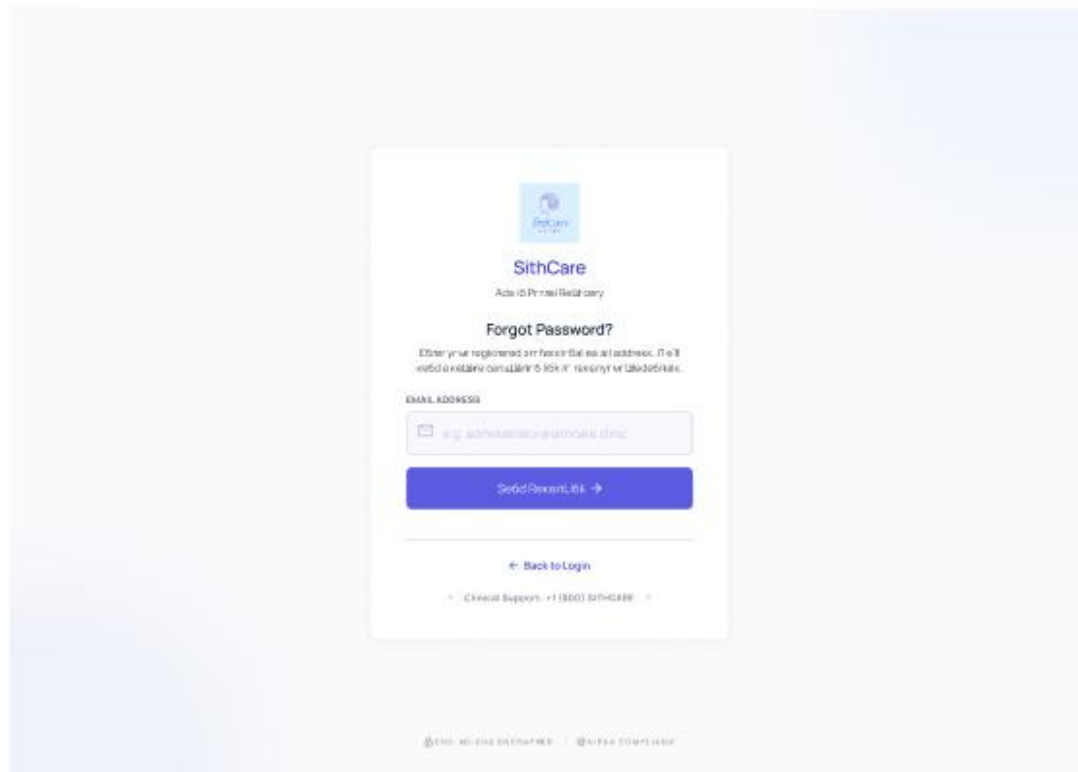


Figure 3.33: Clinic-Admin Dashboard Forgot Password

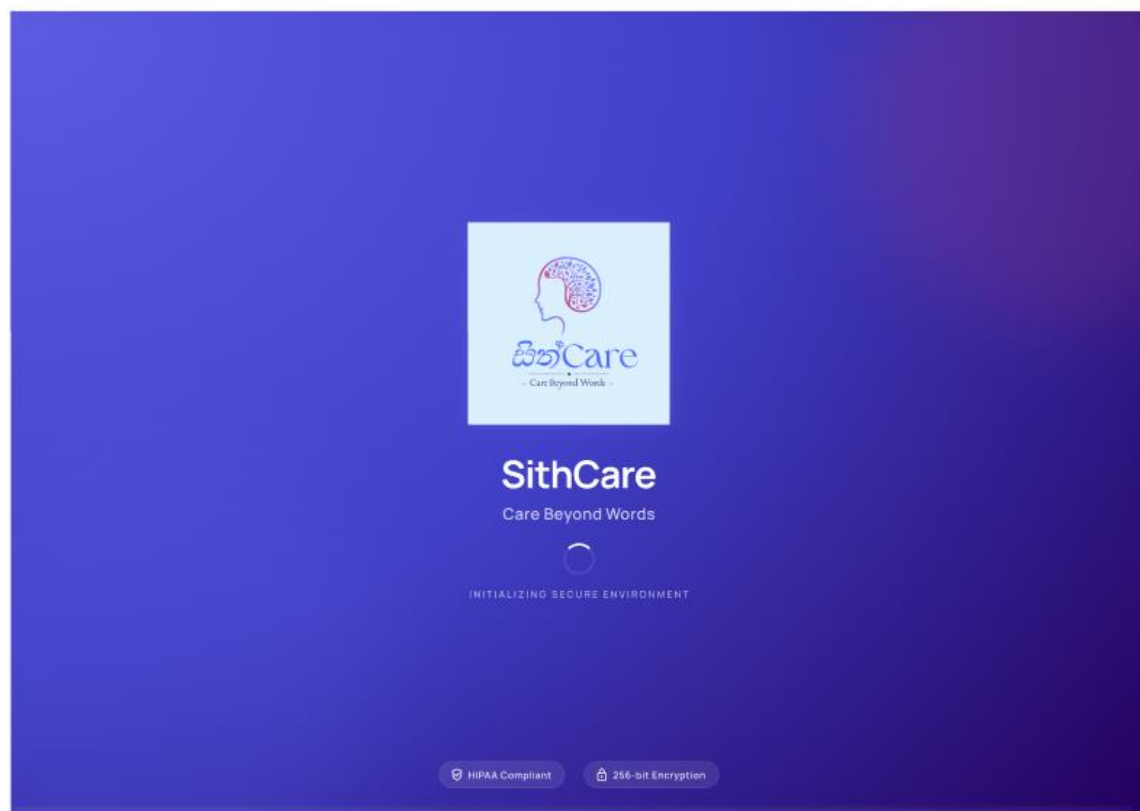


Figure 3.34: Clinic-Admin Dashboard Splash Screen

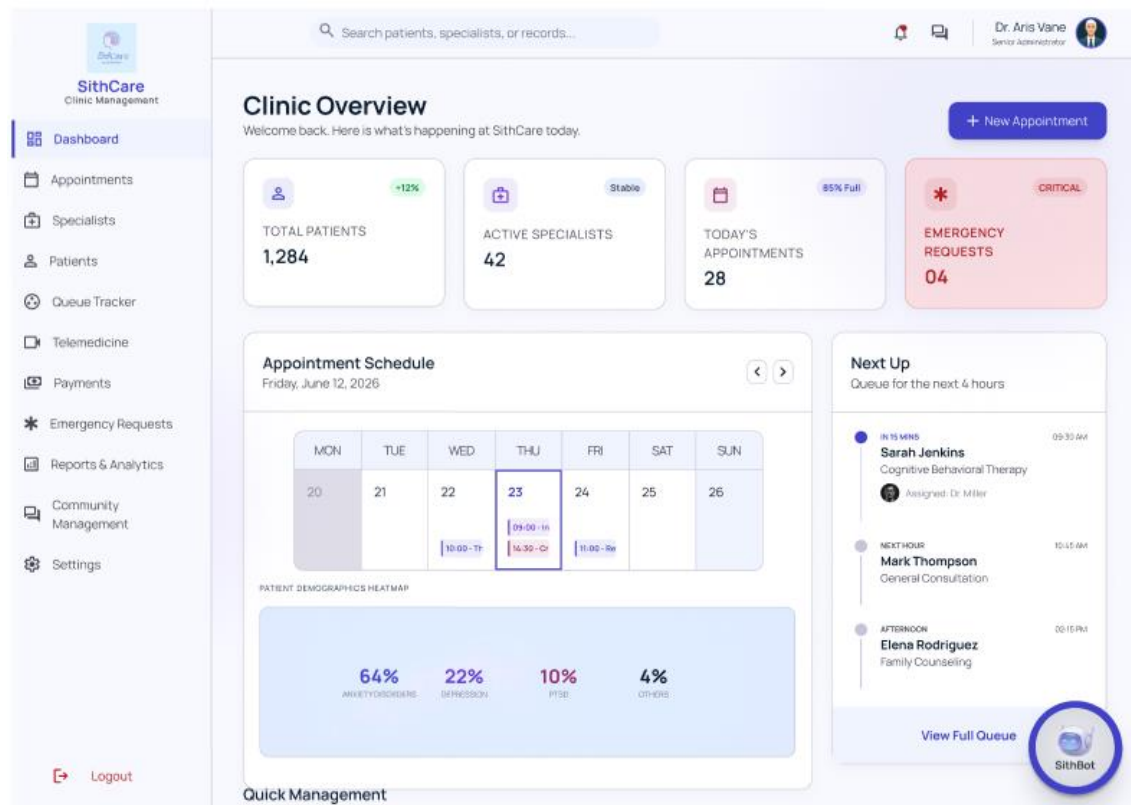


Figure 3.35: Clinic-Admin Dashboard Home Page

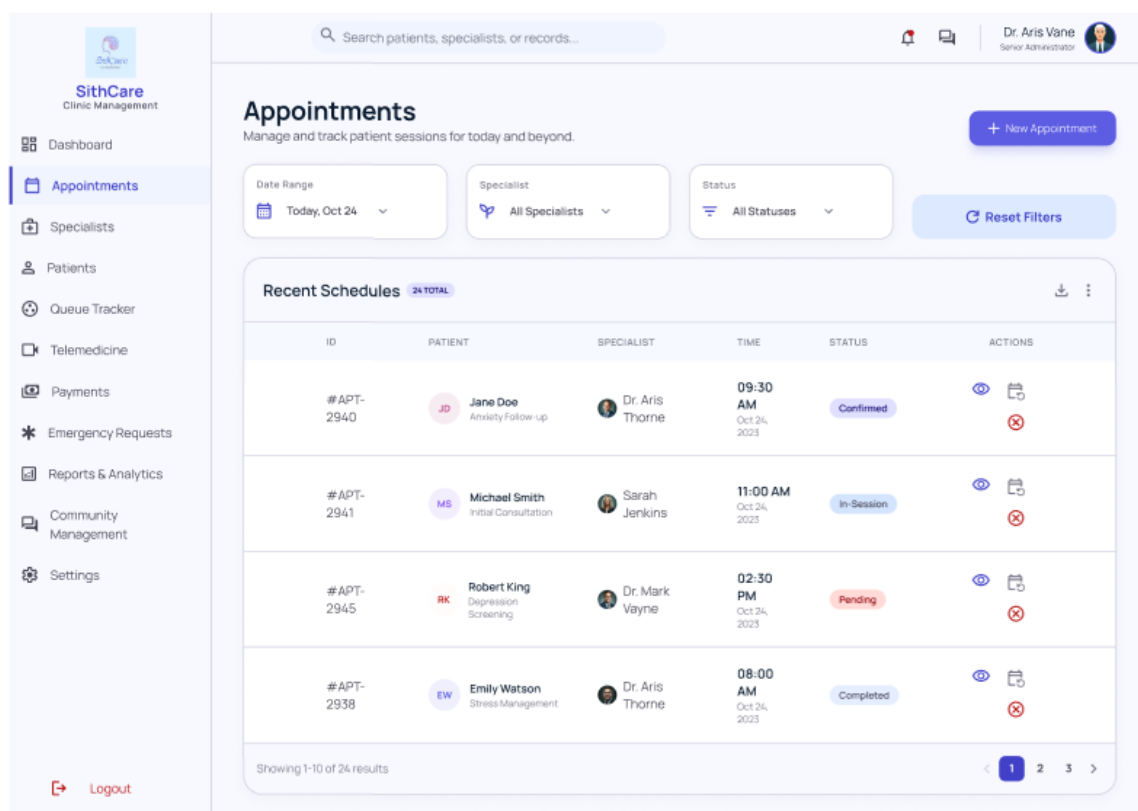


Figure 3.36: Clinic-Admin Dashboard Appointment Management Page



SithCare
Clinic Management

Dashboard
Appointments
Specialists
Patients
Queue Tracker
Telemedicine
Payments
Emergency Requests
Reports & Analytics
Community Management
Settings

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Appointments > Add New Appointment

Schedule Clinical Session

Step 1: Patient Selection & Time Allocation

STEP 01 OF 03

Select Patient [New Patient Record](#)

Search by name, patient ID, or mobile...

Arjuna Wijesekara
ID: SC-2024-9012 • Last visit: 2 weeks ago
[Selected](#) [View History](#)

Nimali Perera
ID: SC-2024-8845 • Last visit: 3 days ago

Kamal Gunawardena
ID: SC-2024-7712 • Last visit: 1 month ago

Shanika Abeywickrama
ID: SC-2024-5541 • Last visit: Never

Schedule Date < May 2024 >

S	M	T	W	T	F	S
28	29	30	1	2	3	4
15	16	17	18	19	20	21
22						

Available Slots

09:00 AM 10:30 AM

11:00 AM 01:30 PM

03:00 PM 04:30 PM

[Cancel Scheduling](#) [Proceed to Specialist →](#)

[Logout](#)

Figure 3.37: Clinic-Admin Dashboard Add Appointment Page

SithCare
Clinic Management

Dashboard
Appointments
Specialists
Patients
Queue Tracker
Telemedicine
Payments
Emergency Requests
Reports & Analytics
Community Management
Settings

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Appointment Successfully Scheduled

Your session has been confirmed and added to the clinic's master queue. A notification has been sent to the patient.

Patient
Arjuna
Wijesekara
ID: PT-6821

Specialist
Dr. Aris Thorne
Clinical Psychologist

Time
11:00 AM
Tuesday, Oct 24

[View Appointment](#) [Return to Dashboard](#)

[Need to change this? Reschedule Appointment](#)

CONFIRMATION REF: SCH-002931-X

[Logout](#)

Figure 3.38: Clinic-Admin Dashboard Appointment Success Page



The screenshot displays the 'Appointment Details' page for a 'Cognitive Behavioral Therapy' session. The left sidebar contains navigation links: Dashboard, Appointments (selected), Specialists, Patients, Queue Tracker, Telemedicine, Payments, Emergency Requests, Reports & Analytics, Community Management, and Settings. The main content area includes a search bar, a session ID (#CBT-9021), and buttons for 'Cancel Session', 'Reschedule', and 'Edit Details'. The 'Session Notes' section describes the patient, Anakin Skywalker, and the session's focus on cognitive distortions. A 'Key Insight' box highlights a protective barrier of aggression. The 'Assigned Specialist' section shows Dr. Obi-Wan Kenobi, a Senior Behavioral Consultant. The 'Payment Status' section indicates a paid session of \$180. The 'Activity Log' shows the session was finalized, payment was processed, and reminders were sent.

SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Appointments > Session ID: #CBT-9021

Cognitive Behavioral Therapy

[View Past Notes](#)

Session Notes

The patient, Anakin Skywalker, presents with chronic stress-induced anxiety and difficulty managing impulsive emotional reactions. This session (4 of 12) focused on identifying cognitive distortions related to past experiences and perceived lack of control.

Key Insight:

"Strong attachment leads to fear of loss, which creates a protective barrier of aggression." – Focus next session on detachment exercises and mindfulness-based stress reduction (MBSR).

Compliance with home practice (meditation intervals) remains at 60%. Discussion held on establishing a consistent routine amidst high-stress professional duties.

ASSIGNED SPECIALIST

Dr. Obi-Wan Kenobi
Senior Behavioral Consultant

PAYMENT STATUS

\$180 / session **PAID**

Invoice ID: #INV-88221
Method: Visa ending in 4421

Activity Log

- Session Finalized
May 23, 04:15 PM by Dr. Aris Vane
- Payment Processed
May 23, 09:30 AM via Stripe
- Reminders Sent

[Cancel Session](#) [Reschedule](#) [Edit Details](#)

Anakin Skywalker
General Anxiety Disorder (F41.1)

Age: **22 Years**

Next Session: **June 02, 2024 at 10:30 AM**

Clinic Records

[Medical History](#) [Treatment Plan](#)

[Logout](#)

Figure 3.39: Clinic-Admin Dashboard Appointment Details

The screenshot displays the 'Specialist Management' page. The left sidebar is identical to the previous figure. The main content area features a search bar, a 'Dr. Aris Vane' profile, and a '+ Add Specialist' button. The 'Our Clinical Network' section shows the status of 24 specialist practitioners, categorized by Psychologists (12 active), Psychiatrists (08 active), and Counsellors (04 active). Below this, four specialist profiles are displayed: Dr. Sarah Johnson (Senior Clinical Psychologist, Trauma Specialist, CBT), Dr. Marcus Thorne (Consultant Psychiatrist, Neuropsychiatry, MD), Emma Vance (Family & Marriage Counsellor, Family Therapy, MSc), and Dr. Leo Vance (Behavioral Specialist, ABA Therapy, PhD). Each profile includes a 'View Profile' button and a 'Logout' button.

SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Our Clinical Network

Manage and monitor the status of 24 specialist practitioners.

[+ Add Specialist](#)

PSYCHOLOGISTS
12 active

PSYCHIATRISTS
08 active

COUNSELLORS
04 active

Dr. Sarah Johnson
Senior Clinical Psychologist
Trauma Specialist CBT
[View Profile](#)

Dr. Marcus Thorne
Consultant Psychiatrist
Neuropsychiatry MD
[View Profile](#)

Emma Vance
Family & Marriage Counsellor
Family Therapy MSc
[View Profile](#)

Dr. Leo Vance
Behavioral Specialist
ABA Therapy PhD
[View Profile](#)

[Logout](#)

Figure 3.40: Clinic-Admin Dashboard Specialist Management



SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Specialists > Edit Profile

Edit Specialist Profile
Modify personal information, medical expertise, and clinical availability.

General Information

Full Name: Dr. Marcus Thorne
Specialty: Clinical Psychology

Professional Biography

Dr. Marcus Thorne is a leading Clinical Psychologist with over 15 years of experience specializing in trauma-informed care and anxiety disorders. He has pioneered several mindfulness-based interventions currently utilized at SithCare. Previously, he served as the Head of Behavioral Sciences at the New Haven Institute. His approach balances clinical rigor with deep empathetic connection, ensuring every patient feels heard and supported through their recovery journey.

Expertise & Qualifications

License Number: PSY-99283-TX
Years of Experience: 15

Clinical Focus Areas

Trauma Recovery X Anxiety Disorders X CBT X + Add Area

Availability

Mon: 09:00 AM - 05:00 PM
Tue: 09:00 AM - 05:00 PM
Wed: 10:00 AM - 04:00 PM
Thu: 09:00 AM - 05:00 PM
Fri: Unavailable

Logout

Figure 3.41: Clinic-Admin Dashboard Edit Specialist Profile

SithCare
Clinic Management

Search patients, specialists, or records...

Dr. Aris Vane
Senior Administrator

Specialists / Availability Management

Dr. Sarah Johnson
Clinical Psychologist • Cognitive Behavioral Therapy Specialist

October 2023
VIEWING SCHEDULE
Week of Oct 16 - Oct 22

AVAILABLE SLOTS
34 +5 vs last week

BOOKED APPOINTMENTS
22 88% Capacity

TOTAL OFF TIME
12h Personal/Break

Available Booked Off

Week of Oct 16 - Oct 22

	MON 16	TUE 17	WED 18	THU 19	FRI 20	SAT 21	SUN 22
08:00 AM	AVAILABLE	PATIENT SESSION Marcus T.	OFF	AVAILABLE	AVAILABLE		
09:00 AM	CBT FOLLOWUP Elena R.	AVAILABLE	AVAILABLE	INITIAL INTAKE Samuel K.	AVAILABLE		
10:00 AM	AVAILABLE	AVAILABLE	AVAILABLE	OFF	AVAILABLE		
11:00 AM	Lunch Break	Lunch Break	Lunch Break	Lunch Break	Lunch Break		
12:00 PM	AVAILABLE	STAFF MEETING Conference	AVAILABLE	AVAILABLE	AVAILABLE		

Default Duration
60 Minutes Session

Buffer Times
Post-session Break: 15 mins

Leave Request
Mark Vacation Time

SithBot

Logout

Figure 3.42: Clinic-Admin Dashboard Specialist Availability

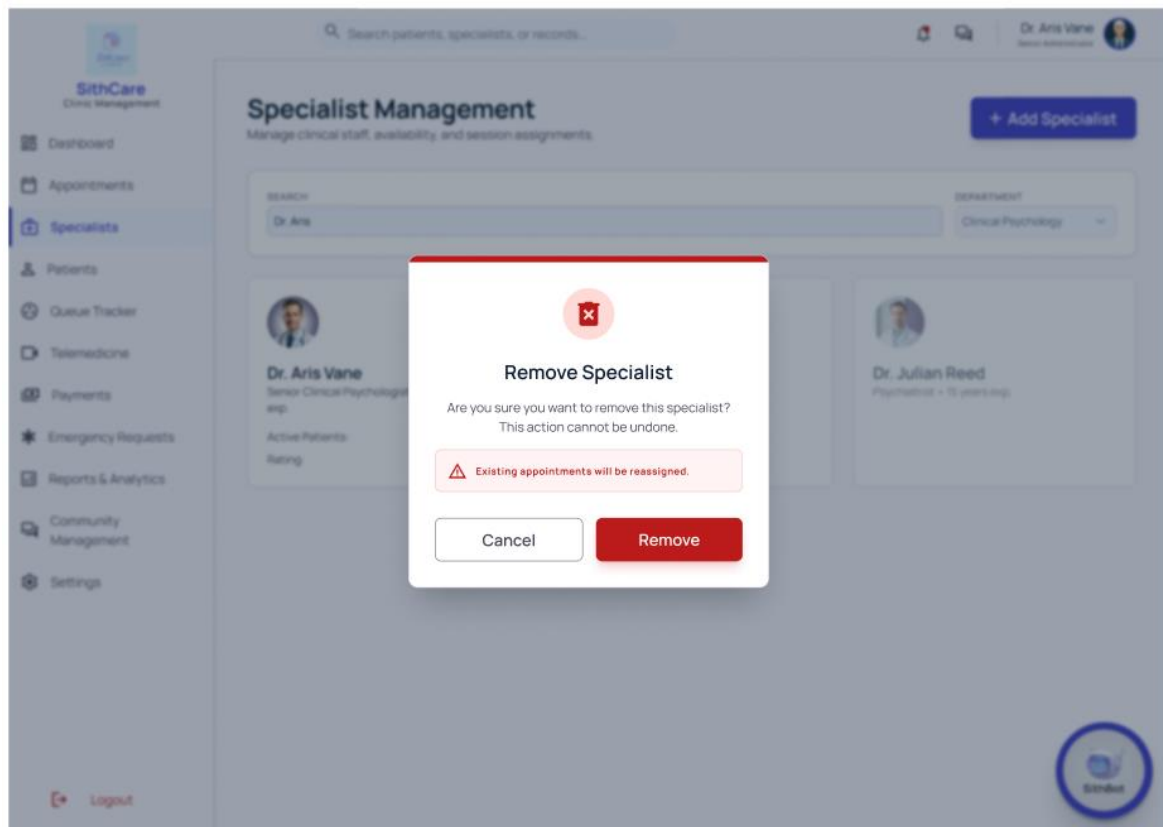


Figure 3.43: Clinic-Admin Dashboard Remove Specialist Confirmation

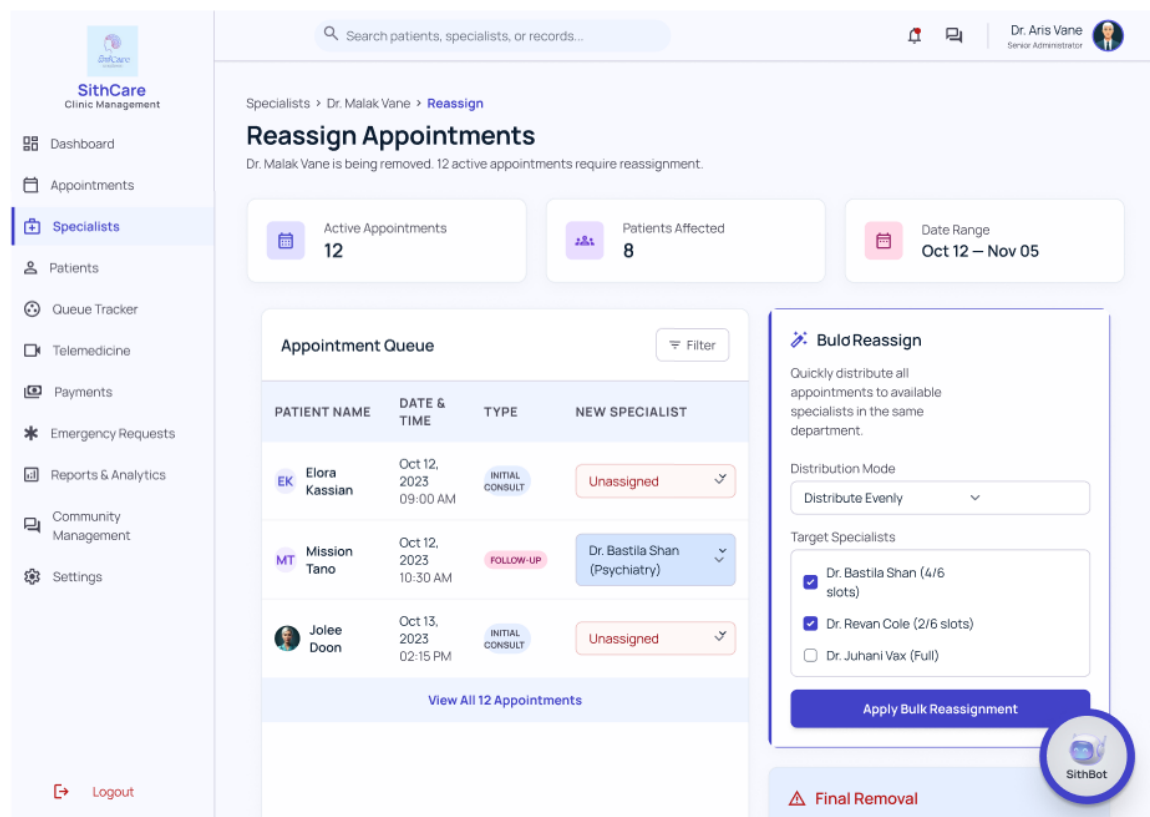


Figure 3.44: Clinic-Admin Dashboard Specialist Appointment Reassignment



The screenshot shows the 'Patient Management' section of the SithCare Clinic Management dashboard. The left sidebar contains navigation links: Dashboard, Appointments, Specialists, Patients (selected), Queue Tracker, Telemedicine, Payments, Emergency Requests, Reports & Analytics, Community Management, and Settings. The main content area has a search bar at the top. Below it, a summary row displays four key metrics: Total Patients (1,284), In-Session Today (42), Urgent Alerts (03), and Reports Pending (12). A table lists the first four patients, showing their names, contact info, status, last visit, assigned specialist, and actions. The patients listed are Elena Rodriguez (In-Session), Marcus Chen (Scheduled), and Sophie Laurent (Emergency). A 'Recent Activity' section shows a log entry for Dr. Sarah Jenkins updating clinical notes for Elena Rodriguez. A 'Telemedicine Upgrade' banner is also present.

Patient Management
View and manage clinical patient records and history.

Summary Metrics:

- TOTAL PATIENTS: 1,284
- IN-SESSION TODAY: 42
- URGENT ALERTS: 03
- REPORTS PENDING: 12

Patient List:

PATIENT NAME	CONTACT INFO	STATUS	LAST VISIT	ASSIGNED SPECIALIST	ACTIONS
Elena Rodriguez ID: #PX-8290	elena.r@example.com +1 (555) 123-4567	In-Session	Today, 10:30 AM	Dr. Sarah Jenkins	[View] [Edit] [Refresh]
Marcus Chen ID: #PX-8112	m.chen@outlook.com +1 (555) 987-6543	Scheduled	Oct 24, 2023	Dr. Michael Aris	[View] [Edit] [Refresh]
Sophie Laurent ID: #PX-7945	sophie.l@protonmail.com +1 (555) 234-8765	Emergency	Oct 25, 2023	Dr. Sarah Jenkins	[View] [Edit] [Refresh]

Showing 1 to 4 of 1,284 patients

Recent Activity [View All](#)

- Dr. Sarah Jenkins updated the clinical notes for Elena Rodriguez. 76 minutes ago

Telemedicine Upgrade
You can now host secure group therapy sessions via the SithCare Portal. Try the

Figure 3.45: Clinic-Admin Dashboard Patient Management

The screenshot shows the 'Patient Profile' for Maya Sterling in the SithCare Clinic Management dashboard. The left sidebar is identical to the previous figure. The main content area displays the patient's profile at the top, including her name, ID, age, gender, and a status indicator (Active Patient). Below the profile, there are four key health metrics: Heart Rate (72 bpm), Blood Pressure (118/76 mmHg), Stress Level (Moderate), and Sleep Quality (6.5 hrs avg). The 'Comprehensive History' section lists two conditions: Generalized Anxiety Disorder (GAD) and Mild Depressive Episodes, each with a brief description and a status label (Chronic and Managed respectively). The 'Medications' section lists Sertraline (Zoloft) and Propranolol. The 'Recent Visit Timeline' shows a visit on October 12, 2023, for a routine follow-up and medication adjustment. The 'Upcoming' section shows a full assessment on November 4 and a group therapy session on November 11.

Patient Profile: Maya Sterling
Patient ID: SC-8829-M • Age: 28 • Gender: Female

Key Metrics:

- Heart Rate: 72 bpm
- Blood Pressure: 118/76 mmHg
- Stress Level: Moderate (6/10)
- Sleep Quality: 6.5 hrs avg

Comprehensive History [View All Records](#)

- Generalized Anxiety Disorder (GAD)** (Chronic)
Diagnosed Jan 2023. Patient reports recurring intrusive thoughts and somatic symptoms including chest tightness and restlessness.
- Mild Depressive Episodes** (Managed)
Secondary diagnosis following major life transition. Managing with Cognitive Behavioral Therapy.

Medications

- Sertraline (Zoloft)**
50mg • Once Daily (Morning)
- Propranolol**
10mg • As needed for anxiety

Recent Visit Timeline

- TODAY, 10:30 AM** (Dr. Sarah Jenkins)
Routine Follow-up & Med Adjustment
Patient reported improved sleep but increased daytime lethargy. Adjusted Sertraline dosage.
- OCT 12, 2023** (Clinical Counselor Mike Ross)
CBT Session - Module 4

Upcoming

- NOV 04** **Full Assessment**
02:00 PM • Room 402B
Dr. Alaric Chen
- NOV 11** **Group Therapy**
05:30 PM • Virtual Session
[TELEMED](#)

[Schedule Next Appointment](#)

Figure 3.46: Clinic-Admin Dashboard Patient Profile

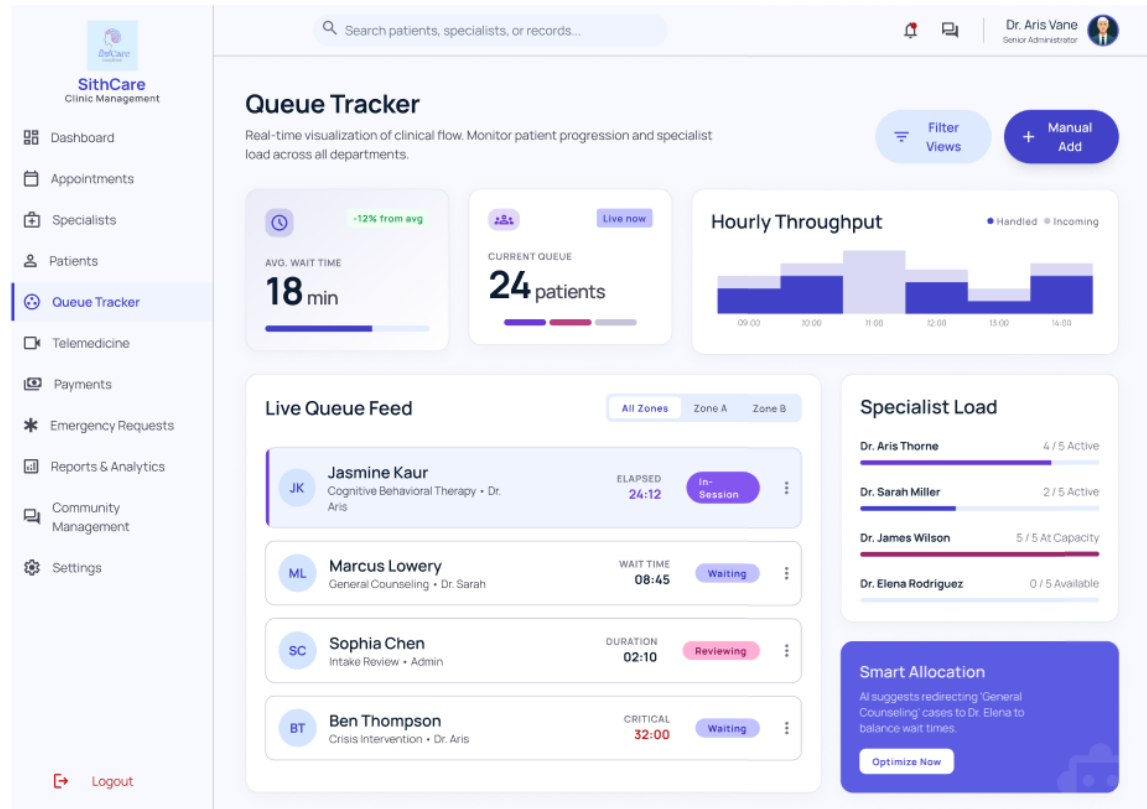


Figure 3.47: Clinic-Admin Dashboard Queue Tracker

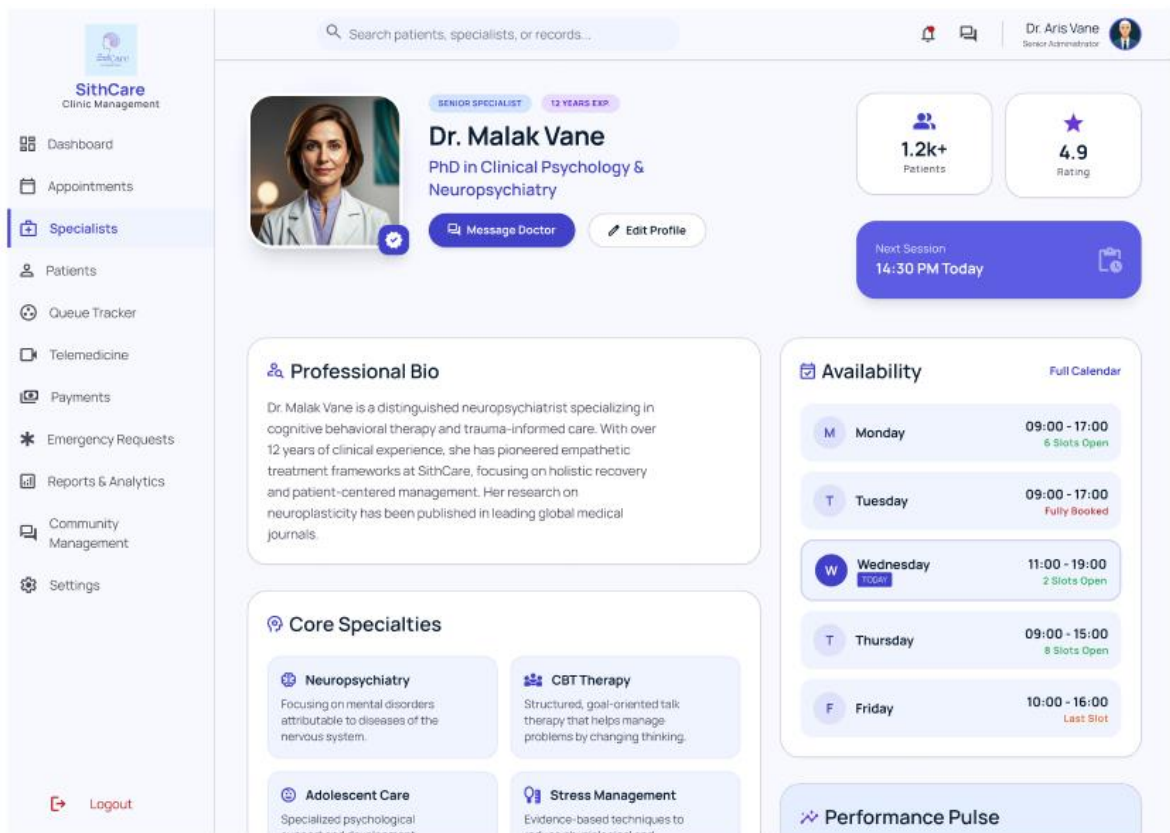


Figure 3.48: Clinic-Admin Dashboard Specialist Profile

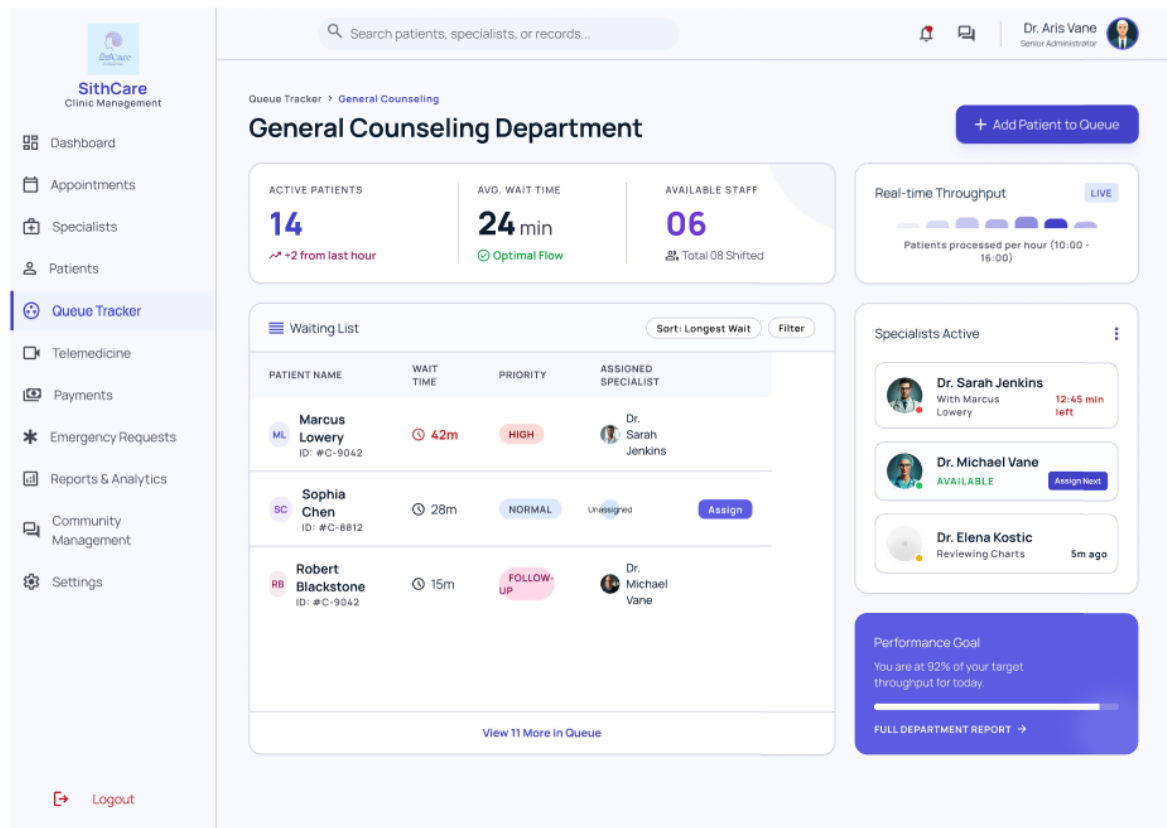


Figure 3.49: Clinic-Admin Dashboard Queue Details

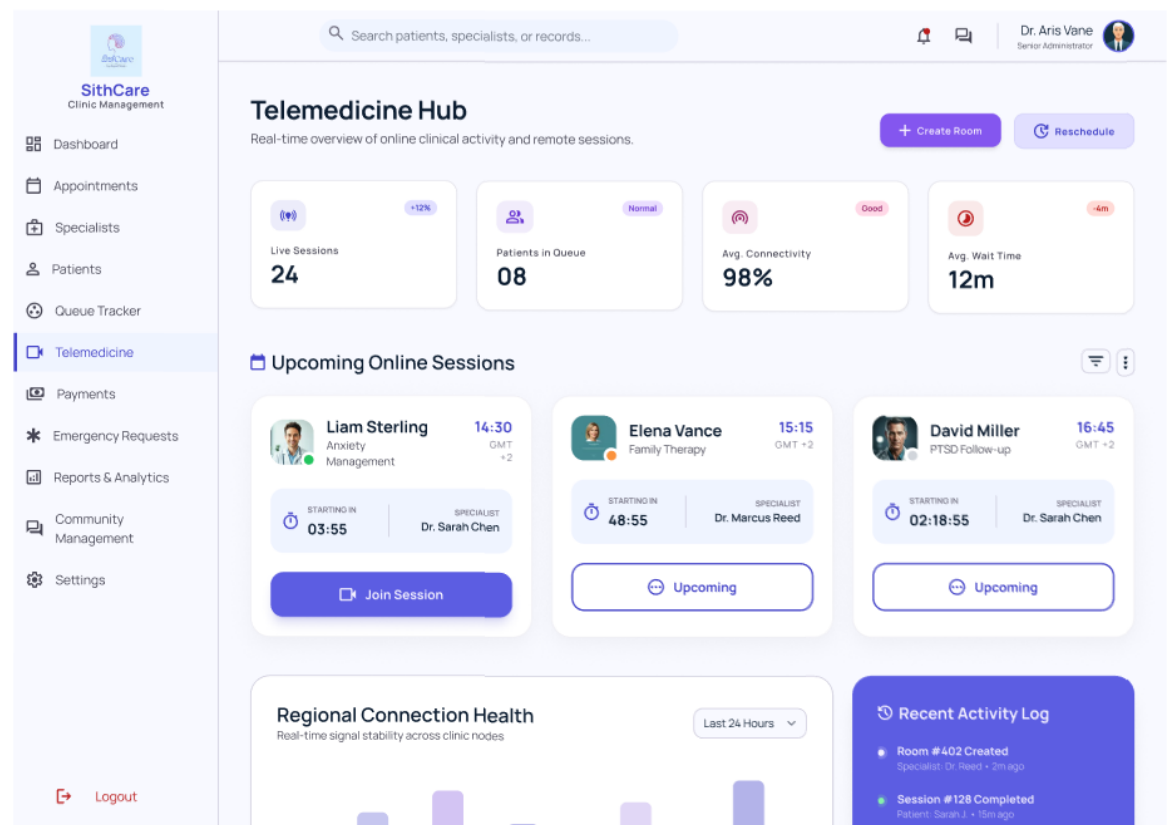


Figure 3.50: Clinic-Admin Dashboard Telemedicine Page

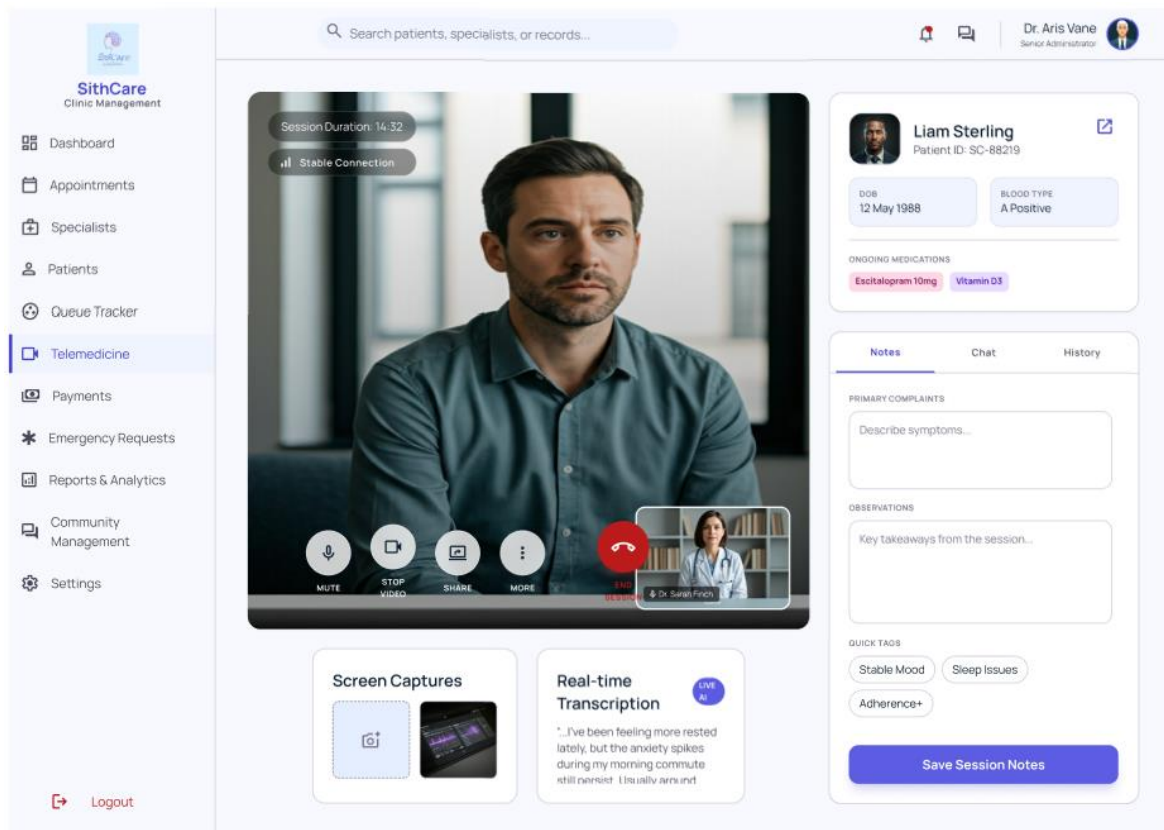


Figure 3.51: Clinic-Admin Dashboard Active Consultation

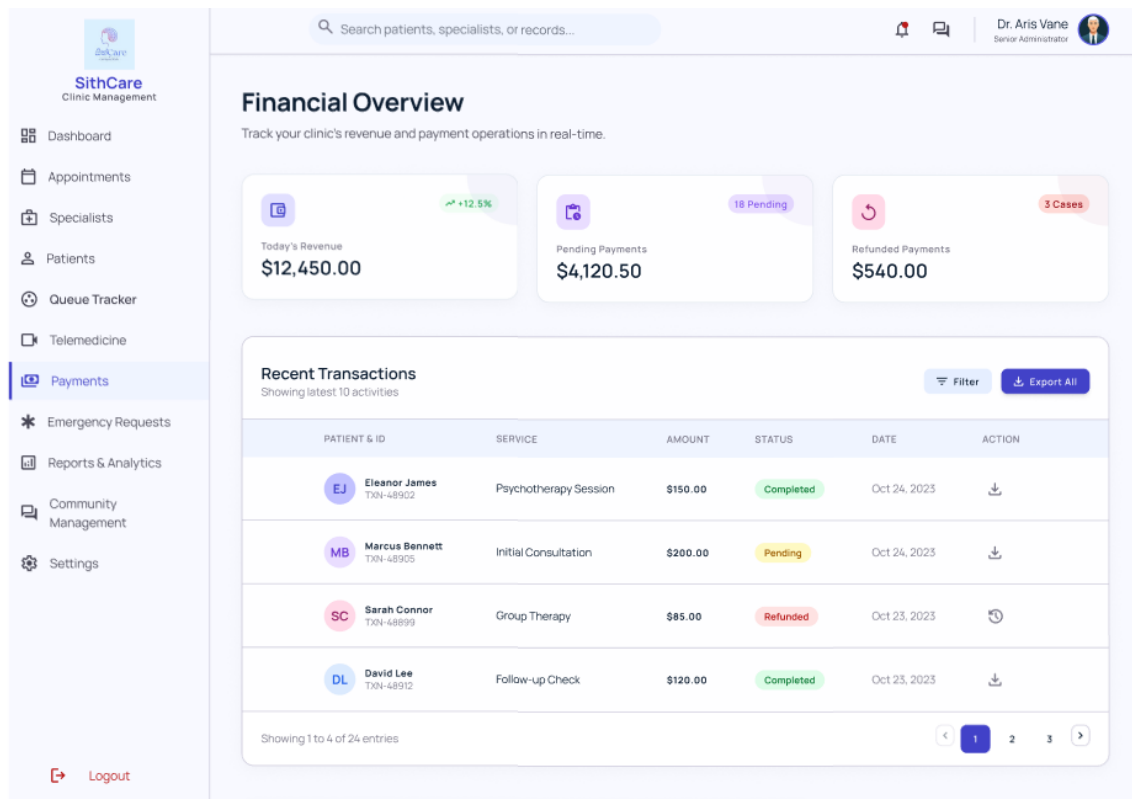


Figure 3.52: Clinic-Admin Dashboard Payment Interface

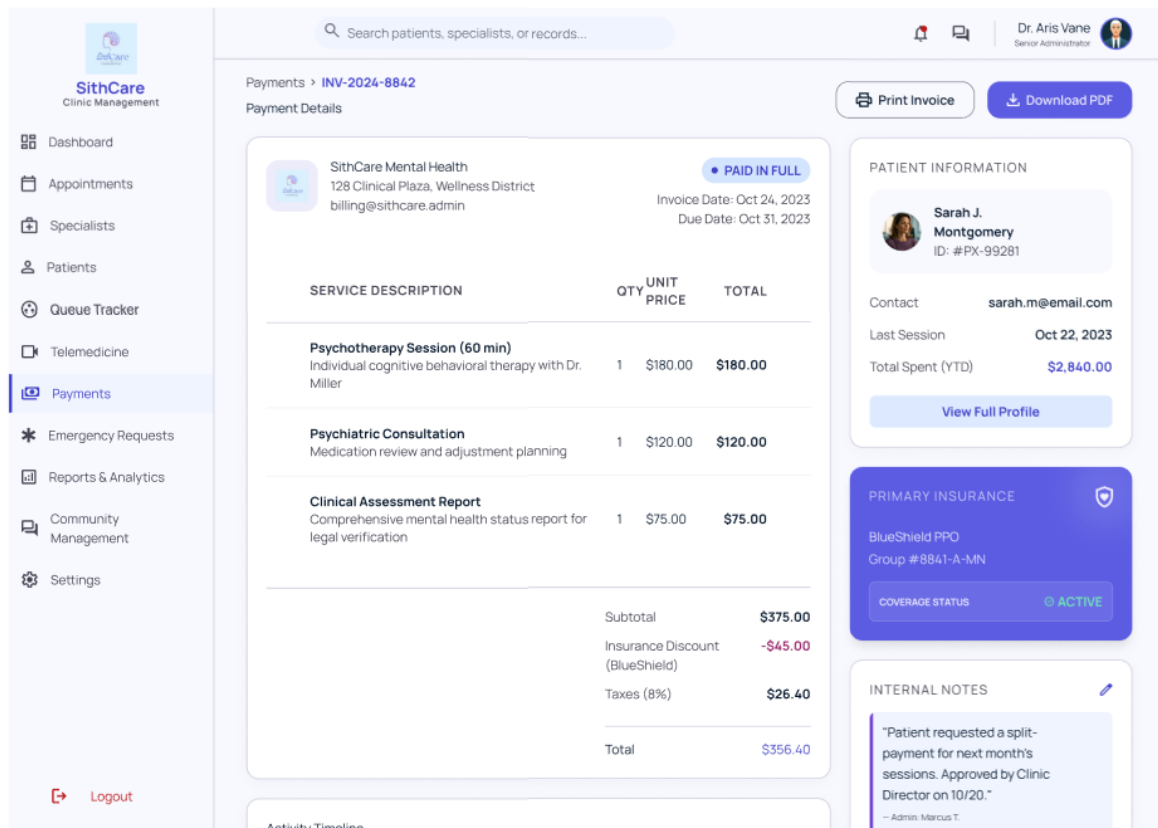


Figure 3.53: Clinic-Admin Dashboard Payment Details

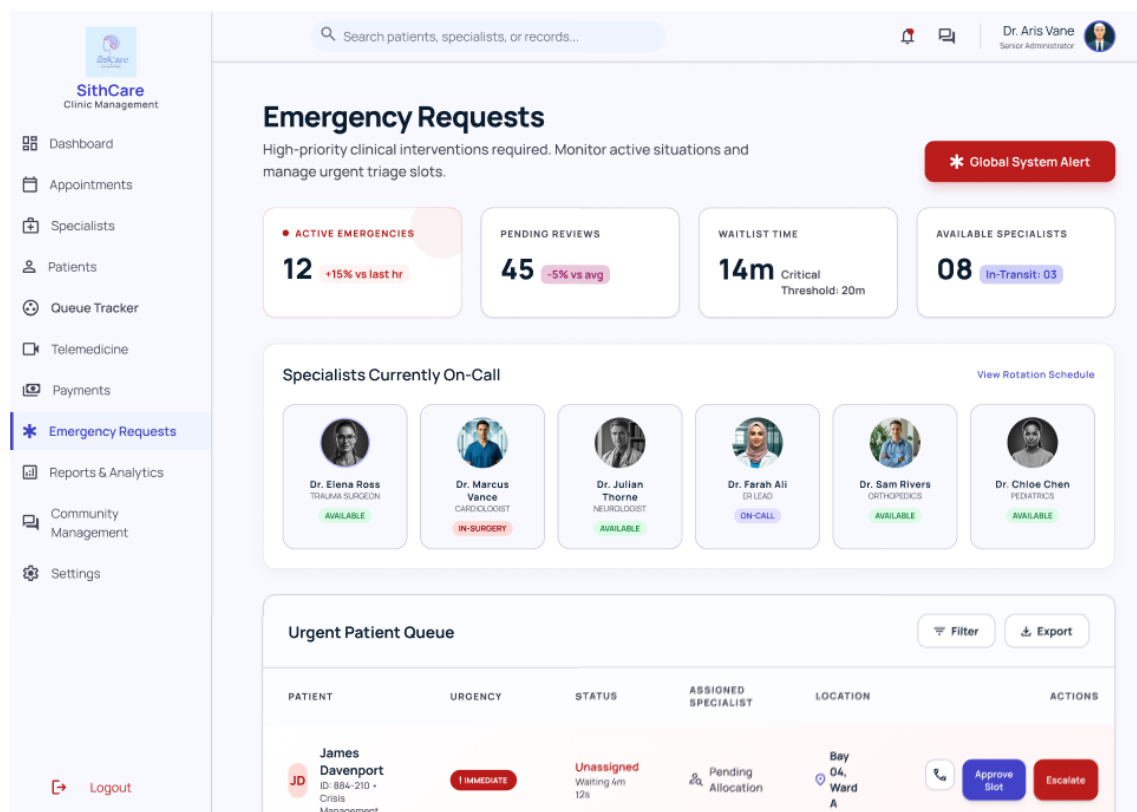


Figure 3.54: Clinic-Admin Dashboard Emergency Requests

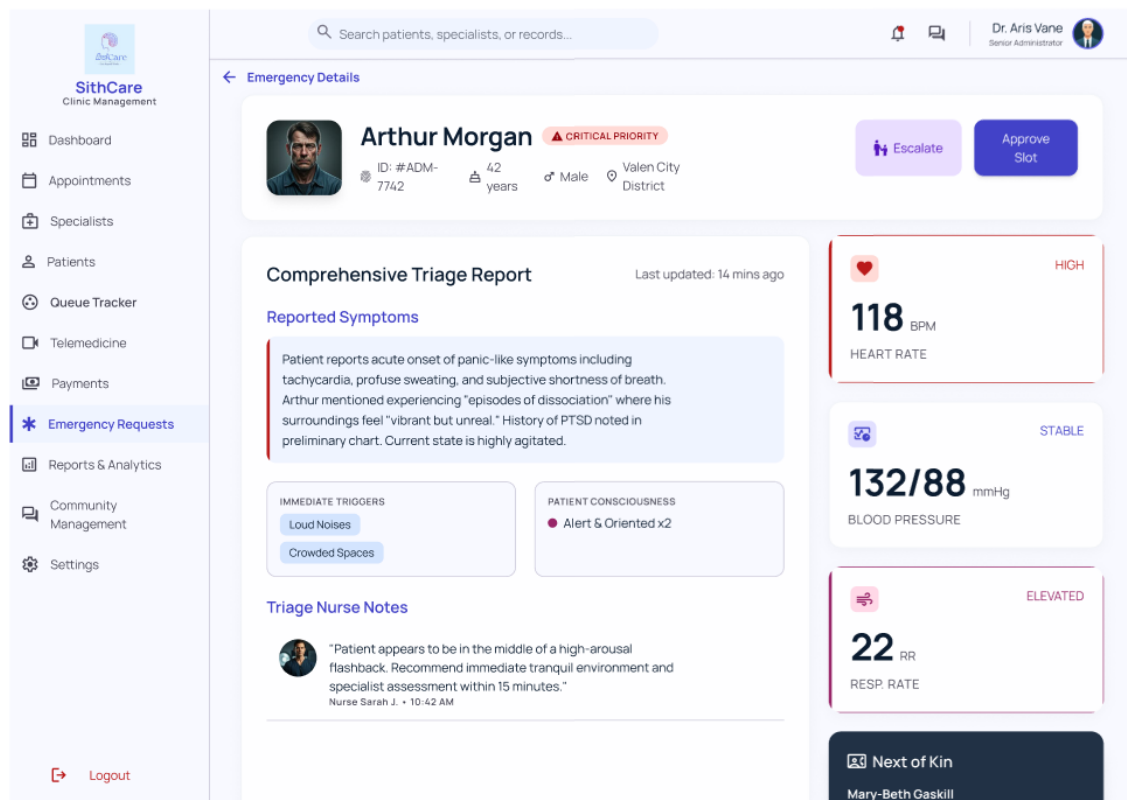


Figure 3.55: Clinic-Admin Dashboard Emergency Request Detail0073

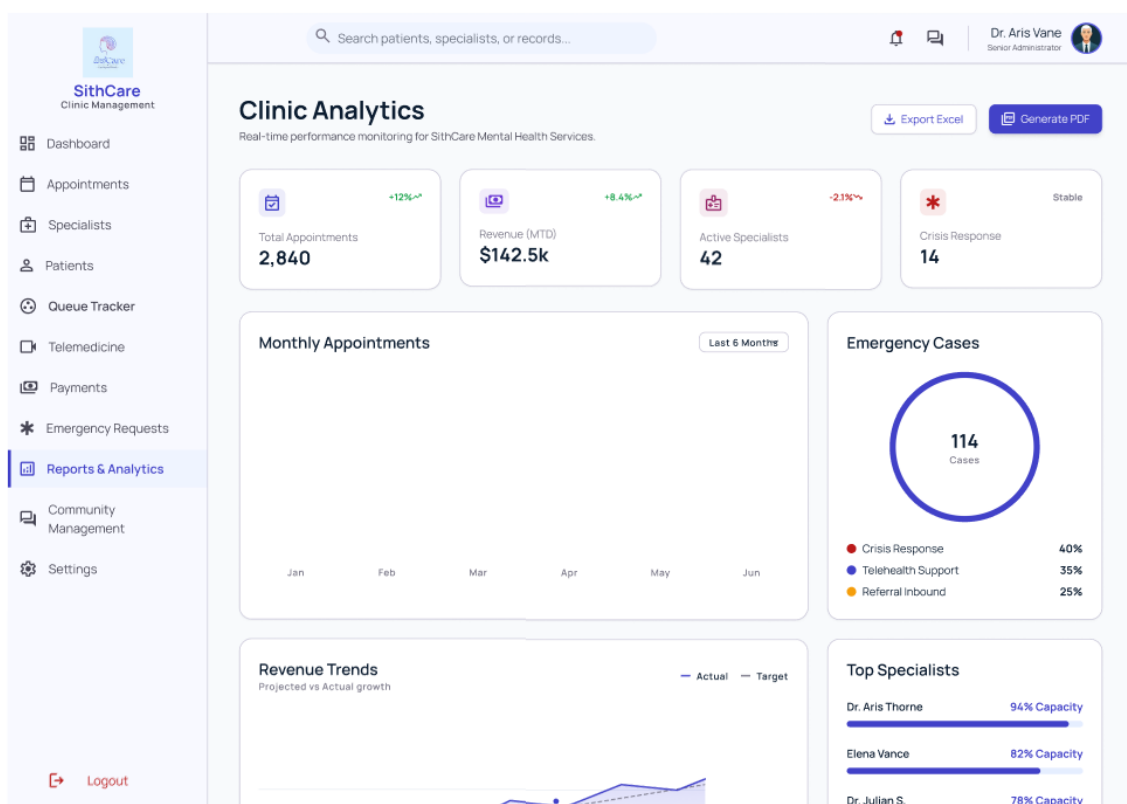


Figure 3.56: Clinic-Admin Dashboard Reports and Analytics

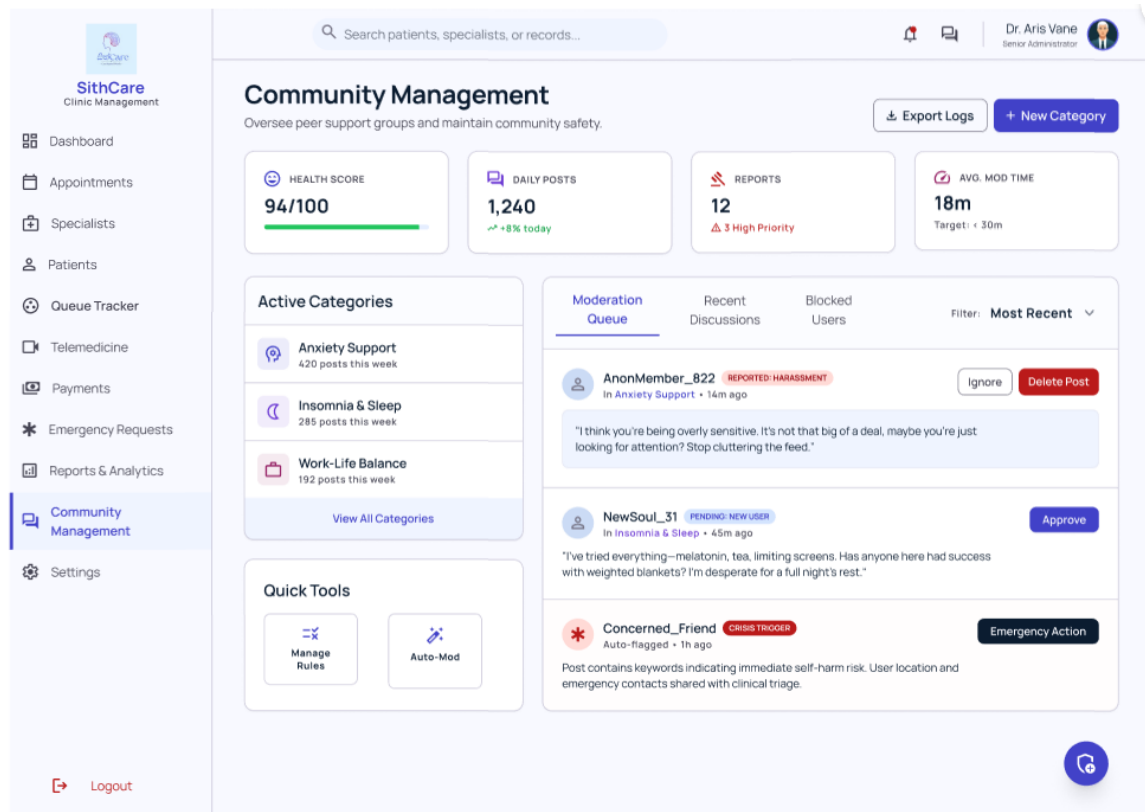


Figure 3.57: Clinic-Admin Dashboard Community Management

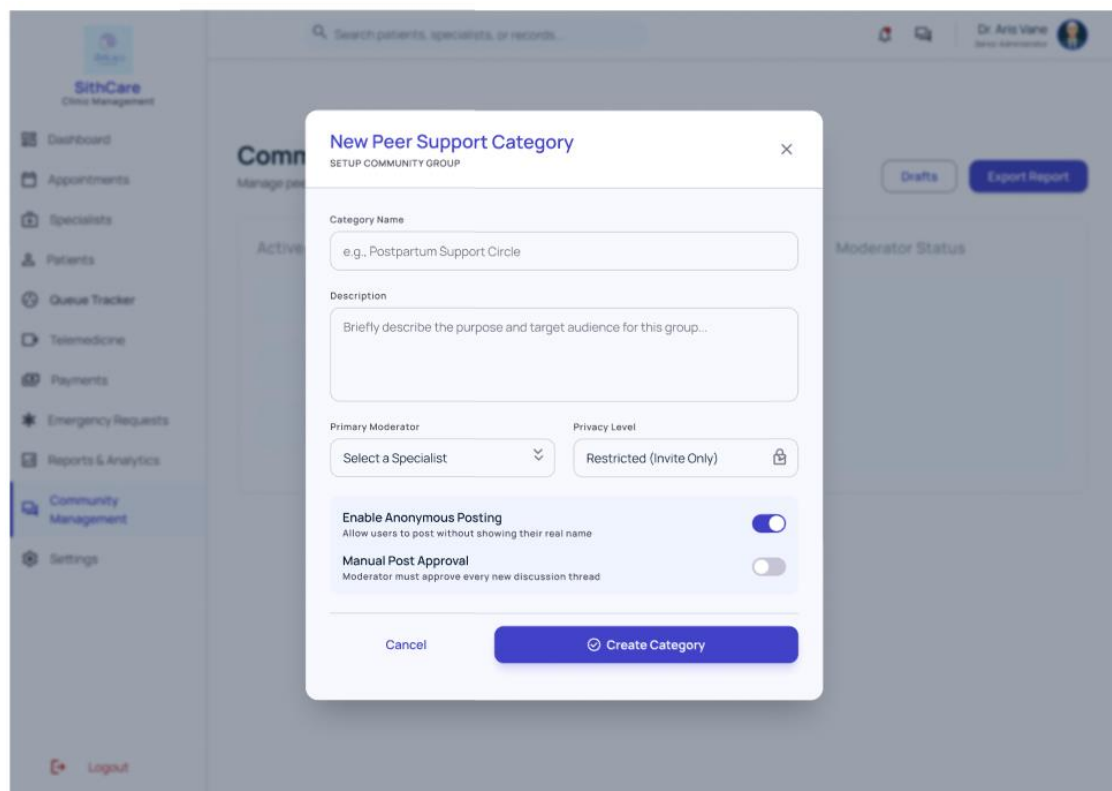


Figure 3.58: Clinic-Admin Dashboard Community Category Creation

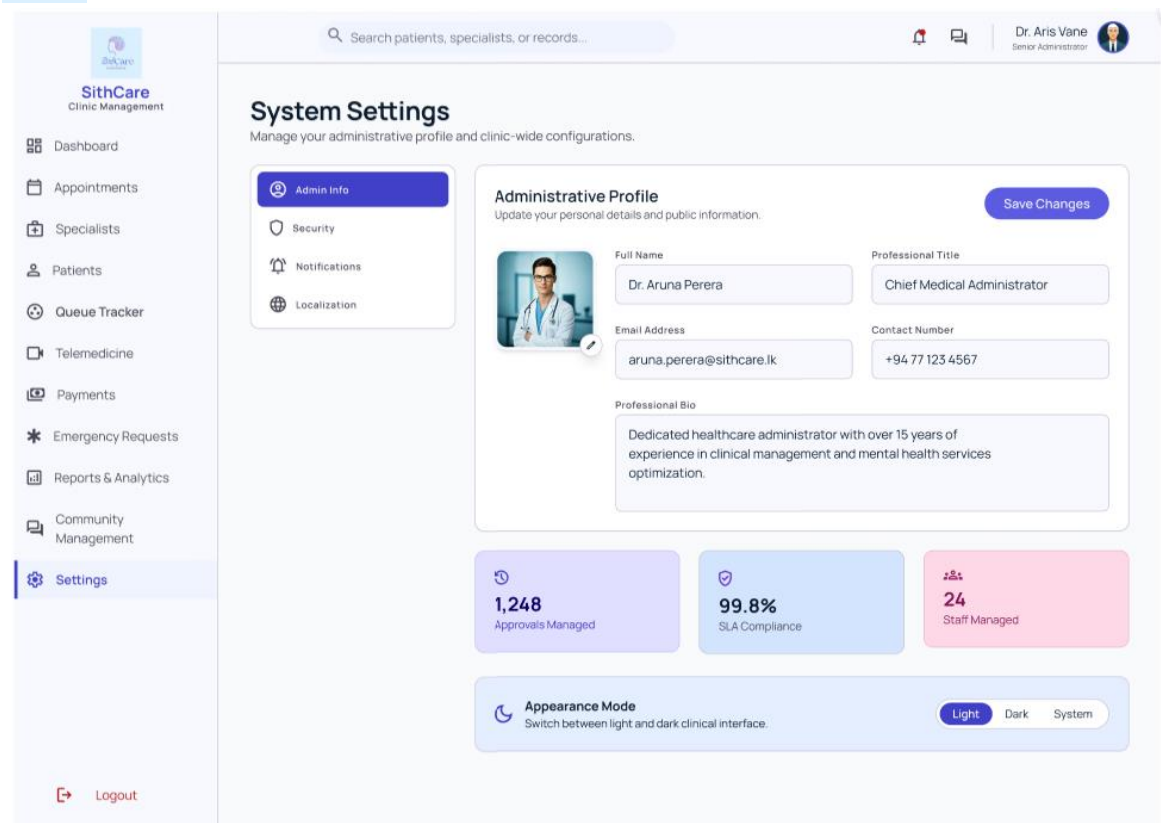


Figure 3.59: Clinic-Admin Dashboard System Settings

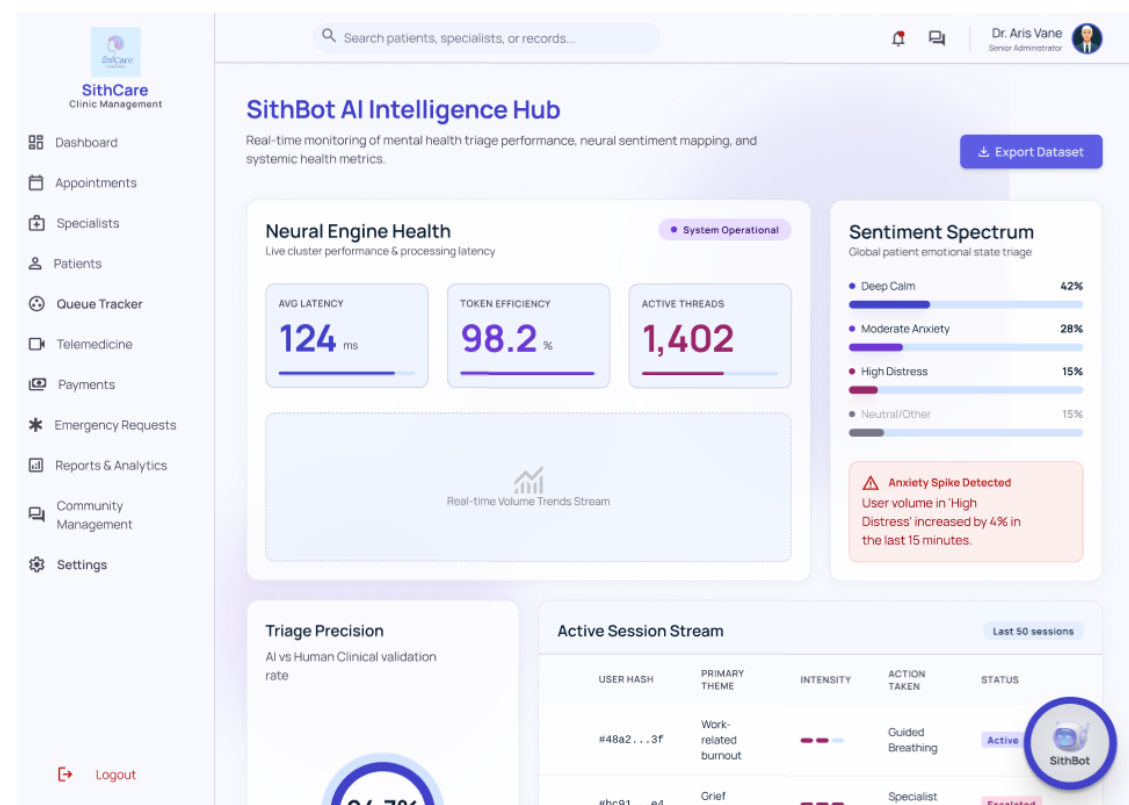


Figure 3.60: Clinic-Admin Dashboard SithBot AI Admin

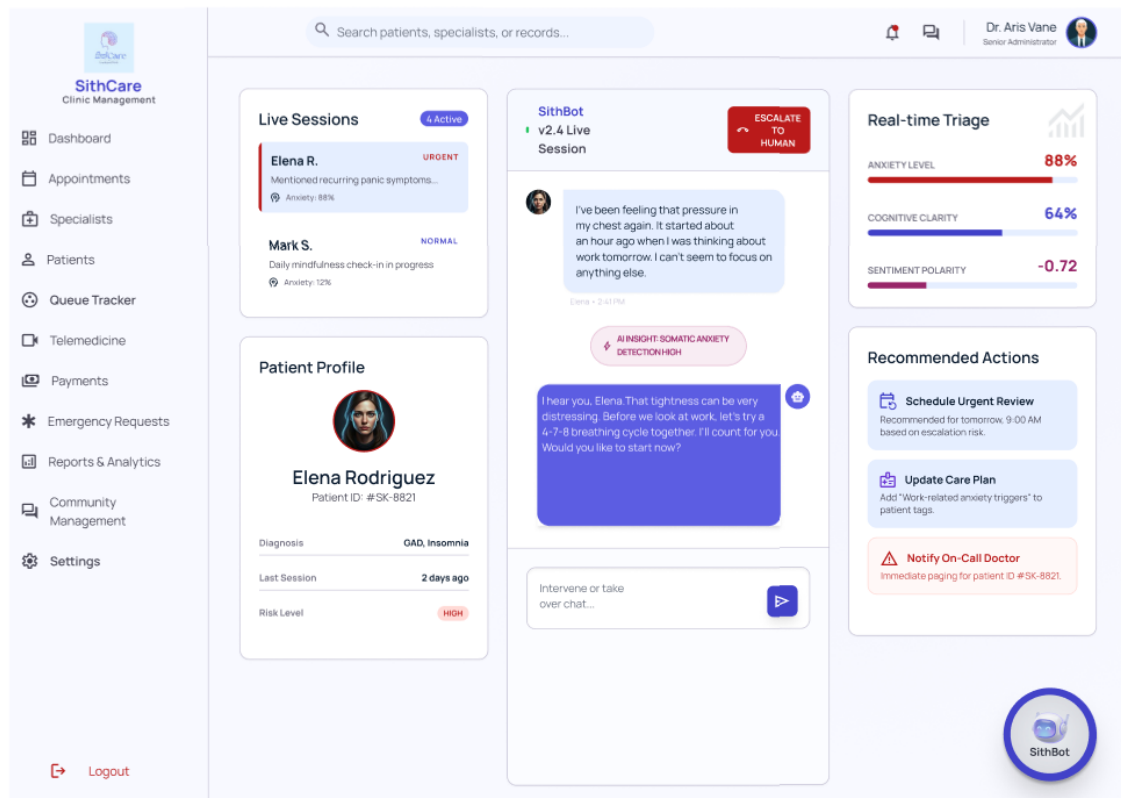


Figure 3.61: Clinic-Admin Dashboard SithBot AI Chat Interface

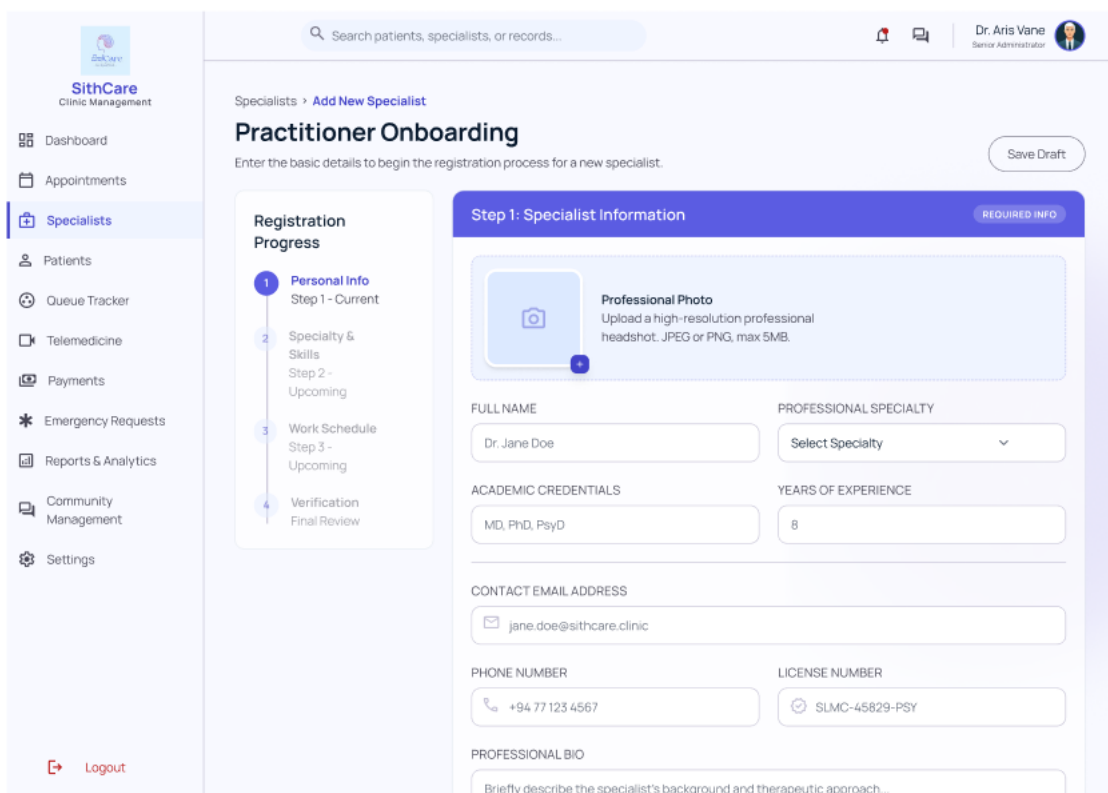


Figure 3.62: Clinic-Admin Dashboard Add Specialist Interface



3.7.5.2. Patients Dashboard

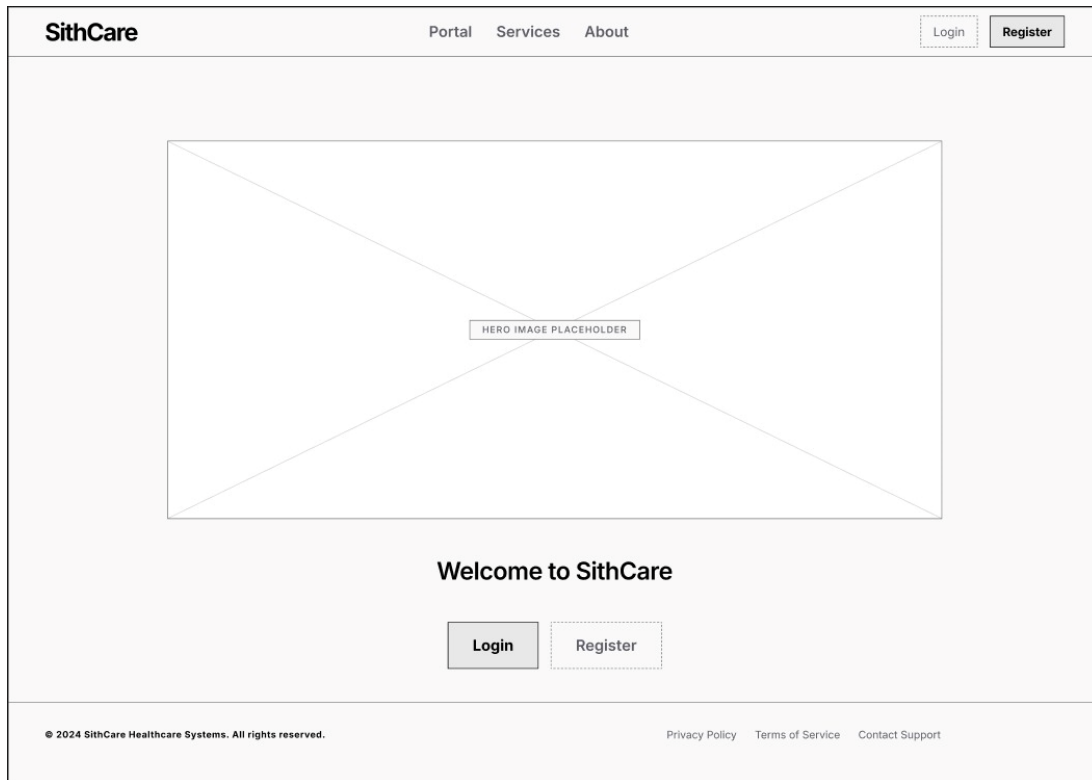


Figure 3.63: Patient Dashboard Landing Page

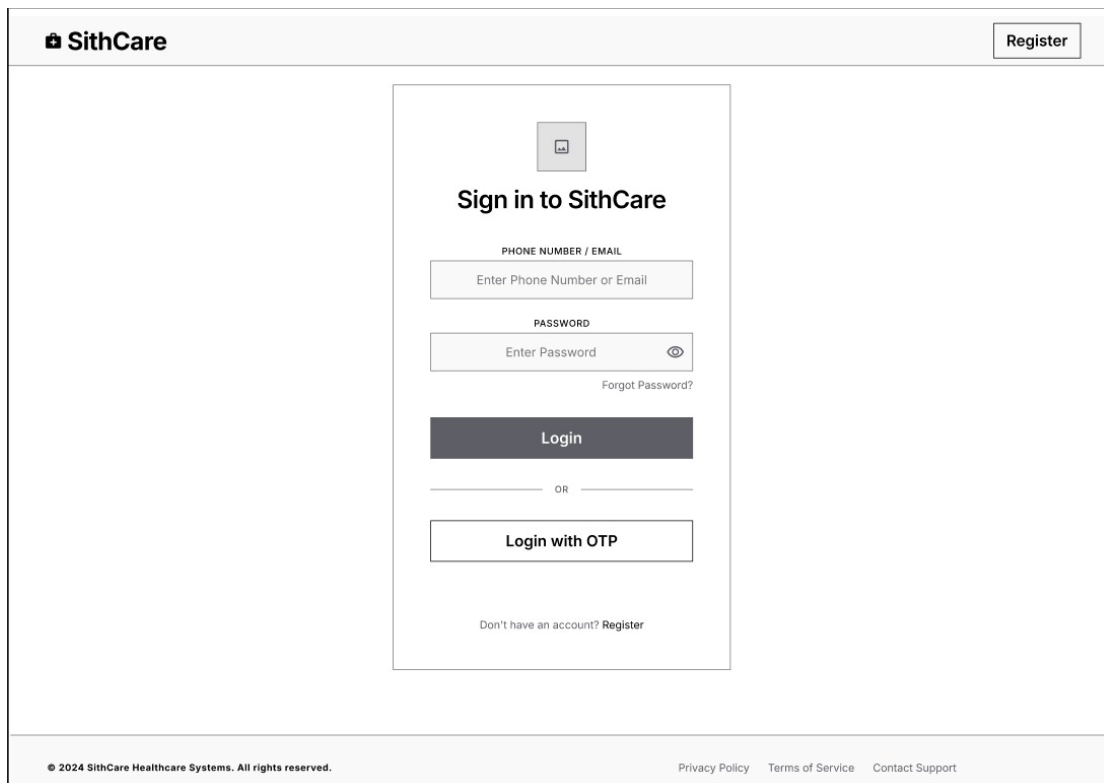


Figure 3.64: Patient Dashboard Login Page



SithCare

LoginRegister

← Create an Account

Create a new structural identity.

FULL NAME

Enter your full name

PHONE NUMBER

Enter phone number

EMAIL ADDRESS (OPTIONAL)

Enter your email

CREATE PASSWORD

Create a password

CONFIRM PASSWORD

Confirm your password

Create Account & Send OTP

Already have an account? Login

SithCare

Privacy PolicyTerms of ServiceContact Support

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Figure 3.65: Patient Dashboard Register Page

←

Verify Your Number

We've sent a 6-digit code to +94 7X XXX XXXX.

Verify & Continue

Didn't receive the code? Resend in 00:28

Figure 3.66: Patient Dashboard OTP Verification Page



SithCare

Welcome to SithCare

Let's set up your experience.

Choose your preferred language

This sets your app language and helps match you with the right therapists.

English

Sinhala

Tamil

Your Privacy Matters

☒ I agree to the Terms of Service and Privacy Policy. *

☐ I consent to sharing my consultation history securely with my booked specialists. *

I agree to the Community Guidelines to participate in anonymous forums.

COMPLETE SETUP

© 2024 SithCare Mental Health Portal. Clinical Wireframe V1.0

[Privacy Policy](#) [Terms of Service](#) [Accessibility](#) [Contact Support](#)

Figure 3.67: Patient Dashboard Complete Setup Page

Patient Portal

Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Hello, [Name]

Here is your health summary for today.

* In a crisis? Get help now.

Emergency Support

Upcoming Consultation Card

Dr. [Name] - Clinical Psychologist

Today, 10:30 AM

Join Video Call

Details

Current Queue: A-10 @Your Token: A-14

Recent Vitals

Heart Rate

72 bpm

Weight

165 lbs

How can we help you today?

Chat with SithBot

Find a Therapist

Anonymous Forum

Video Consultation

Recent Activity

Lab Results Available

View All

Oct 20

Figure 3.68: Patient Dashboard Patient Dashboard Page

117 | Page

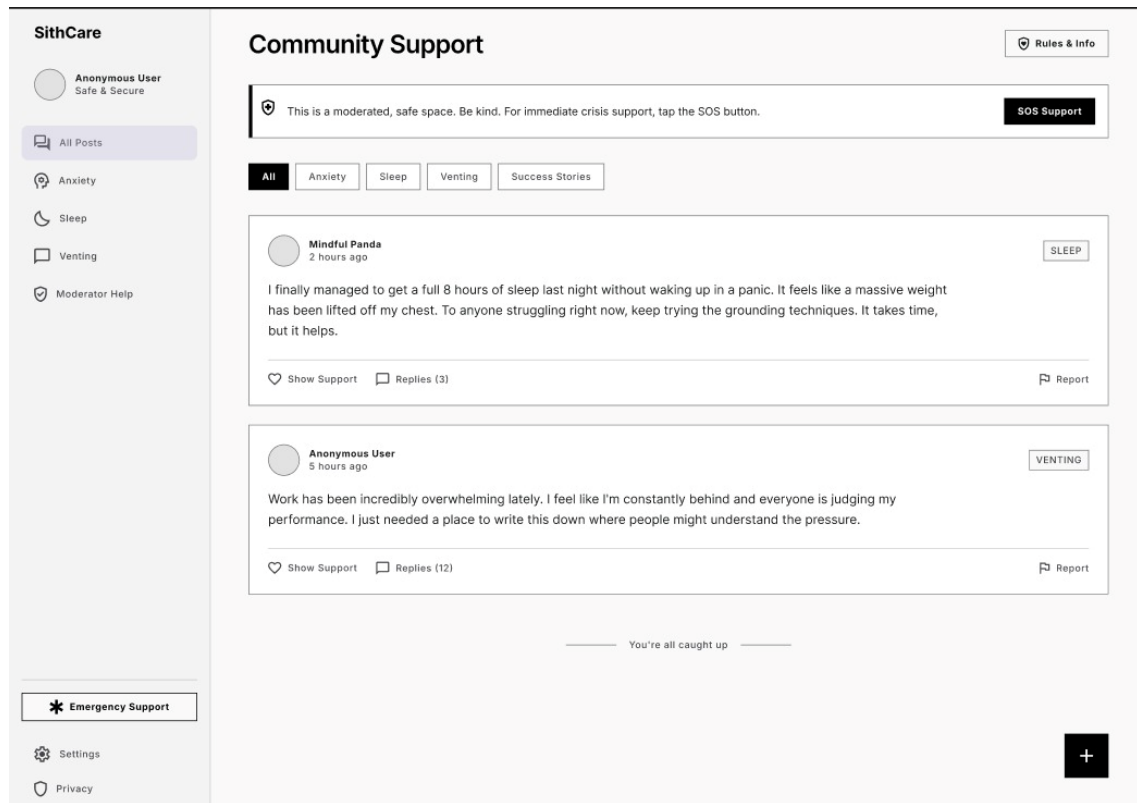


Figure 3.69: Patient Dashboard Anonymous Community Forum Page

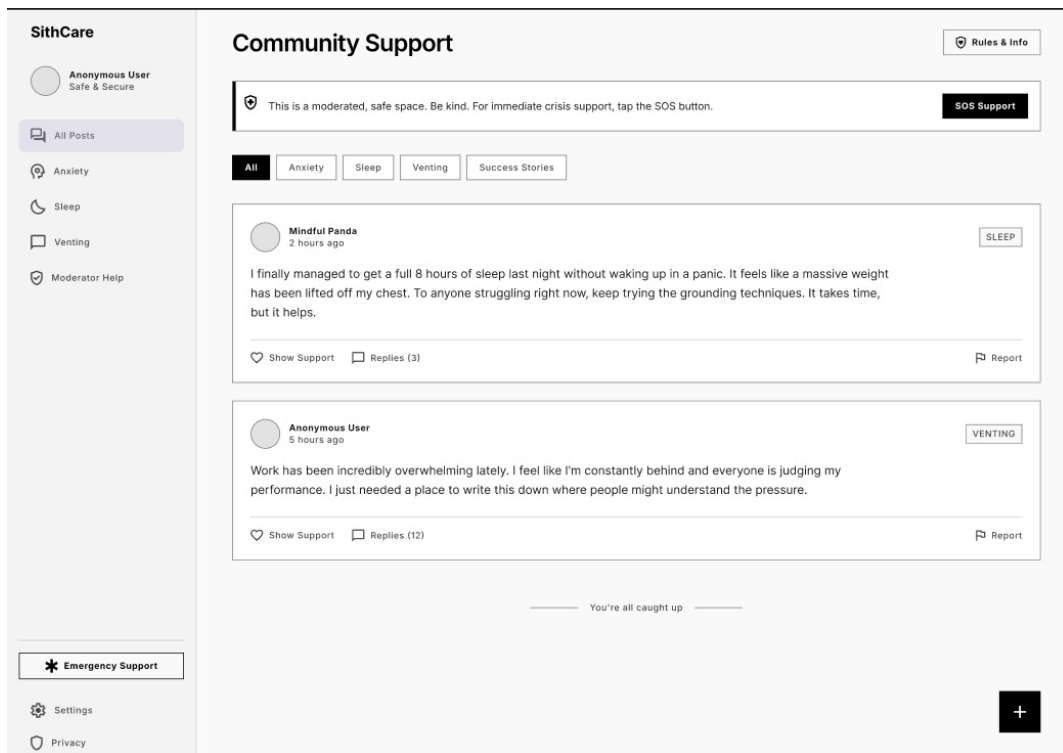


Figure 3.70: Patient Dashboard Community Support Page



The screenshot shows a modal window titled "Share Anonymously" with a close button (X) in the top right corner. Inside the modal, there is a "Category" section with a dropdown menu labeled "Select a topic". Below this is a "Your Message" section with a text input area containing the placeholder text "What's on your mind? Share your experience safely...". Underneath the text area is a small shield icon and a note: "Your real name and profile are hidden. Only moderators can see account details for safety purposes." At the bottom left of the modal is a checkbox labeled "POST ANONYMOUSLY" which is currently checked. At the bottom right is a button labeled "PUBLISH POST" with a right-pointing arrow.

Figure 3.71: Patient Dashboard Share New Anonymous Post Page

The screenshot shows the "SithCare" app interface. At the top, there is a navigation bar with a "Back to Feed" button on the left, the "SithCare" logo in the center, and a menu icon on the right. Below the navigation bar, the main content area displays an anonymous post. The post is from an "Anonymous User" and is categorized under "Topic: Anxiety". It was posted "2h ago". The title of the post is "Struggling with sleep after a long shift". The text of the post reads: "I've been working these 12-hour night shifts lately and it's completely wrecked my sleep schedule. Even when I'm exhausted, I lie in bed for hours just staring at the ceiling. It's starting to affect my mood and I'm feeling constantly on edge. Has anyone dealt with this kind of severe shift-work insomnia? What helped you reset?". Below the post text, there are icons for likes (12) and replies (3). Below the post, there is a section titled "REPLIES". The first reply is from an "Anonymous Peer" posted "1h ago". The text of the reply is: "I feel you. Blackout curtains and a white noise machine saved me when I was doing nights. Also, trying to stick to a consistent wake time even on days off (as painful as it is) helps anchor your circadian rhythm a bit." Below this reply, there is a thumbs-up icon and the number "4". The second reply is from a "Clinical Moderator" (verified) posted "45m ago". The text of the reply is: "Shift work sleep disorder is very common and can significantly impact mental health. While peer advice is great, if this persists for more than a few weeks, consider speaking with an occupational health specialist. In the short term, ensuring your sleep environment is completely dark, cool (around 65°F), and quiet is foundational. Avoid screens for 1 hour before your intended sleep time, even during the day." Below this reply, there is a thumbs-up icon and the number "15". At the bottom of the screen, there is a text input field labeled "Reply anonymously..." and a "Send" button.

Figure 3.72: Patient Dashboard Anonymously Reply Page

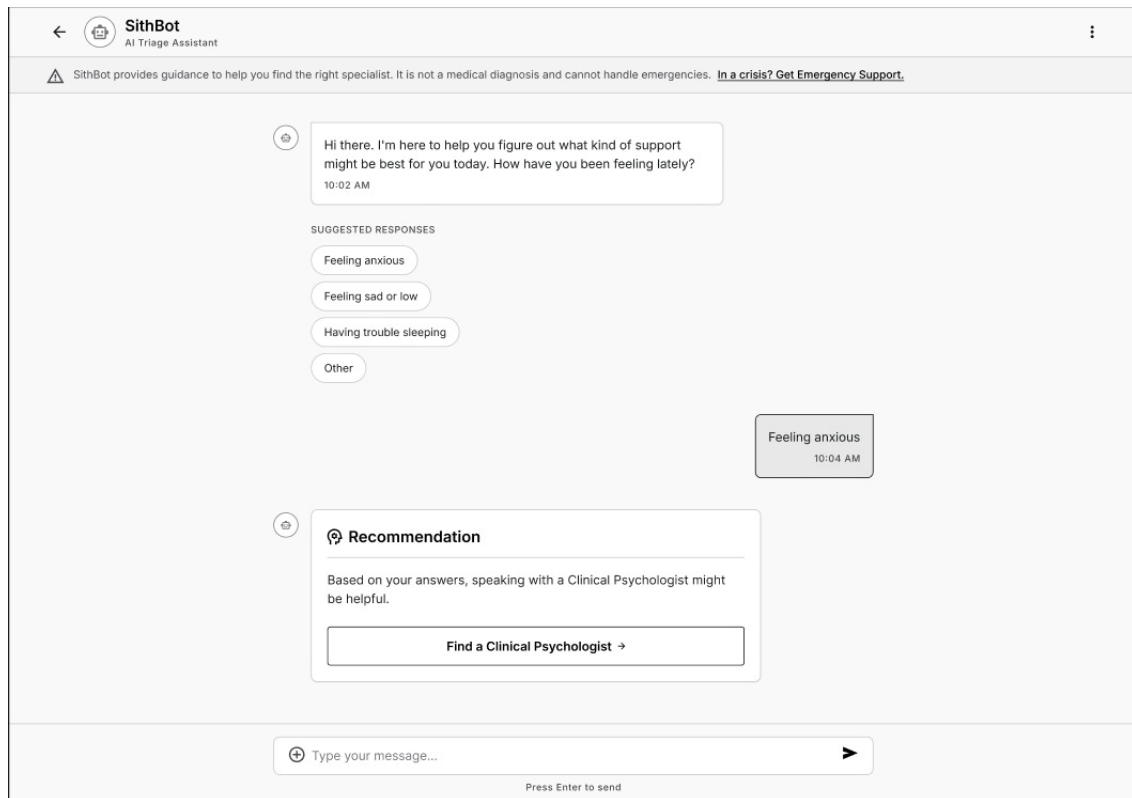


Figure 3.73: Patient Dashboard SithBot Chat Interface

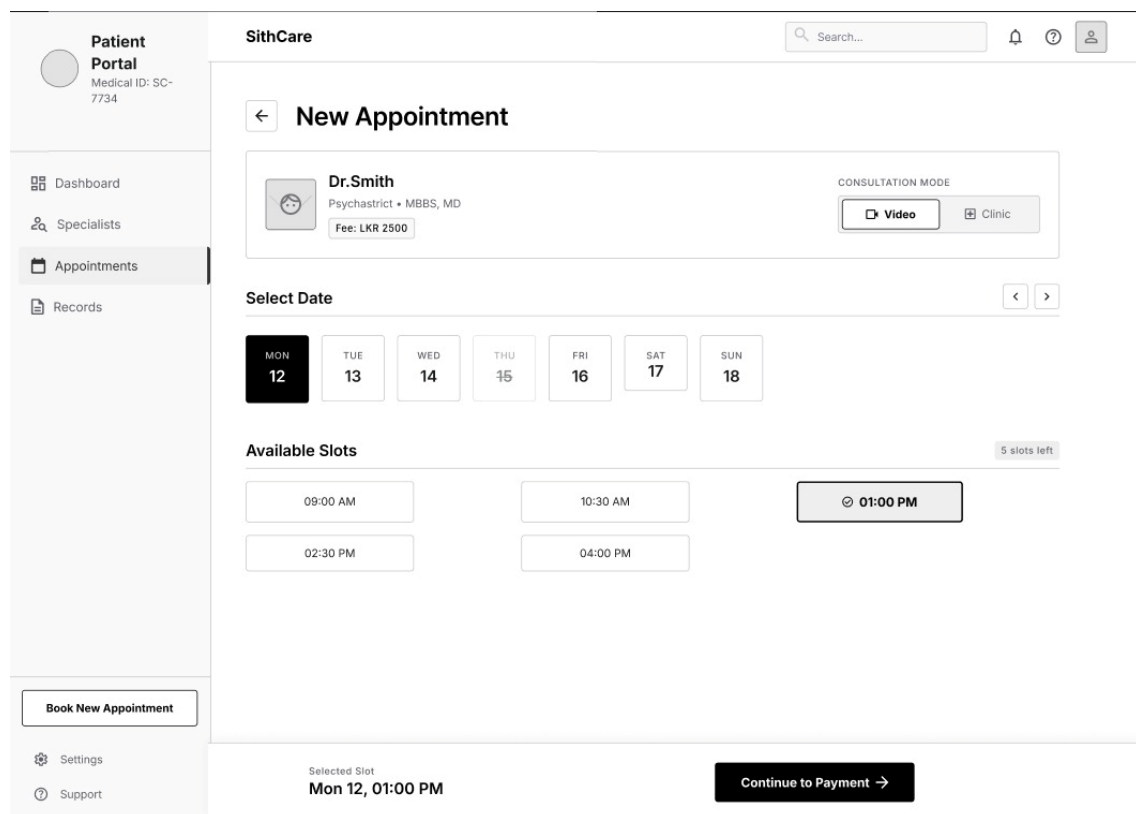


Figure 3.74: Patient Dashboard New Appointment Page



Patient Portal
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Checkout & Payment

Search...

🔔

?

👤

Booking Summary

DATE & TIME

Mon 12 Nov, 10:30 AM

DOCTOR

Dr. Adhithya Jayawardhana

MODE

☒ Video Call

COST BREAKDOWN

Consultation Fee

LKR 2500

Platform Fee

LKR 200

Total Amount

LKR 2700

Choose Payment Method

☒ **Pay Online via PayHere**
Credit/Debit Card, Mobile Wallets

☐ **Pay at Clinic (Cash)**
Unavailable for Video Consultations

☐ I agree to the [cancellation and refund policy](#).

Confirm & Pay LKR 2700

Figure 3.75: Patient Dashboard Checkout and Payment Page

PATIENT PORTAL
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Settings

Support

SithCare

🔔

?

👤

Booking Confirmed!
Your appointment has been successfully scheduled.

DIGITAL RECEIPT

TOKEN

A-14

REFERENCE

#88291

PATIENT

Jane Doe

DATE & TIME

Oct 24, 2023 - 10:00 AM

DEPARTMENT

General Practice

Download PDF Receipt

Add to Calendar

Go to My Dashboard

Figure 3.76: Patient Dashboard Booking Confirmed Page

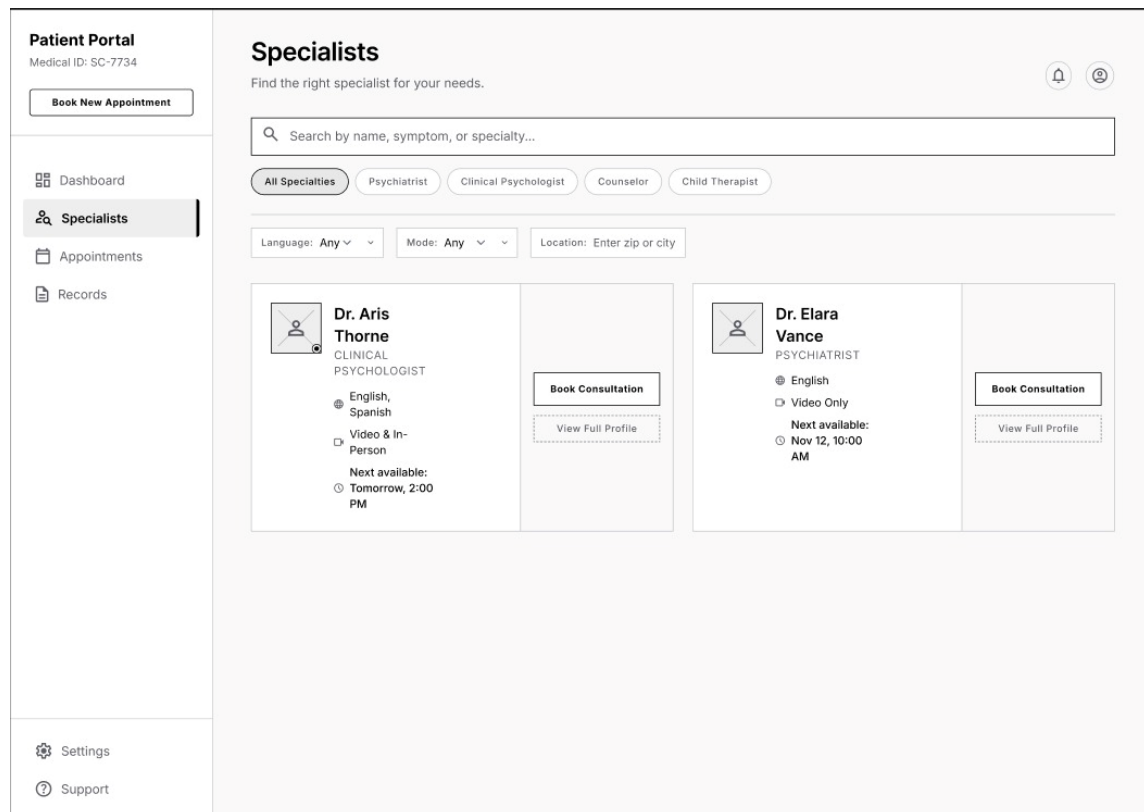


Figure 3.77: Patient Dashboard Specialist Search Page

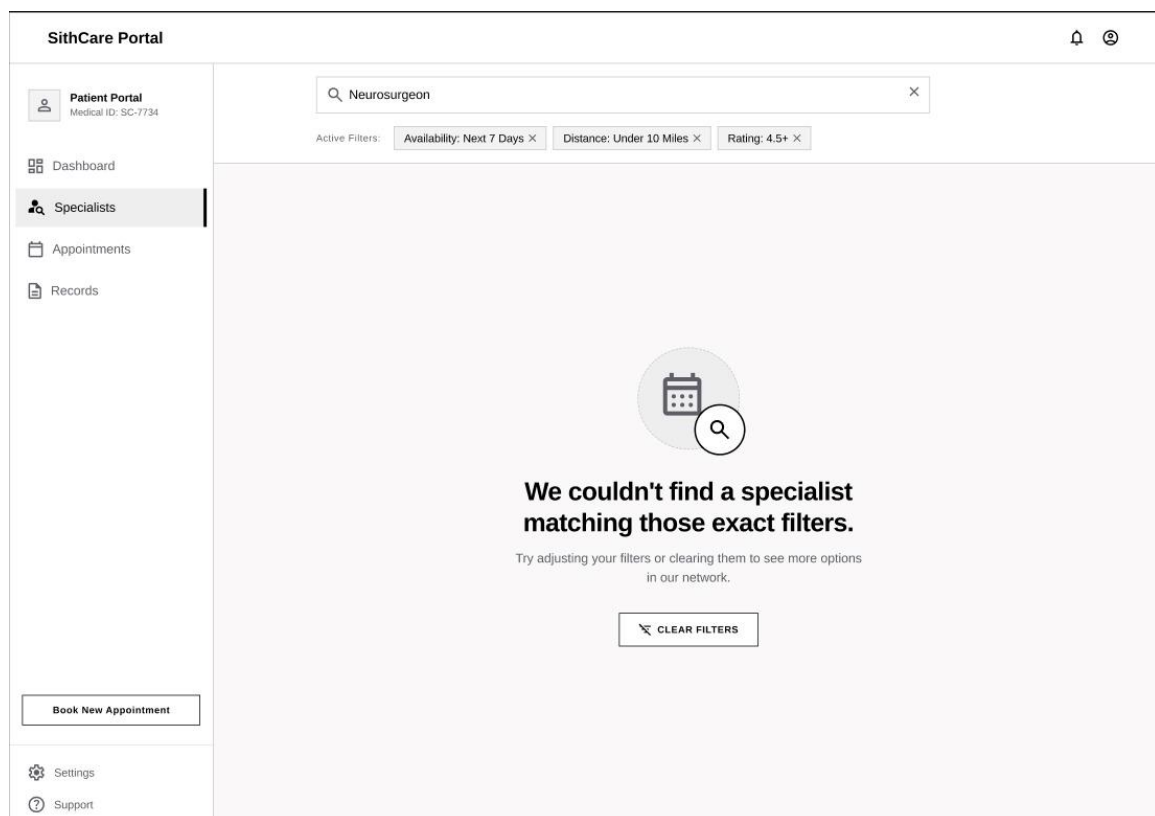


Figure 3.78: Patient Dashboard Specialist Empty Page

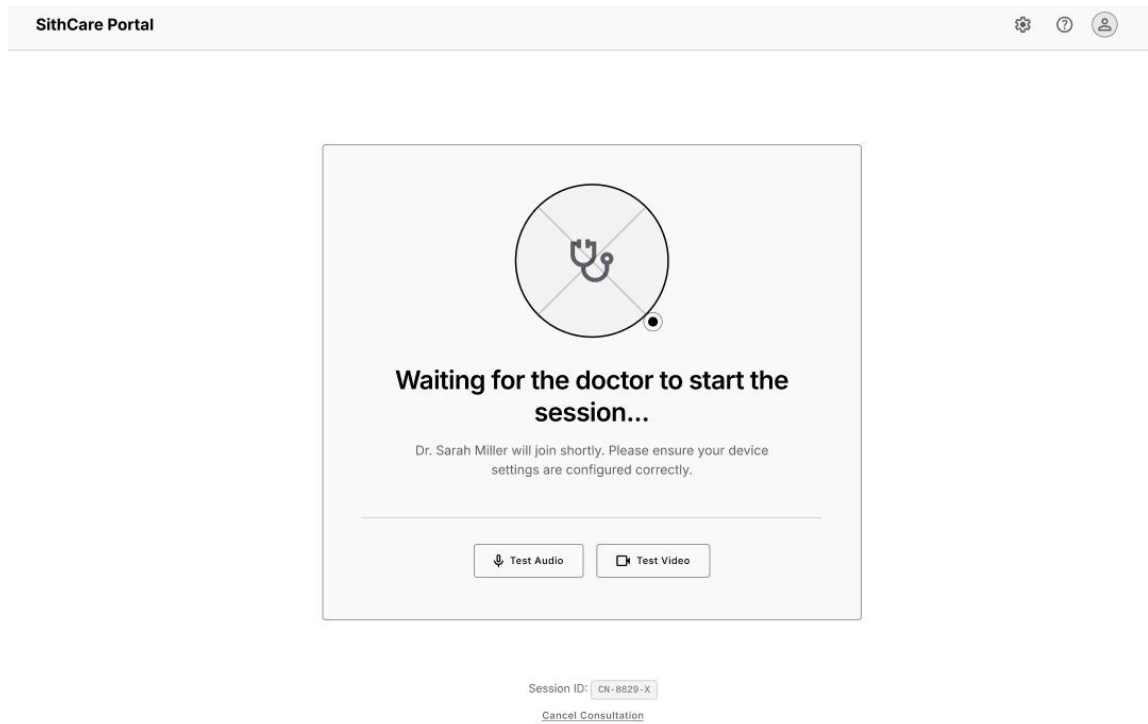


Figure 3.79: Patient Dashboard Video Consultation Waiting Room

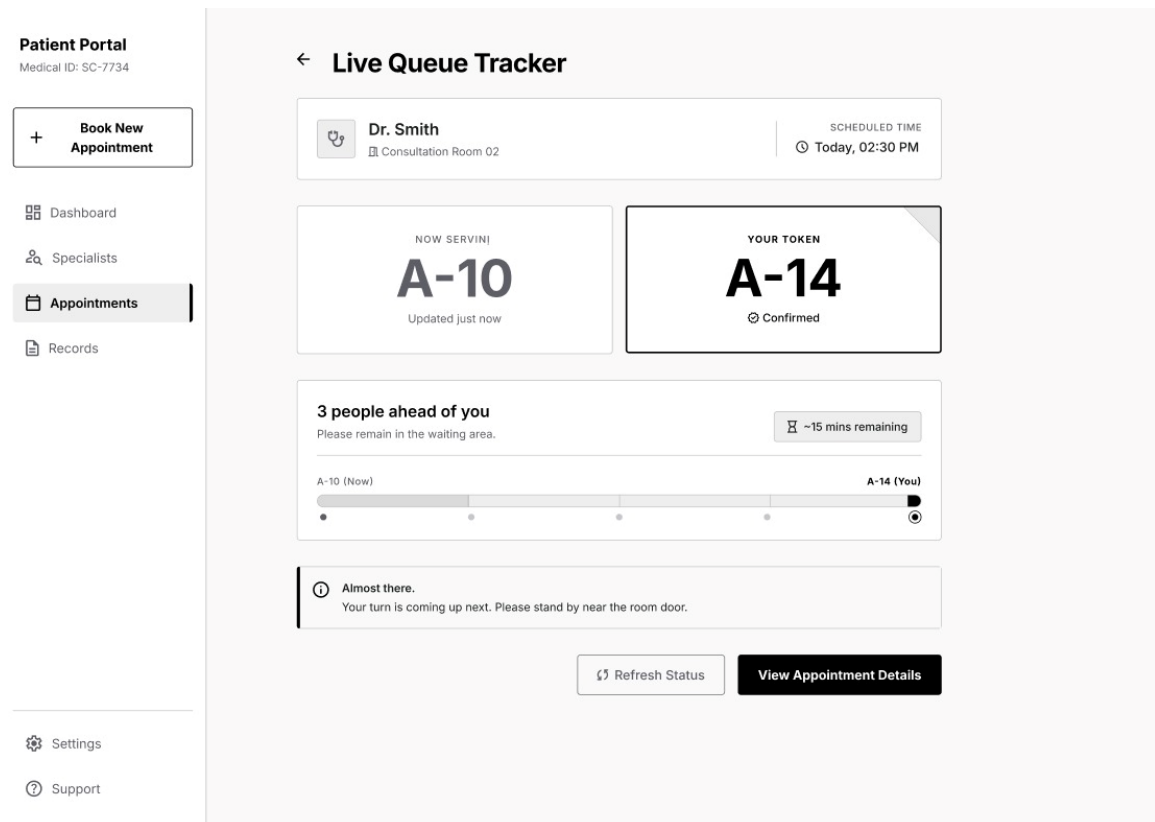


Figure 3.80: Patient Dashboard Live Queue Tracker



Patient Portal
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Settings

Jane Doe | +94 77 XXX XXXX

FORUM ALIAS: MINDFULPANDA

Emergency Contacts

John Doe | Spouse
+94 71 234 5678

+ Add Trusted Contact

Privacy & Data Consent

Share consultation notes with my booked specialists
Required for doctors to view your history

Share my GPS location when I trigger an Emergency
Helps responders find you faster

Share anonymous app usage data to improve SithCare
Optional data collection for analytics

View Full Privacy Policy

Download My Data

App Preferences

App Language
English

Reminders & Alerts

Log Out

Delete Account

Figure 3.81: Patient Dashboard Settings Page

Patient Portal
Medical ID: SC-7734

Dashboard

Specialists

Appointments

Records

Book New Appointment

Settings

Support

Medical Records

Search doctor or date... Upload Existing Record

All Records

Prescriptions

Consultation Notes

Lab Reports

Prescription Oct 24, 2023

Routine Refill

Dr. Sarah Smith • Cardiology

View Online

PDF

Lab Report Sep 12, 2023

Comprehensive Metabolic Panel

CorePath Laboratories

View Online

PDF

Consultation Notes Aug 05, 2023

Annual Physical Summary

Dr. James Wilson • General Practice

View Online

PDF

Figure 3.82: Patient Dashboard Medical Records Page

124 | Page



3.7.5.3. SithCare-Admin Dashboard

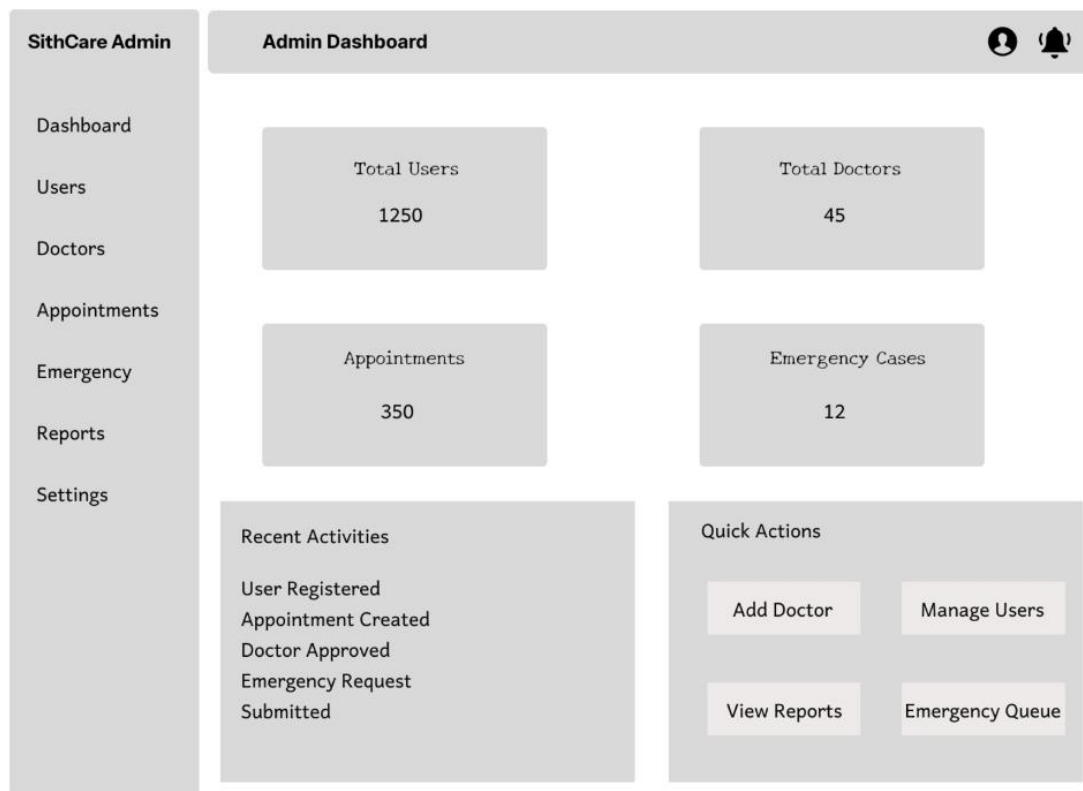


Figure 3.83: SithCare-Admin Dashboard Page

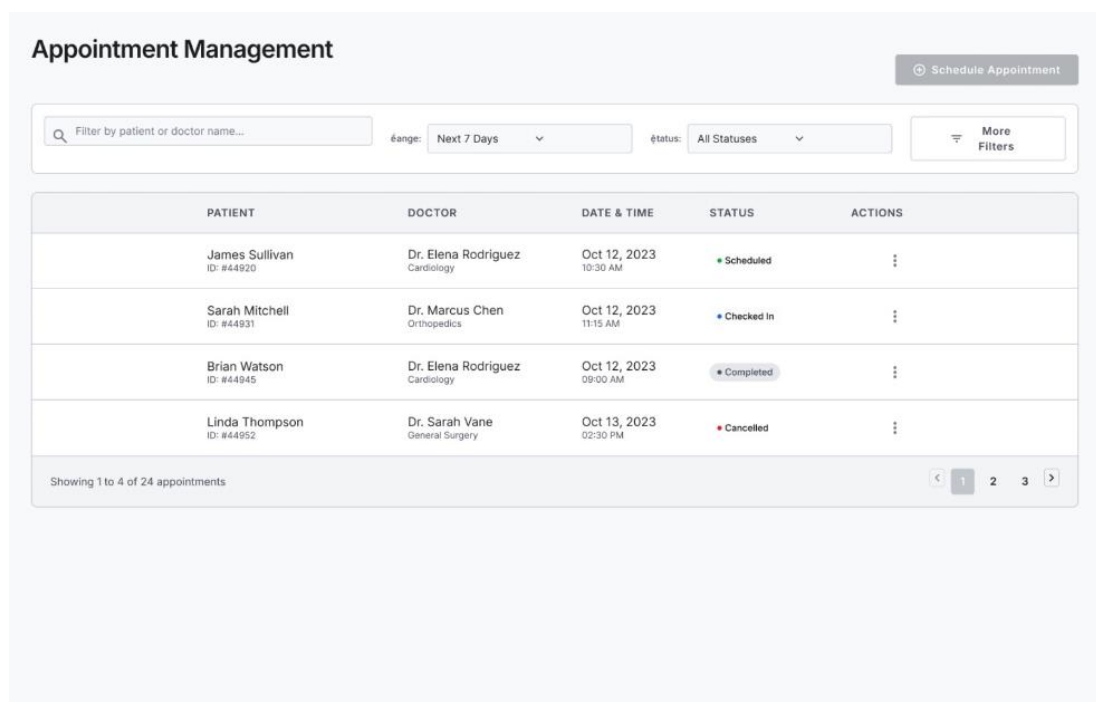


Figure 3.84: SithCare-Admin Dashboard - Appointment Management Page

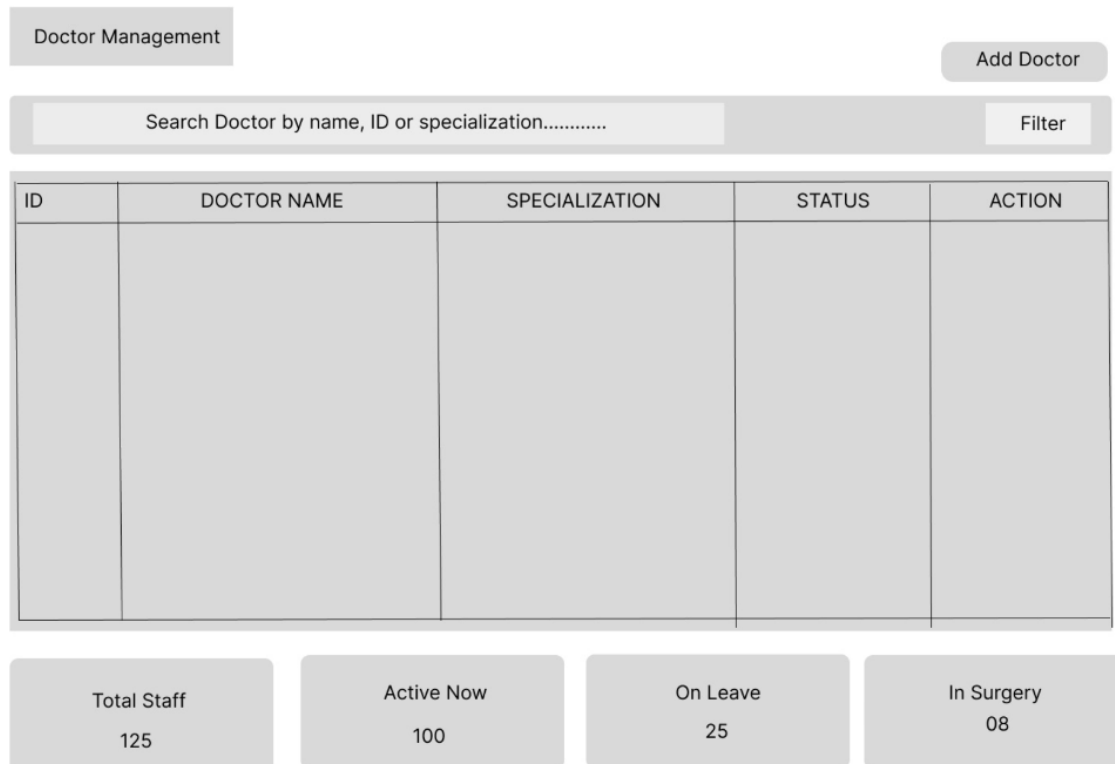


Figure 3.85: SithCare-Admin Dashboard - Doctor Management Page

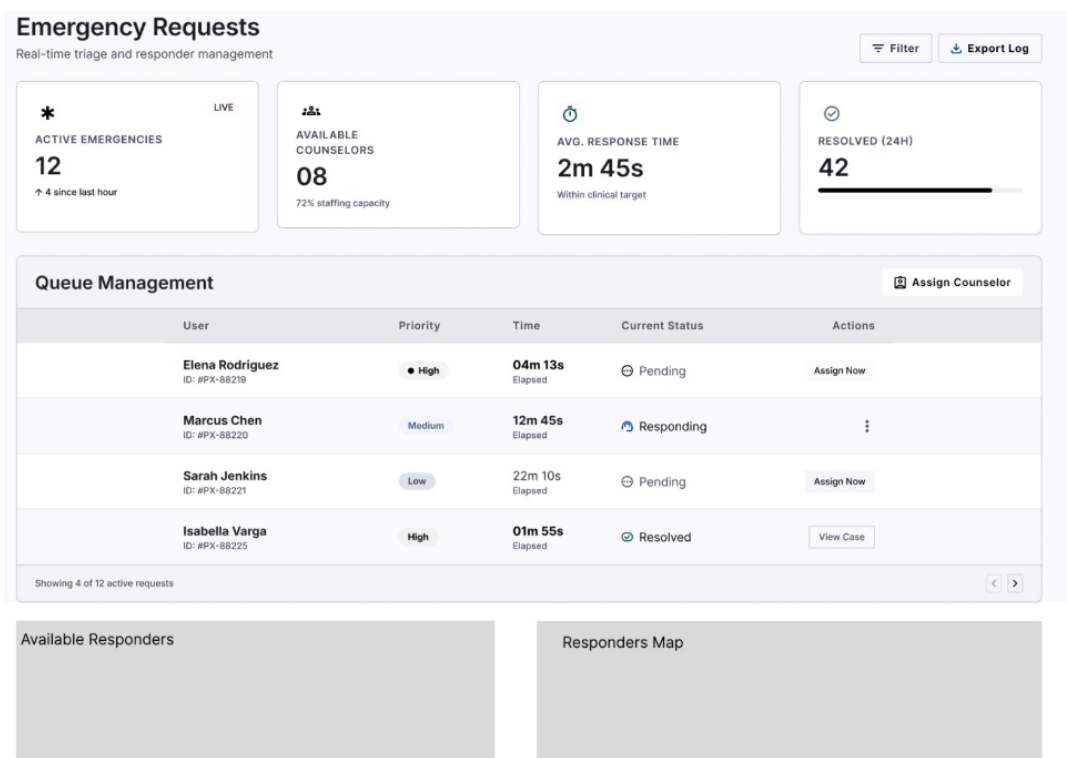


Figure 3.86: SithCare-Admin Dashboard - Emergency Requests Page

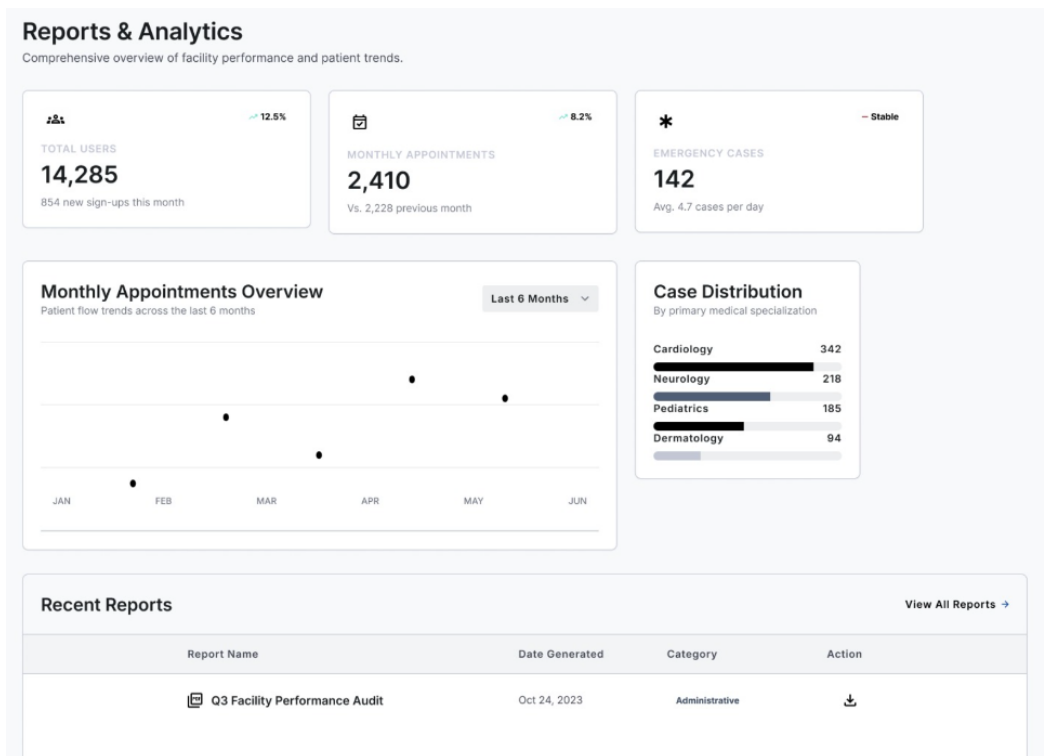


Figure 3.87: SithCare-Admin Dashboard - Reports and Analytics Page

3.7.6. Interface Design

This section defines how SithCare communicates with external services and how the internal modules of the system interact with each other. The interface design supports the role-based web application described in the SRS and follows the layered MVC architecture defined in this SDS.

3.7.6.1. External Interfaces

SithCare uses external services only where they provide clear operational value, such as payments, notifications, telemedicine, maps, and emergency helpline information. All external communication is performed through secure API calls or approved client-side SDK integrations.

External Interface	Purpose	Integration Method	Used By	Design Notes
PayHere Payment Gateway	Online appointment payments, refund handling,	REST API / PayHere Checkout SDK /	Appointment Service, Payment Gateway Adapter, Patient	No card details are stored in SithCare.



	transaction confirmation, and LKR-based payment processing.	Webhook callbacks	Dashboard, Clinic Admin Panel	Payment status is verified using gateway response and webhook validation.
Jitsi Meet API / Video SDK	Telemedicine video consultation rooms for confirmed online appointments.	JavaScript SDK / secure room link generation	Telemedicine Module, Doctor Portal, Patient Dashboard	Room links are generated only for confirmed appointments and are time-limited.
Twilio SMS API / Local SMS Provider	OTP delivery, appointment reminders, queue proximity alerts, and emergency notifications.	HTTPS API request from Notification Service	Authentication Service, Appointment Service, Queue Manager, Emergency Handler	SMS content is sent in the patient's preferred language where possible.
Nodemailer / Email Service	Email OTPs, booking confirmations, password reset messages, appointment reminders, and payment receipts.	SMTP / email service API	Notification Service, Auth Controller, Appointment Service, Payment Service	Email templates avoid exposing sensitive health details in the subject line.
Google Maps API / OpenStreetMap	Displaying clinic locations, directions, and consent-based emergency location support.	Client-side map SDK / location service integration	Patient Dashboard, Emergency Support Module	Location is used only after user permission and explicit consent.
Emergency Helpline Data	Display of Sri Lankan crisis support and emergency contact numbers.	Static configuration maintained by Platform Admin	Emergency Support Module, SithBot	Helpline information is visible even if other non-critical modules fail.

Table 3.2: External Interfaces



3.7.6.2. User Interface Interfaces

The front end is a React Single Page Application. Users interact with SithCare through role-based dashboards and selected modules according to their permissions.

- **Patient Interface:** registration, profile and consent management, specialist search, appointment booking, telemedicine access, payments, queue tracking, community features, SithBot, and emergency support.
- **Doctor / Specialist Interface:** availability management, appointment list, telemedicine session start, patient consultation status updates, and limited access to assigned patient records.
- **Clinic Administrator Interface:** clinic profile management, doctor allocation, appointment approval or rescheduling, queue control, cash payment tracking, and daily appointment monitoring.
- **Platform Administrator Interface:** user and role management, clinic verification, specialist verification, system reports, crisis log review, moderation oversight, and configuration management.
- **Volunteer and Moderator Interfaces:** volunteer listener access, community post review, report handling, and escalation to platform administrators when required.

3.7.6.3. Internal Module Interfaces

Internal communication follows a controller-service-repository structure. The frontend communicates with backend controllers through REST endpoints. Controllers validate requests and call service modules. Service modules apply business rules and use repositories or Mongoose models to access MongoDB Atlas.

Internal Module	Communicates with	Main Interface Methods	Responsibility
Auth Controller	User Controller, Notification Service, User Repository	register(), login(), verifyOTP(), refreshToken(), resetPassword()	Creates secure sessions, verifies OTPs, manages JWT tokens, and routes users to role-based dashboards.



User/Profile Service	Auth Controller, Consent Repository, Emergency Contact Repository	getProfile(), updateProfile(), updateConsent(), updateEmergencyContact()	Maintains profile, preferred language, consent status, and emergency contacts.
Specialist Search Service	Specialist Repository, Clinic Repository, Availability Service	searchSpecialists(), filterByLocation(), getSpecialistProfile()	Returns verified specialists based on category, language, location, consultation type, and availability.
Appointment Service	Availability Service, Payment Service, Notification Service, Queue Manager	createAppointment(), cancelAppointment(), rescheduleAppointment(), updateStatus()	Controls booking, confirmation, cancellation, rescheduling, and appointment state changes.
Availability Service	Doctor Portal, Clinic Admin Panel, Appointment Repository	createSlot(), blockSlot(), updateSchedule(), validateConflict()	Prevents schedule conflicts and exposes available slots to patients.
Payment Service	PayHere Adapter, Appointment Service, Receipt Generator	initiatePayment(), verifyWebhook(), recordCashPayment(), generateReceipt()	Tracks online and cash payments and generates PDF receipts.
Queue Manager	Clinic Admin Panel, Appointment Service, Notification Service	startQueue(), advanceQueue(), pauseQueue(), resetQueue()	Maintains queue tokens and notifies patients when they are close to their turn.
Telemedicine Service	Jitsi Adapter, Appointment Service	generateRoomLink(), validateJoinAccess(), expireRoomLink()	Creates secure video consultation access only for valid online appointments.
Emergency Handler	Notification Service, Location Service,	requestEmergencySupport(), confirmConsent(),	Handles emergency flow, consent-based location sharing,



	Crisis Log Repository, SithBot	logCrisisEvent(), showHelplines()	fallback helpline display, and crisis logging.
Community Service	Moderator Panel, User Repository, Content Filter	createPost(), hidePost(), reportPost(), moderatePost()	Supports pseudonymous posting and moderation without exposing real identities.
SithBot Service	Recommendation Engine, Emergency Handler, Specialist Search Service	startTriage(), detectCrisisKeywords(), recommendCategory()	Provides triage guidance without diagnosis and redirects high-risk input to emergency support.

Table 3.3: Internal Module Interfaces

3.7.6.4. API Design Principles

The backend exposes RESTful JSON APIs over HTTPS. API endpoints are grouped by functional domains to improve maintainability and testing.

- All protected API requests must include a valid JWT access token in the Authorization header.
- Request and response payloads use JSON format with consistent field names and error structures.
- Input validation is performed at controller level before business logic execution.
- Sensitive operations such as payments, emergency actions, role changes, and consent updates are logged for audit purposes.
- API responses must not expose stack traces, database errors, passwords, tokens, or unnecessary patient health data.

API Group	Main Purpose
/api/auth	Registration, login, OTP verification, password reset, token refresh



/api/users	Profile update, consent management, emergency contact management, preferred language update
/api/specialists	Specialist search, specialist profile view, availability summary
/api/appointments	Appointment creation, cancellation, rescheduling, approval, status tracking
/api/payments	Payment initiation, webhook verification, cash payment record, receipt generation
/api/queue	Daily queue start, queue advance, pause, reset, patient queue status
/api/telemedicine	Video room link generation, consultation join validation, room expiry
/api/emergency	Emergency confirmation, location consent, helpline display, crisis event logging
/api/community	Anonymous post creation, reply, report, moderation actions
/api/sithbot	Triage conversation, crisis keyword detection, specialist category recommendation

Table 3.4: API main purposes

3.7.6.5. Hardware and Device Interfaces

SithCare does not require dedicated medical hardware for the MVP. It uses standard browser-accessible device features when required.

- Camera and microphone are accessed only during telemedicine sessions after browser permission is granted.
- Device location is requested only during emergency support and only after explicit user confirmation.
- Mobile and desktop browsers are supported through responsive React and Tailwind CSS layouts.
- If camera, microphone, or location permissions are denied, the system provides fallback instructions or alternative contact options.

3.7.7. Security Design

SithCare handles sensitive mental health-related information. Therefore, the security design focuses on confidentiality, integrity, availability, accountability, and user consent. The



design also supports the security requirements defined in the SRS, including TLS 1.3, AES-256 encryption, JWT expiry rules, RBAC, pseudonymous community identities, and audit logging.

3.7.7.1. Authentication Strategy

Authentication verifies the identity of users before they access protected parts of the system. SithCare uses a token-based authentication mechanism suitable for a web-based MERN application.

- Users register using email or phone number and verify the account using OTP.
- Passwords are stored only as salted hashes using bcrypt. Plain-text passwords are never stored or logged.
- JWT access tokens expire after 15 minutes. Refresh tokens expire after 7 days and are used to request new access tokens securely.
- The system automatically logs users out after 30 minutes of inactivity where applicable.
- Clinic administrators and platform administrators may use optional multi-factor authentication for stronger protection.
- After repeated failed login attempts, the account is temporarily locked to reduce brute-force attacks.

3.7.7.2. Authorization Strategy

Authorization controls what each authenticated user can access or perform. SithCare implements Role-Based Access Control (RBAC) with clear separation between patient, doctor, clinic administrator, platform administrator, volunteer, and moderator functions.

Role	Allowed Access
Patient	Own profile, consent settings, own appointments, own payment receipts, own queue status, community posts under pseudonym, SithBot, emergency support.
Doctor / Specialist	Own availability, assigned appointments, assigned patient consultation information, telemedicine session controls.
Clinic Administrator	Clinic branch details, doctor schedules, appointment approval, queue control, payment tracking, clinic reports.



Platform Administrator	User and role management, clinic onboarding, specialist verification, crisis log review, compliance reports, system settings.
Volunteer Listener	Volunteer session dashboard and assigned listener sessions only. No clinical records or diagnosis features.
Community Moderator	Reported community content, moderation actions, escalation tools. No access to private patient medical details.

Table 3.5: Authorization Strategy

- Every protected endpoint checks the user role before executing business logic.
- Patients can access only their own records. Doctors can access only records related to assigned appointments.
- Administrative actions such as role changes, specialist verification, clinic changes, and crisis log review are audit logged.
- Community posts display pseudonymous identities and never publicly expose real usernames.

3.7.7.3. Data Protection Design

The system protects data throughout its lifecycle: collection, transmission, storage, processing, access, and deletion/minimization.

- Data in transit is protected using HTTPS with TLS 1.3 between browser, frontend hosting, backend APIs, and external services where supported.
- Patient health-related data at rest is protected using AES-256 encryption through database/storage configuration and encryption controls.
- Payment card data is not stored in SithCare. Card handling is delegated to PayHere.
- Consent records are stored with timestamp, user identifier, action, and version of consent text where applicable.
- Emergency location sharing is session-based and requires explicit confirmation each time.
- Backup data is protected using the same access control and encryption policies as production data.
- Logs must avoid storing passwords, OTP values, full JWT tokens, card information, and unnecessary health details.



3.7.7.4. Input Validation and Output Encoding Strategy

Input validation is applied before any data is processed or stored. Validation occurs both in the front end for user experience and in the backend for security enforcement.

- Email, phone number, date of birth, password, appointment time, payment amount, and file uploads are validated against strict rules.
- User-generated community content is checked for harmful content, personal identifying information, and crisis keywords before publishing.
- File uploads such as profile photos are limited by file type and size. Only JPG and PNG formats are accepted for profile images.
- MongoDB queries use controlled model methods and do not directly accept untrusted query objects from users.
- All displayed user-generated content is encoded to prevent script injection in the browser.

3.7.7.5. OWASP Top 10 Considerations

OWASP Risk Area	SithCare Design Response
Broken Access Control	RBAC middleware, ownership checks, route protection, admin action logging, and test cases for each user role.
Cryptographic Failures	TLS 1.3 for transit, AES-256 for stored sensitive data, bcrypt for passwords, and no storage of payment card data.
Injection	Backend validation, Mongoose schema validation, parameterised model access, sanitisation of search filters and community content.
Insecure Design	Security considered in architecture through controller validation, service isolation, audit logging, and emergency fallback design.
Security Misconfiguration	Environment variables for secrets, secure CORS configuration, disabled debug output in production, and secure HTTP headers.
Vulnerable and Outdated Components	Dependency review, package updates, GitHub security alerts, and controlled use of third-party SDKs.
Identification and Authentication Failures	OTP verification, strong password rules, token expiry, refresh token control, MFA for privileged roles, and account lockout.



Software and Data Integrity Failures	GitHub pull requests, code review, CI checks, controlled deployment pipeline, and verification of PayHere webhook responses.
Security Logging and Monitoring Failures	Logging for authentication events, role changes, payment status, emergency actions, failed access attempts, and moderation actions.
Server-Side Request Forgery	Backend does not fetch arbitrary user-provided URLs; external service endpoints are allow-listed and configured through environment variables.

Table 3.6: OWASP Top 10 Considerations

3.7.7.6. Security Logging and Monitoring

Security-related events are logged to support auditing, incident review, and troubleshooting.

- Authentication events: login, logout, failed login, OTP verification, password reset, and account lockout.
- Authorisation events: denied access attempts, role changes, admin actions, and specialist verification changes.
- Healthcare-sensitive events: consent update, emergency support activation, crisis event logging, and location-sharing confirmation.
- Payment events: payment initiation, webhook confirmation, payment failure, cash payment record, and refund status.
- Community events: post creation, report, hide, delete, escalation, and moderation decision.

3.7.8. Non-Functional Design Decisions

This section explains how the SithCare design addresses the non-functional requirements defined in the SRS. Each design decision is linked to the relevant quality attribute so that implementation and testing can be aligned with the original requirements.

3.7.8.1. Performance Design Decisions

NFR ID	Requirement	Design Decision
NFR-P-01	Page load time ≤ 3 seconds for 95% of requests on 4G LTE.	React with Vite is used for fast frontend builds. Static assets are hosted through Vercel/Netlify CDN. Tailwind CSS keeps production CSS lightweight. Images should be compressed before deployment.



NFR-P-02	Appointment booking transaction ≤ 5 seconds excluding PayHere redirect.	Booking logic is handled through a dedicated Appointment Service with simple database operations, slot reservation, and asynchronous notification triggering.
NFR-P-03	Support at least 500 concurrent users.	Node.js non-blocking I/O, stateless REST APIs, and cloud hosting allow horizontal scaling of the backend when traffic increases.
NFR-P-04	Queue updates visible within 60 seconds.	Queue Manager stores the current queue state centrally and patient views refresh automatically within the defined interval. SMS/push proximity notifications are triggered when the patient is near their turn.
NFR-P-05	SithBot response time ≤ 2 seconds for 90% of queries.	SithBot uses a guided triage flow with limited question sets and lightweight recommendation logic for MVP performance.

Table 3.7: Performance Design Decisions

3.7.8.2. Security Design Decisions

NFR ID	Requirement	Design Decision
NFR-S-01	Encrypt data in transit using TLS 1.3.	All frontend-backend and external API communication uses HTTPS. Hosting platforms must provide valid SSL certificates.
NFR-S-02	Encrypt patient health data at rest using AES-256.	MongoDB Atlas/database storage encryption is used for stored sensitive data and backups.
NFR-S-03	JWT expiry: access token 15 minutes, refresh token 7 days.	Auth Controller issues short-lived access tokens and validates refresh tokens before issuing new sessions.
NFR-S-04	Implement OWASP Top 10 mitigations.	Validation, sanitisation, RBAC middleware, rate limiting, secure headers, dependency review, and audit logging are included in the backend design.
NFR-S-05	Patient health/session records restricted to authoring doctor and Platform Administrator.	Ownership checks and role checks are applied before medical/session information is returned by the API.



NFR-S-06	Community posts use pseudonymous IDs.	Community Service stores and displays pseudonymous identifiers instead of real names for forum interactions.
NFR-S-07	Payment card data not stored by SithCare.	PayHere handles card entry and processing. SithCare stores only transaction reference, amount, status, and receipt metadata.
NFR-S-08	Authentication events logged with timestamp and IP address.	Auth events are written to the audit log for security review and incident analysis.

Table 3.8: Security Design Decisions

3.7.8.3. Usability Design Decisions

NFR ID	Requirement	Design Decision
NFR-U-01	Achieve SUS score ≥ 70 .	The interface uses simple role-based dashboards, clear navigation, plain-language labels, and guided workflows for booking, payment, emergency support, and SithBot.
NFR-U-02	New users complete register, find doctor, and book appointment within 5 minutes.	The patient flow is designed as a step-by-step process: register/login, search specialist, select slot, choose payment, and confirm booking.
NFR-U-03	Adult Mode with minimum 18px font and single-level navigation.	Adult Mode provides simplified navigation, larger text, reduced feature set, and direct access to booking and emergency support.
NFR-U-04	WCAG 2.1 Level AA accessibility.	UI design should include sufficient colour contrast, keyboard navigation, semantic labels, screen-reader-friendly forms, and accessibility testing using tools such as axe or WAVE.
NFR-U-05	Plain-language error messages in selected language.	Frontend validation and backend error responses are mapped to user-friendly messages in English, Sinhala, and Tamil where supported.

Table 3.9: Usability Design Decisions



3.7.8.4. Reliability Design Decisions

NFR ID	Requirement	Design Decision
NFR-R-01	System availability $\geq 99.5\%$ excluding scheduled maintenance.	Frontend and backend are deployed on cloud hosting platforms with uptime monitoring. Critical errors are logged and monitored.
NFR-R-02	Maintenance not during 08:00–22:00 SLST.	Deployment and maintenance planning avoids peak patient and clinic operating hours.
NFR-R-03	Automated database backups every 6 hours; RPO = 6 hours.	MongoDB Atlas backup scheduling or equivalent backup strategy is used. Restore testing is planned during system testing.
NFR-R-04	Emergency support available even if non-critical modules fail.	Emergency helpline numbers are stored as static configuration and shown as fallback. Emergency UI is kept independent from optional modules such as community and ratings.

Table 3.10: Reliability Design Decisions

3.7.8.5. Scalability Design Decisions

NFR ID	Requirement	Design Decision
NFR-SC-01	Application layer supports horizontal scaling without code changes.	The backend is designed as stateless REST APIs. Session state is maintained using JWT rather than server memory.
NFR-SC-02	Support up to 200 clinics without schema migration.	MongoDB document-based modelling supports flexible clinic, doctor, availability, and appointment records. Clinic identifiers separate tenant-related data.
NFR-SC-03	Handle $5\times$ traffic spike during public health campaigns.	Frontend CDN delivery, Node.js backend scaling, database indexing, and modular service boundaries support increased traffic.

Table 3.11: Scalability Design Decisions



3.7.8.6. Maintainability Design Decisions

NFR ID	Requirement	Design Decision
NFR-M-01	Consistent coding standards documented in CONTRIBUTING.md.	The project uses GitHub, pull requests, code reviews, linting, and shared coding guidelines.
NFR-M-02	Minimum 70% backend unit test coverage.	Backend services are designed as separate modules, making them easier to test independently.
NFR-M-03	All API endpoints documented using OpenAPI 3.0.	REST endpoints are grouped by domain and documented with request/response schemas for frontend-backend coordination.
NFR-M-04	No hardcoded secrets in source code.	Environment variables are used for JWT secrets, database URLs, PayHere keys, SMS/email credentials, and map API keys.
NFR-M-05	All code changes through pull requests with peer review.	GitHub branch protection and PR workflow ensure review, traceability, and controlled merging.

Table 3.12: Maintainability Design Decisions

3.7.8.7. Summary

The selected design provides a practical and secure foundation for SithCare. The layered MVC architecture, MERN stack, role-based access control, encrypted communication, modular service boundaries, and cloud deployment strategy collectively support the SRS requirements for performance, security, usability, reliability, scalability, and maintainability. These decisions also keep the system feasible for the academic project timeline while allowing future expansion into more advanced healthcare features.



4. CONCLUSION

4.1. Objectives Met

- The SithCare project has successfully achieved all five stated objectives. Comprehensive requirements engineering was conducted through structured interviews with five mental health patients, an online questionnaire with 150 respondents, six specialist interviews, and document analysis of WHO Mental Health Atlas 2024, National Mental Health Policy 2020-2030, and Digital Health Blueprint 2023.
- A professional-grade Software Requirements Specification was produced, documenting twelve use cases with complete descriptions, fifty-two functional requirements with MoSCoW prioritisation, twenty non-functional requirements, and a feasibility assessment with alternative solution analysis. All requirements are traceable to their source and include verification methods.
- A detailed Software Design Specification was developed specifying a Layered MVC architecture with MERN stack implementation, complete class diagram, twelve sequence diagrams, five activity diagrams, twelve state transition diagrams, entity-relationship diagram, UI wireframes for all four dashboard interfaces, and security design addressing all OWASP Top 10 risk areas.
- The project was managed using GitHub Projects with four sprint milestones, epic labels, structured issue backlog, and story point estimation. All artefacts committed to the GitHub organisation repository with version control, branch protection, and conventional commit messages.
- This Final Design Report consolidates all analysis and design artefacts into a single document conforming to the SENG 31242 report specification, providing a complete handoff package for the SENG 34213 development phase.

4.2. Limitations

- Design assumes free-tier cloud services with limitations on bandwidth, storage, and concurrent connections.
- Telemedicine quality depends on Jitsi SDK performance and network conditions in rural Sri Lanka.
- AI chatbot (SithBot) provides triage guidance only and cannot provide clinical diagnoses.
- Emergency support requires active internet; static helplines serve as fallback only.



- Payment processing limited to PayHere gateway; international methods not supported initially.
- No government EHR integration, which may limit continuity of care across providers.

4.3. Future Work

- Native Android and iOS mobile applications for improved accessibility.
- Advanced AI capabilities including NLP for SithBot and sentiment analysis for content moderation.
- Mood tracking and progress monitoring for longitudinal mental health management.
- Comprehensive analytics dashboard with predictive insights for resource allocation.
- Integration with government EHR systems through Health Information Exchange standards.
- Insurance claim processing and fundraising automation.
- Usability testing with target user groups to validate SUS score and refine Adult Mode.
- Performance testing under realistic Sri Lankan network conditions.



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Appendix A: Risk and Mitigation Register

- Video consultation latency: Low likelihood, medium impact. Mitigation: Keep telemedicine optional; SE Asia servers; phone fallback.
- AI chatbot medical liability: Low likelihood, High impact. Mitigation: Triage only; prohibit diagnosis; mandatory disclaimer; professional review.
- Scope too large for timeline: Medium likelihood, High impact. Mitigation: MoSCoW prioritisation; focus MVP; GitHub Projects tracking.
- Patient data privacy breach: Medium likelihood, High impact. Mitigation: AES-256 at rest; TLS 1.3; pseudonymous IDs; strict RBAC; PDPA compliance.
- Crisis features misuse: Low likelihood, Very High impact. Mitigation: Explicit confirmation; per-session consent; non-substitution disclaimer; static fallback helplines.
- Volunteer listener burnout: Medium likelihood, Medium impact. Mitigation: Mandatory training; code of conduct; 60-min limits; report button; moderator review.
- PayHere integration failure: Low likelihood, High impact. Mitigation: Cash payment alternative; slot reservation; webhook retry; status notification.

Appendix B: Glossary of Terms

- SLMC: Sri Lanka Medical Council. Statutory body for registering medical practitioners.
- SLPA: Sri Lanka Psychological Association. Professional body for psychologists.
- Sumithrayo: Volunteer-based Sri Lankan suicide prevention hotline. Helpline: 1926.
- CCCLine: Community Counselling Centre helpline. Helpline: 0112 696 666.
- Sahanaya: Counselling and rehabilitation centre. Helpline: 0115 679 900.
- PDPA: Personal Data Protection Act - proposed Sri Lankan data protection legislation.
- SithBot: AI triage chatbot. Provides specialist category recommendations only.
- Adult Mode: Simplified interface for elderly/low-digital-literacy users.
- Telemedicine: Remote consultation using video/audio communication tools.
- Pseudonymous ID: Platform identifier that hides real identity in community features.



Appendix C: GitHub Repository Artefacts

All project artefacts are maintained in the GitHub organisation repository `seng31242-sithcare-2026`. The repository structure follows the SENG 31242 guidelines:

- `documents/srs/`: SRS draft and final documents, use case descriptions, requirements tables
- `documents/sds/`: SDS document, architecture decisions, design diagrams
- `documents/diagrams/`: Use case, activity, sequence, state, class, ER diagrams (source + exports)
- `documents/pitch/`: Idea pitch deck and project approval form
- `documents/presentation/`: Design presentation slides
- `documents/meetings/`: Meeting notes, stand-up logs, sprint retrospectives
- `research/`: Interview transcripts, survey analysis, competitive analysis, literature review
- `prototypes/`: UI wireframes, Figma exports, low-fidelity prototypes